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INTERNSHIP REPORT

MCDA5587 and MCDA5588

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in partial fulfilment for the awarding of the degree:

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In conclusion, I am profoundly appreciative of all those who have played a part in shaping my internship experience as an ML Engineer. The knowledge and skills acquired during this period will undoubtedly serve as a solid foundation as I continue to explore and contribute to the field of machine learning and artificial intelligence.

CERTIFICATE

This is to certify that Mr. Piyush Priyam (A00464893), a student at Saint Mary's University in the Master of Science in Computing and Data Analytics Program, has completed the projects in fulfilment of the course requirements.

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ABSTRACT

This internship report presents an in-depth exploration of the development of machine learning models employing advanced AI technologies to deliver advanced analytics on e-commerce data. The primary objective of the internship was to design and implement algorithms capable of predicting various essential aspects of e-commerce operations. The scope of this endeavour encompassed customer-centric analyses such as customer lifetime value estimation, customer churn prediction, and product recommendations, along with broader business-focused analytics like sales prediction, time series analysis, and scientific attribution models.

The internship also delved into enhancing interactivity with the e-commerce platform and store using advanced language models, thus elevating the user experience and engagement. Furthermore, the report details the exploration of vector databases and the utilization of Large Language Models (LLMs) to harness their capabilities in enhancing the efficiency and effectiveness of e-commerce operations.

Throughout the internship, hands-on experience was gained in data pre-processing, feature engineering, machine learning algorithm development, model training, and evaluation. The application of machine learning techniques and AI technologies to real-world e-commerce data resulted in the creation of predictive models that offer valuable insights into customer behavior, sales trends, and operational performance.

This report not only highlights the technical aspects of model development but also underscores the practical implications of these advanced analytics in optimizing e-commerce strategies, enhancing customer satisfaction, and fostering business growth. The findings of this internship contribute to the growing body of knowledge at the intersection of AI, machine learning, and e-commerce, offering a glimpse into the immense potential of data-driven decision-making in the digital marketplace.

INTRODUCTION

Company Background

Lifesight Technologies is a pioneering technology company at the forefront of delivering innovative solutions in the realm of data intelligence and analytics. Founded with a vision to harness the power of data to drive meaningful insights, Lifesight is committed to revolutionizing industries through its cutting-edge technologies and expertise.

At Lifesight, our core focus is on transforming raw data into actionable intelligence. Leveraging advanced artificial intelligence (AI) and machine learning (ML) techniques, we empower businesses to uncover hidden patterns, trends, and opportunities within their data. Lifesight's comprehensive suite of data-driven solutions caters to a diverse range of industries, including e-commerce, marketing, retail, finance, and more.

Key Highlights:

- **Advanced Analytics:** Lifesight Technologies specializes in providing advanced analytics solutions that enable businesses to make informed decisions based on data-driven insights. Through the application of sophisticated algorithms and predictive modeling, we help organizations extract maximum value from their data assets.
- **E-commerce Expertise:** Lifesight expertise in the e-commerce domain has been instrumental in assisting businesses to optimize their online operations. Lifesight's solutions encompass a wide spectrum of e-commerce analytics, including customer behavior analysis, sales forecasting, and product recommendation engines.
- **Cutting-edge AI Technologies:** Lifesight is at the forefront of AI research and development. We harness the power of AI technologies, including machine learning, natural language processing, and deep learning, to create intelligent solutions that drive efficiency and innovation.
- **Enhancing Customer Engagement:** Lifesight's innovations extend beyond data analysis. We are committed to enhancing customer engagement through the integration of advanced language models, enabling seamless interactions and personalized experiences.

In conclusion, Lifesight Technologies stands as a trailblazer in the realm of data intelligence and analytics. With a commitment to innovation, advanced AI technologies, and industry expertise,

Lifesight empowers businesses to unlock the true potential of their data assets, drive growth, and stay ahead in an increasingly data-driven world.

Team Composition

Throughout the course of this internship, a dedicated and collaborative team worked collectively to achieve the objectives outlined within the scope. Each member brought their unique expertise and contributions, culminating in the successful development of advanced machine learning models for e-commerce analytics, driven by cutting-edge AI technologies.

- **Mr. Piyush Priyam**, ML Engineer Intern

As the intern, I undertook the pivotal role of researching, designing, implementing, and evaluating various machine learning algorithms to fulfil the scope's objectives. My dedication to learning and applying advanced AI techniques formed the backbone of the project's success.

- **Mr. Rohit Maheswaran**, Chief Product Officer (CPO) & Co-Founder

Rohit Maheswaran provided strategic guidance and valuable insights throughout the internship. His industry expertise and visionary leadership ensured that the developed solutions aligned with Lifesight's overarching goals.

- **Mr. Rajeev Nair**, Co-Founder & VP - Product

Rajeev Nair played a pivotal role in shaping the project's direction. His deep understanding of product requirements and market trends ensured that the developed models met industry standards and user expectations.

- **Mr. Anil Kumar Singh**, VP - Technology

Anil Singh provided crucial guidance in the application of advanced AI technologies. His technical insights and expertise helped in the selection and implementation of appropriate algorithms and methodologies.

- **Mr. Samuel Thiankee**, Principal Engineer

Samuel Thiankee contributed by providing architectural insights and recommendations. His role was instrumental in ensuring that the developed models were integrated seamlessly and effectively within the existing framework.

- **Mr. Shubham Jain, *Product Manager***

Shubham Jain facilitated the alignment of the developed solutions with practical use cases. His understanding of user requirements and market dynamics ensured that the models addressed real-world challenges.

The collective efforts of this diverse and dedicated team resulted in the successful development of advanced machine learning models catering to the specified analytics goals. The collaboration and synergy among team members were essential in achieving the desired outcomes within the internship's scope.

My Role

During my internship, I undertook a pivotal role in the development and implementation of machine learning models aimed at delivering advanced analytics on e-commerce data. The scope of my engagement was aligned with Lifesight Technologies' mission to leverage advanced AI technologies to provide actionable insights and enhance operational efficiency for e-commerce businesses. My role encompassed the following key responsibilities:

- **Algorithm Research and Design:** I conducted in-depth research to understand the intricacies of the various analytical aspects outlined in the scope. This involved studying existing literature, exploring industry best practices, and identifying suitable machine learning algorithms for predicting customer lifetime value, customer churn, product recommendations, sales predictions, time series analytics, and scientific attribution models.
- **Data Pre-processing and Feature Engineering:** I collaborated closely with the data engineering team to pre-process and transform raw e-commerce data into a suitable format for model training. This included data cleaning, feature selection, and engineering relevant features that contribute to accurate predictions.
- **Model Development and Implementation:** Utilizing Python and popular machine learning libraries such as TensorFlow, Robyn, Geolift and PyTorch, I developed, fine-tuned, and implemented predictive models for each of the specified analytics goals. This involved constructing appropriate machine learning pipelines, tuning hyperparameters, and iteratively refining the models to achieve optimal performance.

- **Performance Evaluation and Validation:** I conducted comprehensive evaluations of the developed models to ensure their accuracy and reliability. This encompassed employing various evaluation metrics, cross-validation techniques, and conducting A/B testing where applicable.
- **Integration of Language Models and Vector Databases:** As part of the broader scope, I explored the integration of advanced language models (LLMs) to enhance interactivity with the e-commerce platform and store. This involved experimenting with natural language processing (NLP) techniques to improve customer engagement and satisfaction.
- **Collaboration and Team Coordination:** Throughout the internship, I collaborated closely with cross-functional teams, including data engineers, solution architects, and product managers. Regular discussions and knowledge-sharing sessions ensured a holistic approach to model development and alignment with business objectives.
- **Documentation and Reporting:** I documented the entire process, including algorithm selection rationale, pre-processing steps, model architectures, and outcomes. This comprehensive documentation serves as a valuable resource for future reference and knowledge sharing.

In conclusion, my role within this internship was instrumental in realizing the ambitious goals outlined in the scope. By developing and implementing advanced machine learning models using cutting-edge AI technologies, I contributed to Lifesight Technologies' mission of providing actionable insights and innovative solutions to the e-commerce industry. The experience garnered during this internship has enriched my understanding of machine learning, AI technologies, and their real-world applications.

Tools and Technologies

Software Requirements

The successful execution of the internship project hinged on a carefully curated set of software tools, frameworks, and platforms that facilitated the development, implementation, and evaluation of advanced machine learning models for e-commerce analytics. The following software requirements formed the core of the project's technical infrastructure:

- Python (Version 2.11)
- R (Version 4.1)
- Jupyter Notebook (7.0.1)
- TensorFlow (Version 2.12)

- PyTorch (Version 2.0.1)
- Google Cloud Platform (GCP)
- Advanced Language Models (LLMs):
 - GPT-3
 - BERT
- Visualization Libraries:
 - Matplotlib
 - Seaborn
 - Plotty
- Integrated Development Environment (IDE):
 - PyCharm
- Version Control:
 - Bitbucket
- Docker
- Postman

The seamless integration of these software tools and platforms formed the technical backbone of the project, enabling the development of sophisticated machine learning models, the analysis of e-commerce data, and the realization of advanced analytics objectives. The judicious selection and utilization of these software resources were integral to the successful execution of the project's goals.

Hardware Requirements

The successful execution of the internship project was made possible through a well-configured hardware setup that provided the necessary computational power and resources to develop, train, and deploy advanced machine learning models. The following hardware specifications were integral components of the project's technical infrastructure:

- Processor: An Intel Core i7 processor
- Memory (RAM): With a substantial 32GB of RAM
- Graphics Processing Unit (GPU): NVIDIA GeForce RTX 3070 GPU with CUDA
- Storage: 1TB Solid State Drive (SSD)
- Cloud Computing Resources: Google Cloud Platform (GCP)

- Internet Connection: A stable high-speed internet connection
- Display: A high-resolution display, boasting Full HD
- Input Devices: Standard input devices, including a keyboard, mouse, and extended monitor.

The meticulous selection and integration of these hardware components established a powerful and responsive technical framework. This configuration not only supported the intricate computations required for advanced machine learning but also fostered efficient development, analysis, and deployment of e-commerce analytics models. The collaborative interplay of hardware and software resources was pivotal in achieving the objectives of the internship project.

Project Development Methodology

The successful execution of the internship project was guided by a structured and adaptive project development methodology. This methodology facilitated a systematic approach to achieving the project's objectives, ensuring efficient collaboration, effective management, and the delivery of high-quality results. The following development methodology was adopted for the internship project:

- **Agile Framework:** The project followed an Agile methodology, which emphasized flexibility, iterative development, and close collaboration among team members. This approach allowed for regular feedback and adjustments, ensuring alignment with evolving requirements.
- **Requirement Gathering and Analysis:**
 - The initial phase involved detailed discussions with stakeholders to precisely define project goals and scope.
 - Clear requirements were documented, encompassing the development of advanced machine learning models for e-commerce analytics.
- **Planning:**
 - A comprehensive project plan was formulated, outlining tasks, milestones, timelines, and resource allocation.
 - Prioritization of tasks based on their significance in achieving project objectives was established.
- **Design and Architecture:**
 - The design phase involved the conceptualization of algorithms, model architectures, and data pipelines.

- Architectural decisions were made, considering factors such as model integration, data flow, and technology stack.
- **Implementation and Iterative Development:**
 - The development process began with the implementation of foundational components, including data pre-processing, feature engineering, and model training.
 - Iterative cycles of development were adopted, enabling incremental improvements and adjustments based on ongoing feedback.
- **Testing and Validation:**
 - Rigorous testing was conducted to validate the correctness and functionality of implemented algorithms and models.
 - Unit tests, integration tests, and performance tests were performed to ensure reliable and accurate results.
- **Integration and Deployment:**
 - The developed models were integrated into the e-commerce analytics ecosystem, ensuring seamless interaction with other components.
 - Model deployment was carried out on the Google Cloud Platform's Vertex AI, enabling efficient serving and scalability.
- **Monitoring and Optimization:**
 - Continuous monitoring and evaluation of the deployed models were performed to identify and address performance bottlenecks.
 - Optimizations were implemented to enhance model accuracy, efficiency, and response times.
- **Documentation:**
 - Comprehensive documentation was maintained throughout the project lifecycle, encompassing code documentation, model descriptions, and usage guidelines.
- **Presentation and Reporting:**
 - The project's outcomes, insights, and results were presented to stakeholders through detailed reports and presentations.
 - Achievements, challenges, and recommendations for future enhancements were communicated effectively.

By adhering to this Agile project development methodology, the internship project benefited from flexibility, adaptability, and continuous improvement. Regular communication, iterative cycles of development, and a focus on delivering value ensured that the project successfully met its goals of developing advanced machine learning models for e-commerce analytics.

Goals

Learning Goals

The internship presented a unique opportunity for me to acquire valuable knowledge, skills, and practical experience in the fields of machine learning, data analytics, artificial intelligence, and advanced AI technologies. My learning goals for this internship were centred around:

- **Advanced Machine Learning Techniques:** To gain an in-depth understanding of advanced machine learning algorithms, including deep learning and ensemble methods, and their applications in real-world scenarios.
- **E-commerce Analytics:** To develop expertise in e-commerce data analysis, including customer behavior analysis, sales prediction, and recommendation systems, and to explore the challenges and opportunities in this domain.
- **Natural Language Processing (NLP):** To delve into the intricacies of NLP techniques, particularly with the integration of advanced language models such as GPT-3 and BERT for enhancing interactivity and user engagement.
- **Artificial Intelligence Fundamentals:** To grasp the foundational concepts of artificial intelligence, including its principles, applications, and ethical considerations, and to understand its role in shaping modern technologies.
- **Cloud Computing and Deployment:** To become proficient in deploying machine learning models on cloud platforms, particularly Google Cloud's Vertex AI, and to understand the nuances of scalability, maintenance, and resource management.
- **Collaboration and Communication:** To enhance my teamwork and communication skills by actively collaborating with cross-functional teams, sharing ideas, and effectively presenting complex technical concepts.

Internship Goals

In addition to my learning objectives, the internship was structured to achieve specific project goals that aligned with Lifesight Technologies' objectives. My internship goals encompassed:

- **Development of Advanced Analytics Models:** To design, implement, and evaluate machine learning models capable of delivering advanced analytics insights on e-commerce data, including customer lifetime value prediction, churn prediction, product recommendations, sales forecasting, and attribution modeling.
- **Integration of Language Models:** To successfully integrate and leverage advanced language models such as GPT-3 and BERT to enhance the interactivity and responsiveness of the e-commerce platform, thereby improving user engagement.
- **Utilization of Cloud Services:** To effectively deploy developed machine learning models using Google Cloud's Vertex AI, ensuring efficient model serving, scalability, and optimization for real-time use.
- **Data Visualization and Interpretation:** To employ visualization techniques using tools like Matplotlib, Seaborn, and Plotty to present analytical findings clearly and intuitively, enabling stakeholders to make informed decisions.
- **Documentation and Reporting:** To create comprehensive documentation detailing the project's methodology, implementation, and results, and to deliver well-structured reports and presentations that convey the project's achievements and insights.

By achieving these learning and internship goals, I aimed to contribute to Lifesight Technologies' projects, enhance my professional skill set, and lay a solid foundation for my future career in machine learning, artificial intelligence, and data analytics.

CHRONOLOGICAL TIMELINE

Welcoming Week

April 25th, 2023 - Orientation:

- Introduction to the company's culture, goals, and expectations.
- Overview of the internship program and objectives.

Task	Description	Hours
Orientation	During orientation, I observed the company's culture promoting collaboration, diversity, and innovation. The emphasis on these values was clear in the interactions among employees. The orientation also emphasized the company's goal of delivering exceptional client solutions, underscoring their commitment to exceeding client expectations and maintaining high-quality services. <u>As a result</u> • Attending the company's orientation provided me with insights into the organization's culture that emphasizes collaboration, diversity, and innovation. • This understanding will guide my interactions and contributions within the company. Furthermore, learning about the company's emphasis on delivering exceptional client solutions will drive me to align my efforts with this goal and strive for excellence in my work.	2
Team Introduction	I attended a team meeting aimed at facilitating introductions and fostering familiarity among team members. Each member took a moment to provide a concise introduction, outlining their respective roles and offering insights into their backgrounds and areas of expertise. This interactive session proved effective in promoting a deeper understanding of the team's composition and capabilities. <u>As a result</u> • Participating in the team meeting allowed me to establish connections and increase familiarity with my colleagues. This understanding of their roles and expertise will contribute to more effective collaboration and communication within the team.	2
Platform Access Setup	I took the initiative to ensure my access to essential platforms, ensuring that I had all the necessary tools to carry out my tasks efficiently. I also assisted myself in setting up and configuring these platforms, obtaining	2

Task	Description	Hours
	required permissions, and resolving any access-related issues. This proactive approach significantly improved the overall efficiency of my workflow. <u>As a result</u> • By securing access to the essential platforms and effectively configuring them, I am now equipped with the necessary platforms to carry out my tasks efficiently. • Overcoming access-related challenges has streamlined my workflow, enabling me to focus more on my responsibilities and contribute effectively to the team's goals.	
Familiarization	I actively participated in an exploration of the resources and tools that were provided, with the goal of acquainting the team with their functionalities and potential applications. Through a collaborative effort, we evaluated the capabilities of these resources and tools, pinpointing ways in which they could be seamlessly integrated into our projects and tasks for optimal results. <u>As a result</u> • By delving into the available resources and tools, I facilitated a deeper understanding of their functionalities. The collaborative assessment enabled us to identify innovative ways to integrate these resources and tools into our projects and tasks, enhancing our overall efficiency and productivity.	2

Table 1 – Orientation

April 26th, 2023 - Platform Orientation:

- Detailed walkthrough of the platforms and tools used for the project.

Task	Description	Hours
Platform Orientation	Received comprehensive platform orientation, gaining a thorough understanding of their interfaces, features, and navigation. <u>As a result</u> • Acquiring a comprehensive understanding of the platform interfaces, features, and navigation has equipped me with the knowledge and confidence to navigate and utilize these tools effectively.	2

Task	Description	Hours
	This orientation will significantly contribute to my ability to work efficiently and contribute to projects seamlessly, ensuring optimal outcomes for the tasks at hand.	
Accessing Data	Facilitated by team members, I learned how to access project datasets, ensuring I had the necessary knowledge to retrieve crucial data. <u>As a result</u> - This guidance from my team members has equipped me with the skills and understanding needed to confidently access and retrieve essential data from project datasets. - This newfound capability will enable me to contribute more effectively to data-driven tasks, make informed decisions, and support the success of our projects.	2
Environment Setup	Successfully configured the development environment, ensuring all required software and settings were properly established. <u>As a result</u> - Accomplished the successful configuration of the development environment has set the foundation for smooth and efficient workflow. - With all necessary software and settings in place, I am now well-prepared to engage in productive development tasks, contribute effectively to projects, and collaborate seamlessly with the team, ultimately leading to enhanced project outcomes.	3
Collaborative Tools Setup	Established communication and collaboration channels through platforms like Slack, ensuring seamless interactions and efficient information sharing among team members. Also, created Bitbucket account and configured Git for streamlined code collaboration, fostering a collaborative development environment and enabling effective version control. <u>As a result</u> - Creating robust communication and collaboration channels via Slack will enhance the ability to exchange information, share insights, and work together seamlessly. - The integration of Bitbucket and Git setup for code collaboration has further bolstered our development processes, allowing for efficient teamwork, version control, and code management. - These efforts collectively contribute to improved coordination, increased productivity, and better project outcomes.	1

Table 2 - Platform Orientation

April 27th, 2023 - Project Scope:

- Detailed understanding of the project's scope, objectives, and deliverables.

Task	Description	Hours
Project Scope Overview	Obtained a comprehensive breakdown of the project's overarching objectives, which provided clarity on both the desired outcomes and specific targets that needed to be accomplished. This detailed overview highlighted the key milestones and deliverables that were intricately linked to the project's core objectives, effectively charting a clear and concise roadmap for the project's progression. During this communication, the importance of each individual objective was underscored in relation to the project's overarching mission. This emphasis on context ensured that every team member possessed a thorough understanding of the vital work that lay ahead and the role that each objective played in driving the project's success. <u>As a result</u> • Having a comprehensive understanding of the project's overarching objectives provided me strong foundation for effective planning and execution. • The highlighted milestones and deliverables create a clear roadmap, enabling the team to track progress and stay aligned with project goals. • By emphasizing the importance of each objective in the context of the project's mission, team members are better equipped to contribute meaningfully and work collectively toward the successful realization of our project's mission.	2
Task Breakdown	Systematically deconstructed the project into well-defined tasks, outlining the necessary steps to achieve the overarching objectives. To ensure effective project management, established precise milestones within the project timeline, strategically positioned as checkpoints to assess progress and maintain alignment with the project's overarching goals. This involved collaborative efforts with team members, as we collectively set deadlines and structured a plan. This approach not only facilitated efficient task completion but also served as a navigational guide to keep the project seamlessly on track. <u>As a result</u> • Breaking down the project into manageable tasks provides a systematic approach to achieving the overarching objectives. The establishment of clear milestones empowers me to monitor progress and make necessary adjustments to stay on course.	2

Task	Description	Hours
	- Through collaborative effort, setting deadlines and creating a structured plan ensures that each team member understands their role, promotes accountability, and fosters efficient task completion. - This approach enhances our collective ability to meet project goals within the set timeframe while maintaining quality and alignment with our mission.	
Expectation Alignment	Actively participated in a comprehensive discussion aimed at defining the precise goals of the internship. This encompassed a detailed overview of the skills I would acquire during my time as an intern. Furthermore, the internship objectives were presented with measurable outcomes, ensuring that expectations were transparently communicated. In addition, a structured framework was outlined to assess my progress and accomplishments throughout the internship period. This holistic approach not only provided clarity regarding my role but also laid the foundation for a productive and goal-oriented internship experience. <u>As a result</u> - Participating in the comprehensive goal-setting discussion has provided a clear direction for my internship, encompassing the skills and knowledge I aim to acquire. - The establishment of measurable outcomes ensures that I have a tangible framework to guide my efforts and gauge my performance throughout the internship. - This structured approach will enable me to focus on targeted skill development and contribute effectively to the team's objectives, ultimately enhancing my learning experience and overall impact as an intern.	2
Task Assignment	Collaboratively identified my initial tasks, ensuring a timeline distribution of responsibilities. <u>As a result</u> - Through collaborative efforts, the identification of my initial tasks and their timeline distribution has established a structured plan for my responsibilities. - This approach enhances my clarity, accountability, and efficient task execution, ultimately contributing to a more organized and productive work environment.	2

Table 3 - Project Scope

April 28th, 2023 - Technical Setup:

- Setting up the technical environment required for the project.

Task	Description	Hours
Software Installation	Successfully installed the essential software and tools required for the project, guaranteeing a smooth workflow. <u>As a result</u> • Completing the installation of necessary software and tools ensures a seamless workflow, enabling me to work efficiently and contribute effectively to the project's progress. • This proactive step sets the foundation for productive tasks and supports the team's goals, ultimately leading to successful project outcomes.	2
Platform Integration	Collaboratively integrated various platforms to streamline data access, allowing us to retrieve and utilize project datasets seamlessly. This integration established a harmonious synchronization between platforms, thereby enhancing data sharing efficiency and minimizing the time needed for accessing crucial information. To ensure a seamless process, we verified the compatibility and functionality of the integrated platforms, proactively addressing any potential issues. This diligence ensured consistent and reliable data access, contributing to the overall smooth functioning of our workflow. <u>As a result</u> • The collaborative integration of diverse platforms has significantly improved data accessibility, enabling me to retrieve and leverage project datasets seamlessly. • The seamless synchronization between these platforms enhances data sharing efficiency, saving valuable time, and contributing to more effective collaboration. • By verifying compatibility and functionality, I established a reliable data access system.	2
Data Exploration	Conducted an initial exploration of the provided datasets to gain insights into their structure, contents, and potential relevance to the project. Through this analysis, I discerned patterns, anomalies, and preliminary trends that could serve as guideposts for the project's trajectory and informed decision-making. Engaging in collaboration with my fellow team members, I openly shared the insights and observations gleaned from this dataset exploration. This collaborative exchange nurtured a collective	2

Task	Description	Hours
	comprehension of the data's nuances and laid the foundation for subsequent analyses and actions. <u>As a result</u> • The initial exploration of the datasets has provided a foundational understanding of their structure and contents, offering valuable insights into their potential relevance to the project's objectives. • By identifying patterns, anomalies, and initial trends, we have laid the groundwork for informed decision-making and strategic project direction. • Collaborating with team members to share our findings enhances our collective knowledge and creates a solid basis for deeper analysis and exploration. • This collaborative effort will contribute to more effective decision-making, better project outcomes, and a cohesive team approach towards achieving our goals.	
Coding Environment	Set up the project's coding environment by configuring vital Integrated Development Environments (IDEs), libraries, and dependencies. To maintain uniformity, I installed foundational packages specific to each IDE, ensuring the coding environment's stability and reliability. <u>As a result</u> • The successful setup of the coding environment, complete with configured IDEs, libraries, and dependencies, lays a solid foundation for streamlined development. • Installed basic packages for specific IDEs guarantees a consistent and well-prepared environment. • This proactive approach reduces potential roadblocks, and empowers to focus on productive coding, ultimately leading to effective development processes and successful project outcomes.	2

Table 4 - Technical Setup

Project 1 - Product Recommendations & Consumer Basket Analysis

Week 1: May 1st - May 5th, 2023 - Project Initialization and Data Exploration:

- Setting up the project environment and exploring the datasets.

Task	Description	Hours
Project Kick-off	Provided a comprehensive introduction to the project's overarching goals and scope, detailing the intended outcomes and specific project boundaries. During discussions, we deliberated on scope limitations to proactively prevent potential scope creep. This concerted effort aimed to guarantee that all team members held a unified understanding of the project's defined boundaries, as well as a clear grasp of the tasks encompassed within the stipulated scope. <u>As a result</u> • Delivering a thorough overview of the project's goals and scope has established a strong foundation for understanding its purpose and direction. • Open discussions about scope limitations have effectively mitigated the risk of scope creep, ensuring that all team members are on the same page regarding the project's boundaries. • This shared clarity enhances collaboration, prevents unnecessary deviations, and contributes to the efficient and focused execution of tasks within the defined project scope.	2
Environment Setup	Created a dedicated virtual environment for the project, effectively isolating dependencies and maintaining a controlled development environment. Within this virtual environment, I adeptly installed the required packages, enabling smooth utilization of specific libraries and tools without encountering any conflicts. This approach promoted efficient and organized development practices. <u>As a result</u> • By creating a specialized virtual environment, dependencies are isolated, promoted a controlled and stable development environment. • The successful installation of required packages within this environment enables efficient utilization of specific libraries and tools, will foster a productive workflow devoid of compatibility issues. • This approach will enhance the quality of work and minimize disruption.	1

Task	Description	Hours
Data Collection	Collected raw e-commerce datasets from BigQuery, sourcing the foundational data essential for analysis and subsequent processing. Employing SQL queries, I executed the retrieval process to extract pertinent information from BigQuery. This meticulous approach guaranteed that the acquired datasets contained the requisite data to align with the project's objectives effectively. <u>As a result</u> • The successful retrieval of raw e-commerce datasets from BigQuery has provided the foundational data required for analysis and subsequent processing. • The execution of SQL queries ensured that the acquired datasets are comprehensive and aligned with the project's specific objectives. • This data acquisition sets the stage for informed decision-making and enables me to move forward with data-driven insights and analyses.	1
Data Loading and Inspection	Utilized Pandas to load the obtained datasets, transforming them into a structured format within Python to facilitate convenient manipulation and analysis. I initiated an initial inspection of the loaded data, conducting data profiling to glean insights into its structure, quality, and attributes. Leveraging Pandas' features, I conducted fundamental exploratory data analysis (EDA), including generating summary statistics, detecting missing values, and creating visualizations to understand data distributions. Through collaboration with my team, I shared findings and insights from this preliminary data examination. This collaborative effort laid the groundwork for well-informed decision-making and guided subsequent analyses. <u>As a result</u> • The utilization of Pandas to load and structure of acquired datasets in Python has streamlined data manipulation and analysis. Through an initial examination and data profiling, valuable insights into the data's structure, quality, and attributes were gained. • By employing Pandas' capabilities for exploratory data analysis (EDA), including summarizing statistics and visualizing distributions, we've begun to uncover meaningful patterns and trends. • Collaborative sharing of observations and insights with team members enhanced our collective understanding and paved the way for well-informed decision-making and more in-depth analysis.	2

Task	Description	Hours
Data Pre-processing	Identified missing values in the dataset through strategies like imputation or removal, ensuring data completeness. Using statistical methods and visualization techniques, I identified outliers and assessed their impact on the analysis. I made informed decisions, choosing to retain, transform, or exclude outliers based on their relevance. Working collaboratively with team members, we ensured uniformity and accuracy in addressing missing values and outliers. Employing suitable techniques, I imputed missing values while considering factors like data type, distribution, and domain expertise. This approach bolstered the integrity of the analysis. <u>As a result</u> • The identification of missing values and the implementation of strategies like imputation and removal have contributed to enhancing data completeness, ensuring the integrity of our analysis. • The detection of outliers through statistical methods and visualization has provided insights into potential anomalies and their impact on the analysis. • Collaborative decision-making within the team has fostered consistent and accurate handling of missing values and outliers, further strengthening the reliability of our findings. • By applying suitable techniques for imputing missing values, we have taken a data-driven approach that considers various factors, contributing to more accurate and insightful analysis of our dataset.	4
Exploratory Data Analysis	Created a variety of data visualizations, employing techniques such as bar charts, scatter plots, and histograms to vividly portray the dataset's patterns and interrelationships. Leveraging libraries like Matplotlib, Seaborn, and Plotly, I crafted these visualizations to provide a comprehensive understanding of the data's distribution and trends. In close collaboration with team members, we carefully selected the most suitable visualization methods to effectively communicate insights and align with the project's goals. This collaborative effort enriched our interpretation of the data and bolstered our overall project objectives. <u>As a result</u> • The creation of diverse data visualizations using techniques like bar charts, scatter plots, and histograms has provided an intuitive representation of the dataset's underlying patterns and relationships. By leveraging libraries such as Matplotlib and Seaborn, we have enriched the team's comprehension of the data's distribution and trends.	2

Task	Description	Hours
	- Collaborative discussions within the team have led to the selection of the most impactful visualization methods, ensuring that our insights are effectively conveyed and align with the project's objectives and conveyed in a standard format. - These visualizations not only facilitate clearer understanding but also contribute to well-informed decision-making, enhancing the overall quality and value of our analysis.	
Feature Identification	Explored an in-depth exploration of the dataset to pinpoint potential features that could enhance the analysis and align with the project's objectives. Through a preliminary feature selection process, I assessed factors such as relevance, uniqueness, and potential influence on the desired outcomes. Collaborating closely with domain experts and fellow team members, we validated the chosen features to ensure their alignment with the analysis objectives. This collaborative effort guaranteed the inclusion of meaningful attributes that would contribute effectively to our analysis. <u>As a result</u> - Thoroughly exploring the dataset to pinpoint potential features that align with the project's goals has laid a strong foundation for our analysis. - The preliminary feature selection process, incorporating criteria such as relevance, uniqueness, and potential impact, ensures that our chosen features are well-suited for achieving the desired outcomes. - Collaborated with domain experts and team members to validate these selected features guaranteed that our analysis is grounded in expertise and aligned with project objectives. - This meticulous feature selection process enhanced the accuracy and effectiveness of our analysis, contributing to more meaningful insights and informed decision-making.	2
Initial Model Exploration	In the pursuit of selecting suitable Machine Learning (ML) algorithms for collaborative filtering, an extensive research endeavor was undertaken. Various techniques were explored, including matrix factorization, user-item embeddings, and neighborhood-based methods, while also delving into advanced algorithms like Singular Value Decomposition (SVD), Alternating Least Squares (ALS), and factorization machines. This investigation involved a comprehensive evaluation of their respective strengths and limitations, tailored to the specific context of the project at hand.	8

Task	Description	Hours
	<u>As a result</u> - The research conducted on a range of Machine Learning algorithms tailored for collaborative and hierarchical filtering has provided a solid understanding of the landscape. - Exploring various techniques, from matrix factorization to deep learning models, and assessing their applicability, strengths, and weaknesses has paved the way for well-informed decisions.	
Model Exploration	Explored hierarchical filtering methods, focusing on decision tree-based algorithms like Random Forests and Gradient Boosting. These algorithms showed potential in capturing complex relationships within data containing hierarchical structures. Additionally, investigated the applicability of deep learning models such as Recurrent Neural Networks (RNNs) and Transformers for hierarchical filtering tasks. Special emphasis was placed on their ability to capture sequential dependencies in the data. <u>As a result</u> - These explorations provided valuable insights into the potential of decision tree-based algorithms and deep learning models for hierarchical filtering tasks. - The investigation paved the way for informed decisions regarding the selection and implementation of appropriate methods within the project.	8
Model Exploration	Had discussions revolved around the feasibility and suitability of different ML algorithms. Factors like data size, complexity, interpretability, and scalability for large datasets were carefully considered. Algorithm performance metrics including accuracy, precision, recall, and F1-score were evaluated to gauge the effectiveness of diverse models in achieving the intended outcomes. Regular review of research papers, articles, and online resources ensured the incorporation of the latest advancements in both collaborative and hierarchical filtering techniques. <u>As a result</u> - Collaborative discussions within the team, considering factors such as scalability and performance metrics, ensure that the chosen algorithms align with project needs. - By documenting the findings and sharing them with the team, we have established a comprehensive reference that guides future	6

Task	Description	Hours
	implementation and fosters a deeper understanding of algorithmic choices. Ultimately, this comprehensive research supports the selection of the most suitable algorithms, enabling us to advance confidently toward the project's goals while leveraging the latest advancements in ML techniques.	
Task Review and Planning	At the end of Week 1, a comprehensive review assessed completed tasks and outcomes, identifying challenges and adjustments for the following week. Collaboratively planning Week 2 tasks factored in project timelines and priorities, optimizing efficiency and alignment with goals. <u>As a result</u> - The thorough review of Week 1's tasks and outcomes has provided valuable insights into our progress and potential areas for improvement. - By identifying challenges and necessary adjustments, we are well-equipped to address any hurdles and enhance our efficiency in the upcoming week. - The collaborative planning for Week 2 considered the project's timeline and priorities, ensuring a strategic allocation of tasks that align with our objectives. - This proactive approach promotes better coordination, optimized workflow, and a higher likelihood of achieving our project milestones successfully.	1
Model Selection Research Documentation	The investigation spanned a spectrum of models including regression, classification, clustering, and neural networks, with a critical evaluation of their relevance to the dataset and problem domain. The research encompassed both traditional statistical models and contemporary deep learning architectures, factoring in considerations like model complexity, interpretability, and computational demands. In collaboration with domain experts and team members, the identified models were validated, drawing from their insights and expertise to ascertain their appropriateness. The outcome of this research was documented, capturing the characteristics, benefits, and limitations of each model. This documentation served as a valuable guide for decision-making, offering transparency in the model selection process. By methodically considering a diverse range of models, their specific attributes, and expert input, the research ensured that the chosen model aligns optimally with the	4

Task	Description	Hours
	project's requirements, leading to informed and effective decision-making. <u>As a result</u> • The extensive research conducted to identify suitable machine learning models has equipped us with a comprehensive understanding of our options in addressing the project's objectives and challenges. • By exploring a variety of models spanning regression, classification, clustering, and neural networks, we've considered a diverse range of approaches tailored to our specific dataset and problem domain. • Balancing traditional statistical models with modern deep learning architectures ensured a well-informed perspective on model complexity, interpretability, and computational demands. • Collaborating with domain experts and team members has further validated the appropriateness of our chosen models. • Documenting the research findings fosters a transparent decision-making process, providing a solid foundation for selecting the most fitting model that aligns with our project's goals and requirements.	
Communication and Updates	Attended a team meeting to provide updates on project progress, discussing achievements, challenges, and solutions, while collaborating with team members to ensure alignment and consider adjustments for an optimized plan. <u>As a result</u> • The team meeting served as a crucial platform to communicate updates, fostering transparency and a shared understanding of the project's advancement. • By sharing insights into completed tasks, challenges encountered, and the corresponding solutions, we promoted a comprehensive awareness of our collective efforts. • Collaborative discussions further ensured that all team members were on the same page regarding the project's status. Additionally, exploring potential adjustments to the plan based on achieved progress allowed us to adapt proactively and optimize our approach for optimal outcomes.	1

Table 5 - Week 1 (Product Recommendations & Consumer Basket Analysis)

Week 2: May 8th - May 12th, 2023 - Feature Engineering and Model Selection:

- Extracting relevant features and selecting initial machine learning models.

Task	Description	Hours
Feature Extraction	Analyzed available data to pinpoint pertinent features like purchase history, browsing patterns, and interaction frequencies, offering insights into customer behavior. Employing data transformation techniques, I extracted customer behavior information from raw data, creating a structured format conducive to analysis. Techniques including cohort analysis, sessionization, and user segmentation were employed to delve into customer behavior patterns and trends. Collaborating with data engineers, I ensured the efficient retrieval of necessary data while addressing technical hurdles that emerged. The extraction process was documented in detail, outlining methods, data sources, and preprocessing steps, ensuring transparency and reproducibility. <u>As a result</u> • The analysis of available data has enabled the identification of pertinent features and attributes, shedding light on customer behavior patterns such as purchase history and interaction frequencies. • Through strategic data transformation, I extracted and structured customer behavior information for meaningful analysis. By employing techniques like cohort analysis, sessionization, and user segmentation, I delved deeper into understanding complex behavior trends. • Collaborating with data engineers has ensured smooth data extraction, effectively addressing technical complexities.	4
Data Transformation	Conducted scaling of numerical features to establish uniformity for effective variable comparison. Observed heightened model performance, particularly with algorithms sensitive to varying scales, after implementing scaling techniques. Safeguarded inherent data patterns during scaling procedures, upholding data integrity and minimizing potential distortions. Adhered to algorithm specifications by selecting appropriate scaling methods like normalization and standardization, fostering optimal model interaction. Ensured seamless integration of categorical variables into the model by performing essential encoding (one-hot/label), enabling accurate representation in numerical form.	4

Task	Description	Hours
	<u>As a result,</u> • The scaling of numerical features has promoted uniformity, facilitating accurate variable comparison within the analysis. The observed improved model performance, especially with algorithms sensitive to scale, signifies the significance of implementing scaling techniques. • Throughout the process, safeguarding inherent data patterns has maintained the authenticity of our data, reducing the risk of introducing inaccuracies. • By adhering to algorithm specifications and selecting suitable scaling methods, we've enhanced the interaction between our models and the data. • Furthermore, the seamless integration of categorical variables through essential encoding ensures accurate numerical representation, enabling a holistic approach to model development that accommodates both numerical and categorical inputs.	
Model Selection	Explored a range of recommendation algorithms to pinpoint the best options for distinct use cases and data attributes. This involved scrutinizing the strengths and limitations of algorithms, taking scalability, personalization, and data sparsity into account. Collaborative filtering, content-based, and hybrid methods were thoroughly examined to ascertain the most fitting approaches for generating precise and impactful recommendations. This comprehensive analysis aimed to ensure optimal recommendation strategies that align seamlessly with the project's goals. <u>As a result</u> • The comprehensive investigation into a range of recommendation algorithms has provided valuable insights for tailoring our choices to specific use cases and data attributes. • Evaluating the strengths and weaknesses of various algorithms in terms of scalability, personalization, and data sparsity ensures that our selection aligns with the project's requirements. • The exploration of collaborative filtering, content-based, and hybrid approaches enables us to strategically choose the most fitting strategies for delivering recommendations that are accurate, effective, and tailored to the unique characteristics of our data and objectives.	2

Task	Description	Hours
Model Implementation	Imported necessary libraries such as NumPy, Pandas, and Scikit-learn for data manipulation and computations. Loaded the dataset and pre-processed it. This involved handling missing values, transforming data into a user-item interaction matrix, and splitting data into training and testing sets. Chose a similarity metric coefficient to measure the similarity between users and items. Calculated the similarity matrix based on the chosen metric. This involved computing similarities between users and items using the interaction matrix. <u>As a result</u> - The strategic import of essential libraries streamlined data manipulation and computation tasks. Loading and pre-processing the dataset, including handling missing values and data transformation, ensured a clean and organized foundation for analysis. - The choice and calculation of similarity metrics facilitated the quantification of user and item similarities, essential for collaborative filtering.	8
Model Implementation	In user-based collaborative filtering, I predicted ratings and preferences for uninteracted items by calculating weighted averages of ratings from similar users. For item-based collaborative filtering, I predicted ratings for items by considering ratings and interactions of similar items. To gauge model performance, I evaluated it using suitable metrics like Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). I also fine-tuned the model by experimenting with hyperparameters like neighborhood size (for user-based CF) and the number of similar items (for item-based CF) to enhance performance. Furthermore, I addressed the "cold start" challenge for new users and items with limited interaction data by implementing relevant solutions. These steps collectively enriched the collaborative filtering model's accuracy and robustness. <u>As a result</u> - Through user-based and item-based approaches, the model efficiently predicted user preferences and item ratings. - Rigorous evaluation using metrics like MAE and RMSE verified the model's effectiveness. - Hyperparameter tuning optimized model performance by iteratively experimenting with key parameters. Additionally, solutions were	8

Task	Description	Hours
	thoughtfully implemented to address the "cold start" problem, enhancing the model's robustness in handling new users and items with limited interaction data.	
Model Evaluation	Calculated accuracy to assess the overall correctness of predictions about actual outcomes. Also, computed precision, evaluating the proportion of correctly predicted positive instances among all predicted positives. Furthermore, evaluated recall, indicating the fraction of actual positive instances that were correctly predicted among all actual positives. <u>As a result</u> - By calculating accuracy, precision, and recall, we gained a comprehensive understanding of the model's performance in predicting outcomes. Accuracy provides an overview of the overall correctness of predictions, while precision and recall delve deeper into the specifics of positive predictions. Precision measures the accuracy of positive predictions, and recall highlights the model's ability to capture all relevant positive instances. - These metrics collectively offer insights into different aspects of the model's effectiveness, aiding informed decision-making and refinement of predictive capabilities.	2
Model Tuning	Delved into a range of hyperparameters to enhance model performance, utilizing methods like grid and random search to optimize their values. Employing techniques such as cross-validation, I determined the most suitable hyperparameter configurations, ensuring the model was fine-tuned to achieve optimal performance. These approaches collectively contributed to refining the model's accuracy and effectiveness in generating reliable predictions. <u>As a result</u> - The exploration of diverse hyperparameters has played a crucial role in refining model performance. Leveraging techniques such as grid and random search has allowed us to systematically explore a range of hyperparameter values, optimizing the model's configuration. - Employing cross-validation adds rigor to the process, enabling the selection of hyperparameters that consistently perform well across different data subsets. - This comprehensive approach to hyperparameter tuning enhances our model's accuracy, robustness, and generalization capabilities.	2

Task	Description	Hours
Cross-Validation	Cross-validation was employed to rigorously assess the model's performance across diverse subsets of the data. The dataset was partitioned into multiple folds, with the model being trained iteratively on some folds while tested on others. For each fold, evaluation metrics such as accuracy, precision, recall, and F1-score were computed, ensuring a comprehensive evaluation of the model's consistency. By aggregating these performance metrics, a holistic assessment of the model's overall effectiveness was obtained, affirming its reliability across varying data scenarios. <u>As a result</u> • By leveraging cross-validation, we have established a robust method for assessing the model's performance across various data subsets. • The dataset was split into multiple folds, allowing us to train and test the model iteratively on different partitions. By calculating evaluation metrics like accuracy, precision, recall, and F1-score for each fold, we obtained a clear understanding of the model's consistency. • Aggregating the performance metrics provided an encompassing view of the model's effectiveness, enabling us to confidently gauge its capabilities and make informed decisions based on reliable performance assessments.	2
Model Updates	Analyzed monitoring data to extract valuable insights into the system's behavior and performance. This involved recognizing recurring patterns, emerging trends, and irregular anomalies present within the amassed data. Based on this analysis, I formulated targeted recommendations for updates aimed at improving system efficiency, reliability, and security. It was ensured that these suggested updates were precisely aligned with the identified issues, providing substantial contributions to the overall enhancement endeavors. <u>As a result</u> • Thoroughly examining the monitoring data has afforded valuable insights into the system's dynamics and performance. The identification of patterns, trends, and anomalies within the dataset enhanced our understanding of operational intricacies. • Through data-driven analysis, we have proposed updates aimed at elevating system efficiency, bolstering reliability. • Crucially, these recommendations are intricately aligned with the specific issues identified, ensuring that the proposed updates	2

Task	Description	Hours
	substantively contribute to the broader objective of enhancing the model's overall functionality and resilience.	
Performance Optimization	Integrated the suggested modifications into the system, following the guidance provided by the identified areas for improvement. This encompassed the implementation of changes designed to elevate functionality, fine-tune performance, and enhance the overall user experience. A crucial aspect of this process involved ongoing monitoring of the effects of the implemented enhancements, verifying their impact and confirming the attainment of improved outcomes. This approach ensured that the system remained dynamic and responsive to evolving needs while consistently delivering enhanced results. <u>As a result</u> - Through the seamless integration of recommended modifications, the system has undergone transformative changes in alignment with pinpointed areas for improvement. - Implementation efforts have been dedicated to elevating functionality, bolstering performance, and creating a more user-centric recommendations. - By actively monitoring the effects of these enhancements, we have established a feedback loop that validates their positive impact, ensuring that our modifications are effectively contributing to the overarching goal of delivering better results and a more refined system.	2
Evaluation of the Model	Evaluated the model's performance following hyperparameter optimization to measure its progress. Utilizing relevant evaluation metrics such as accuracy, precision, recall, and F1-score, I quantified the enhanced effectiveness achieved through the optimized model. By conducting a comparative analysis between the outcomes post-optimization and the initial baseline, I substantiated the extent of the improvements attained. This process validated the efficacy of the optimization efforts and provided concrete evidence of the model's advancement. <u>As a result</u> The evaluation of the model's performance after hyperparameter optimization has provided a quantifiable measure of its enhanced capabilities. By employing relevant evaluation metrics like accuracy, precision, recall, and F1-score, we've effectively gauged the increased effectiveness of the optimized model.	2

Task	Description	Hours
	A direct comparison between the post-optimization results and the baseline has confirmed the substantial improvements achieved, solidifying the significance of the model's refinement through meticulous hyperparameter tuning.	
Task Review and Planning	Conducted a comprehensive review of tasks completed during Week 2, enabling effective progress tracking and identification of any remaining items. Through collaborative discussions, I participated in the formulation of a task plan for Week 3. This plan outlined distinct objectives, prioritized tasks, and set specific timelines, fostering a structured and efficient workflow. <u>As a result</u> - The review of Week 2's completed tasks enabled us to monitor our advancement and recognize any tasks that require attention. - Through collaborative discussions, we have laid out a comprehensive task plan for Week 3, ensuring that our objectives are well-defined, tasks are appropriately prioritized, and timelines are established. - This strategic approach set the stage for an efficient and organized workflow.	1
Communication and Updates	Participated in the team meeting to engage in discussions about ongoing projects and deliver status updates. <u>As a result</u> Actively engaged in productive discussions concerning ongoing projects and contributed by providing comprehensive status updates.	1

Table 6 - Week 2 (Product Recommendations & Consumer Basket Analysis)

Week 3: May 15th - May 19th, 2023 - Model Tuning and Optimization:

- Fine-tuning the models and addressing issues.

Task	Description	Hours
Model Hyperparameter Tuning	Identified crucial hyperparameters within the model that wield significant influence over its behavior and results. Utilizing techniques like grid search and random search, I methodically explored a range of hyperparameter values. Model performance was evaluated through metrics such as cross-validation accuracy and loss, analyzed across a variety of hyperparameter setups. By assessing these performance indicators, I identified the optimal hyperparameters that best elevated the model's effectiveness and enhanced its ability to generalize and adapt. <u>As a result</u> • By pinpointing critical hyperparameters that wield significant influence over the model's behavior, we've laid the groundwork for precise optimization. • Through the strategic utilization of techniques like grid search and random search, we systematically investigated a wide array of hyperparameter values. • The evaluation of model performance, spanning metrics such as cross-validation accuracy and loss, across diverse hyperparameter setups provided the necessary insights to inform our selections. • The identification of the best-performing hyperparameters amplifies the model's prowess, enhancing its overall performance.	2
Bug Fixes and Improvements	Reviewed the model's codebase, diligently identifying and rectifying errors and bugs that hindered its functionality. Through precise examination of error logs, exception messages, and debugging tools, I accurately diagnosed the underlying causes of encountered issues. Data pre-processing procedures were enhanced to guarantee pristine input quality for the model. I conducted exploratory data analysis to extract insights into potential data patterns and anomalies, thereby refining subsequent modeling decisions. Exploring a variety of algorithms and architectures, I identified the optimal solution tailored to the specific problem. <u>As a result</u> • By rigorously scrutinizing the model's codebase, diagnosing errors, and rectifying bugs, we have fostered an environment of heightened functionality.	4

Task	Description	Hours
	Leveraging analysis of error logs, exception messages, and debugging tools, we have pinpointed and addressed the root causes of discrepancies. The enhancement of data pre-processing steps ensures pristine input quality, while the deployment of exploratory data analysis unearths invaluable insights.	
Bug Fixes and Improvements	Fine-tuning hyperparameters was executed judiciously to optimize model behavior and enhance performance. To mitigate overfitting, strategic techniques like regularization and dropout were implemented, bolstering the model's generalization capabilities. I executed feature engineering strategies to distill meaningful insights from the data, thereby elevating the model's predictive capacity. I validated modifications using cross-validation and separate test datasets, enabling measurable assessment of the achieved enhancements. <u>As a result</u> - Diverse algorithms and architectures have been explored to arrive at the most suitable solution, hyperparameters have been fine-tuned for optimal performance, and protective measures like regularization and dropout have been implemented. - Strategic feature engineering has further bolstered the model's predictive capabilities. Validation through cross-validation and distinct test datasets guaranteed the quantifiable impact of changes, and diligent documentation safeguards a clear roadmap for future endeavors.	8
Cross-Validation	Partitioned the dataset into multiple subsets using cross-validation methodology. This involved iteratively training the model on one subset while validating its performance on the remaining subsets. I then computed performance metrics to assess the model's consistency and its ability to generalize effectively across different data partitions. <u>As a result</u> - By segmenting the dataset through cross-validation, we have established a robust mechanism for training and validating the model. The iterative process of training on one subset and validating on others enabled us to thoroughly assess the model's performance across various data partitions. - Performance metrics provided valuable insights into the model's consistency and its ability to generalize its learnings beyond the training subset.	2

Task	Description	Hours
	This comprehensive evaluation approach enhances our understanding of the model's predictive capabilities and its behavior across diverse data scenarios.	
Model Evaluation	Calculated accuracy to determine the overall correctness of predictions in relation to actual outcomes. Additionally, I computed precision to gauge the proportion of correctly predicted positive instances among all instances predicted as positive. Furthermore, I evaluated recall, indicating the fraction of actual positive instances that were correctly predicted among all actual positive instances. These metrics collectively provided a comprehensive understanding of the model's performance in terms of correctness, precision, and recall, contributing to a holistic assessment of its effectiveness. <u>As a result</u> - By computing accuracy, precision, and recall, we have gained a comprehensive understanding of the model's predictive performance. - Together, these metrics offer a well-rounded evaluation of the model's effectiveness in making predictions and identifying positive cases, contributing to informed decision-making and performance optimization.	2
Model Updates	Carried out an in-depth analysis of the collected monitoring data to extract valuable insights into the behavior and performance of the system. This involved identifying patterns, anomalies, and potential areas for improvement present within the dataset. Subsequently, I formulated well-informed recommendations for updates and adjustments, with the goal of enhancing system efficiency, reinforcing reliability, and strengthening security measures. <u>As a result</u> - Through the review of collected monitoring data, we've acquired valuable insights into the system's operational dynamics and performance. By identifying noteworthy patterns, anomalies, and potential areas for enhancement within the data, we've cultivated a deep understanding of the system's intricacies. - Leveraging this analysis, we've formulated well-considered recommendations that advocate for updates and adjustments. These proposed changes were aligned with the goal of enhancing system efficiency, and fostering greater reliability, thus laying the groundwork for a more robust and resilient operational framework.	2

Task	Description	Hours
Performance Optimization	Integrated the suggested enhancements and updates, aligning pertinent elements of the system with the enhancement strategy. This process involved seamlessly integrating refined components within the existing system architecture. I conducted thorough testing procedures to confirm the successful achievement of intended outcomes. Subsequently, I continuously monitored the post-implementation performance of the system to validate the effectiveness of the enhancements. In addition, I actively engaged with users and stakeholders, gathering their feedback to assess their satisfaction and gain insights for potential further refinements. This collaborative approach ensured that the enhancements were meeting the needs and expectations of the users and stakeholders, contributing to the iterative improvement of the system. <u>As a result</u> • By seamlessly integrating the suggested enhancements into the system, we have embraced a proactive approach to refining its functionality. The modifications, aligning with the enhancement plan, have been carefully woven into relevant components, harmonizing the existing architecture. • Rigorous testing and performance monitoring have affirmed the attainment of desired outcomes. Engaging users and stakeholders in the feedback loop has provided valuable insights, facilitated satisfaction assessment, and drove continuous improvement. This holistic approach ensures the realization of an enhanced system that not only meets but exceeds expectations.	8
Task Review and Planning	At the conclusion of Week 3, I conducted a meticulous evaluation of the tasks completed and the progress made. This involved a comprehensive analysis of outstanding items, allowing for the identification of areas requiring additional attention and resolution. Engaging in collaborative discussions, I contributed to the formulation of a thorough task plan for Week 4. This plan outlined clear objectives, prioritized tasks, and established well-defined timelines. <u>As a result</u> • The review of the tasks completed during Week 3 has allowed us to gauge our progress and achievements accurately. Through a diligent analysis of pending items, we have pinpointed areas requiring dedicated attention and further effort. • Collaboratively devising a well-structured task plan for Week 4, replete with clear objectives, task priorities, and established	2

Task	Description	Hours
	timelines, facilitated an organized and efficient allocation of work. This strategic roadmap positioned us to seamlessly move forward, confidently work toward the successful progression of the project.	
Communication and Updates	Participated in a team meeting to deliver a detailed update on the project's ongoing progress and recent advancements. During the meeting, I shared comprehensive insights encompassing achieved milestones, challenges faced, and upcoming tasks. <u>As a result</u> - The dissemination of information pertaining to achieved milestones, encountered challenges, and forthcoming tasks served to strengthen team cohesion and ensured that all members are well-informed and aligned. - This proactive approach contributed to an informed and motivated team, enhancing our collective ability to navigate the project's trajectory with unity and precision.	1
Model Deployment Preparation	The finalization of the trained model by preserving its parameters and architecture for future use. I compiled all necessary code, libraries, and dependencies into a unified package to ensure the model's smooth functionality within a deployment environment. To ensure the model's compatibility and responsiveness, I rigorously tested it within the designated deployment environment, ensuring optimal performance before granting user access. <u>As a result</u> - The comprehensive finalization of the trained model, encompassing parameter retention and architectural preservation, laid the foundation for future application. - By bundling all requisite code, libraries, and dependencies, we ensure the model's consistent and reliable functionality within a deployment context. - Rigorous testing within the deployment environment affirmed compatibility and responsiveness, granting users access to a refined and fully operational model that meets the highest standards of performance and user experience.	2

Task	Description	Hours
Model Deployment	Leveraged Google Cloud's Vertex AI platform to simplify the deployment process of the model. I proficiently packaged the trained model, integrating vital pre-processing and post-processing scripts. By customizing model deployment settings, including hardware specifications and scaling preferences, I fine-tuned performance for optimal results. I diligently monitored the operational performance and usage patterns of the deployed model, promptly making adjustments as necessary to ensure consistent peak functionality. <u>As a result</u> • By harnessing the capabilities of Google Cloud's Vertex AI platform, we've orchestrated a seamless model deployment process. • The inclusion of crucial pre-processing and post-processing scripts within the packaged model guarantees a holistic deployment. Tailoring deployment settings ensures optimal utilization of hardware resources and enables efficient scaling. • Ongoing monitoring and timely adjustments bolster the deployed model's efficacy, culminating in a robust and high-performing solution that effectively caters to real-world usage demands.	2
Real-Time Inference Testing	Submitted sample data inputs to the deployed model, generating real-time predictions. Through this process, I rigorously evaluated the accuracy and timeliness of the model's predictions, effectively validating its reliability and effectiveness in a real-world context. <u>As a result</u> • By subjecting the deployed model to sample data inputs, we have effectively tested its predictive capabilities in a real-time environment. • The meticulous evaluation of prediction accuracy and timeliness affirmed the model's robustness and reliability, demonstrating its potential to perform effectively and consistently within real-world scenarios.	2

Task	Description	Hours
Bug Fixes and Enhancements	Conducted a thorough analysis of deployment logs and performance metrics, pinpointing potential bottlenecks and inefficiencies. This guided me in refining deployment settings, adjusting resource allocation and scaling parameters to optimize operational efficiency based on observed usage patterns. By conducting load testing procedures, I simulated varying levels of user activity to assess the deployed model's ability to handle increased demand effectively. Maintained a vigilant and continuous monitoring of the deployed model's performance and responsiveness. This iterative approach enabled me to finely adjust deployment configurations over time, ensuring the attainment of optimal operational efficiency. <u>As a result</u> • By scrutinizing deployment logs and performance metrics, we have been able to uncover potential bottlenecks and inefficiencies, subsequently addressing these issues for optimized performance. • Adjustments to deployment settings in alignment with observed usage patterns ensure efficient resource allocation and scaling. • Rigorous load testing provides insights into the model's capacity to handle varying user demands. The continuous monitoring and iterative refinement of deployment configurations underscore our commitment to sustaining optimal operational efficiency and ensuring a seamless user experience.	3

Table 7 - Week 3 (Product Recommendations & Consumer Basket Analysis)

Week 4: May 22nd - May 26th, 2023 – Deployment modifications to Vertex AI:

- Preparing and deploying the selected model to Google Cloud's Vertex AI.

Task	Description	Hours
Model Export and Packaging	Successfully completed model training, subjecting it to rigorous validation to ensure its adherence to specified performance benchmarks. I preserved essential components of the trained model, including weights, architecture, and critical pre-processing procedures. By establishing APIs and functions, I created a seamless communication link between the deployed model and external applications. To facilitate deployment, I compiled model files, vital dependencies, and essential environment configurations into a comprehensive package.	3

Task	Description	Hours
	Furthermore, I executed comprehensive testing protocols to confirm the model's robust functionality, unwavering stability, and impeccable compatibility within the designated deployment environment. <u>As a result</u> • By successfully finalizing the model training phase and ensuring its compliance with desired performance criteria, we solidified our commitment to excellence. Preservation of the model's fundamental elements guaranteed its integrity and efficacy. • The establishment of APIs and functions served as a conduit for effective communication between the model and external applications. • Thoughtful packaging of model assets, dependencies, and environment configurations streamlined the deployment process. Thorough testing reinforces the model's dependability, enhances its stability, and verified its harmonious integration within the deployment environment, setting the stage for a seamless user experience.	
Model Deployment	Adeptly utilized Google Cloud's Vertex AI platform to facilitate the seamless deployment of the model. Through strategic configuration of deployment settings, I encompassed critical factors such as instance type selection, autoscaling implementation, and robust authentication protocols. <u>As a result</u> • Through the strategic utilization of Google Cloud's Vertex AI platform, we've streamlined the deployment process, ensuring a smooth transition of the model to its operational state. • The configuration of deployment settings, covering essential elements like instance type, autoscaling dynamics, and robust authentication, further fortified the model's hosting environment. This dual focus on security and efficiency culminates in a deployment that is not only reliable but also optimally aligned with industry best practices.	2
Real-Time Inference Testing	Generated sample data inputs that accurately mirrored the complexities of real-world scenarios. These carefully prepared inputs were then sent to the deployed model's API, resulting in real-time predictions. I scrutinized the prediction outputs to ensure their accuracy and alignment with expected outcomes. Additionally, I evaluated the	4

Task	Description	Hours
	response times of the predictions, confirming their prompt and efficient delivery. Engaging in iterative cycles of refinement and testing as necessary, I consistently validated the reliability and consistency of the real-time predictions. <u>As a result</u> • By generating sample data inputs that accurately mirror real-world scenarios, we've laid the foundation for a rigorous assessment of the deployed model's capabilities. The process of transmitting these samples to the model's API for real-time predictions allowed for a tangible evaluation of its performance. • Scrutinizing prediction outputs, assessing response times, and iteratively refining based on the results ensured the reliability and relevance of real-time predictions in actual usage contexts. • This comprehensive testing approach served to fortify our confidence in the model's aptitude for delivering timely, accurate, and dependable predictions.	
Bug Fixes and Enhancements	Engaged in vigilant performance monitoring of the deployed model, tracking key metrics such as response time, throughput, and error rates. By delving into deployment logs and usage patterns, I extracted valuable insights that uncovered potential avenues for optimization. I honed resource allocation with precision, making calibrated adjustments to instance types and scaling parameters, achieving optimal operational efficiency. Strategic caching mechanisms were strategically implemented to mitigate redundant computations, elevating prediction speed and enhancing user experience. I executed comprehensive load testing procedures, simulating varying user loads to comprehensively evaluate the system's behavior under different conditions. Leveraging the capabilities of cloud platform features, I facilitated seamless auto-scaling mechanisms to maintain impeccable performance during peak usage scenarios. Conducting periodic reviews and updates of model dependencies ensured unwavering compatibility and robust security. Collaborating seamlessly with DevOps teams, I fostered a continuous cycle of refinement for deployment strategies. <u>As a result</u>	8

Task	Description	Hours
	- The monitoring of the deployed model's performance through vigilant metric tracking underscored our commitment to maintaining optimal output. Analyzing deployment logs and usage patterns has enabled us to uncover optimization opportunities and make informed enhancements. - Through precise resource allocation, strategic caching, and thorough load testing, we have solidified the model's operational efficiency and reliability. - Leveraging cloud platform features and regular dependency updates ensures scalability and security. - Collaboration with DevOps teams adds a layer of synergy to deployment strategies, ultimately contributing to a consistently high-performing and resilient model.	
Model Monitoring	Implemented robust monitoring tools to continually assess the performance and status of the deployed model. By strategically configuring alerts, especially for crucial metrics like response time, error rates, and resource usage, I enabled prompt attention to critical conditions. Incorporating sophisticated logging mechanisms, I systematically captured detailed information about incoming requests and their corresponding responses. Furthermore, I established a routine reporting and visualization system, providing comprehensive insights into monitoring data. This empowered timely decision-making and enabled proactive resolution of potential issues <u>As a result</u> - The strategic implementation of monitoring tools helped in continuous vigilance over the deployed model's performance and health. With configured alerts for vital metrics and detailed logging mechanisms, we fortified our capacity to identify anomalies and potential concerns swiftly. - Regular reporting and visualization of monitoring data ensure that decision-making is well-informed and issue resolution is proactive. This holistic approach to monitoring fosters a resilient and high-performing deployment, enabling us to confidently navigate real-world scenarios.	2
Model Updates	Collected and subjected monitoring data to a thorough analysis, uncovering valuable insights into the operational dynamics and performance of the deployed system. I sensitively detected patterns,	3

Task	Description	Hours
	anomalies, and areas requiring improvement within the extensive tapestry of monitoring data. Using this information, I strategically formulated actionable recommendations that outlined precise updates and adjustments aimed at enhancing system efficiency, bolstering reliability, and reinforcing security. Promoting a collaborative environment, I engaged with relevant teams, facilitating discussions and deliberations on the proposed updates. By prioritizing these recommendations, we ensured a seamless implementation process that aligns with the broader objectives of the system's optimization and refinement. <u>As a result</u> • By collecting and scrutinizing monitoring data, we've harnessed an invaluable source of intelligence regarding the deployed system's behavior. The identification of trends, anomalies, and areas warranting improvement establishes the groundwork for evidence-based decision-making. • Our actionable recommendations, aligned with observed issues, offered a roadmap for enhancing system efficiency, reliability, and security. • Through collaboration and prioritization, we ensure that these proposed updates are methodically integrated, culminating in a well-refined and fortified system that confidently navigates the complexities of its operational landscape.	
Performance Optimization	Reviewed both monitoring data and product team feedback, identifying areas that were ready for refinement. Collaborating closely with the development team, I orchestrated a strategic plan to prioritize the impending enhancements. Rigorous testing was then conducted to ensure that these changes maintained the existing system's integrity without introducing unintended bugs. I also refactored existing code and introduced innovative features, ensuring alignment with the overarching enhancement goals. This included fine-tuning algorithms, data structures, and processes to elevate the system's overall performance. I integrated enhancements to the user interface and user experience as dictated by the context, ensuring an improved interaction for users. Thorough validation of these enhancements was carried out using real-world data and actual user scenarios, assuring their tangible effectiveness. Documented each implemented alteration and update,	8

Task	Description	Hours
	creating a comprehensive reference that serves both present considerations and future maintenance needs. <u>As a result</u> • Through the orchestration of a planned and executed series of enhancements, we fortified our system's architecture and functionality. • The seamless integration of code changes, optimization of processes, and incorporation of user-centric improvements underscore our commitment to delivering a high-performing solution. • The iterative validation process, complemented by product team's feedback, ensured the ongoing relevance and effectiveness of these enhancements, ultimately leading to a refined and resilient system ready to navigate the challenges of real-world usage.	
Evaluation	Performed a thorough evaluation of the model's performance subsequent to the implementation of optimization measures, effectively assessing the impact of the executed changes. By employing relevant evaluation metrics such as accuracy, precision, recall, and F1-score, I quantitatively measured the discernible improvements achieved. Engaging in a comparative analysis, I juxtaposed the results post-optimization with the established benchmarks from the pre-optimization phase. This systematic approach allowed for a precise quantification of the extent of enhancements realized. <u>As a result</u> • The rigorous assessment of the model's performance post-optimization affirms our commitment to continuous improvement. • By harnessing well-established evaluation metrics, we've objectively measured the efficacy of our enhancements. The comparison with pre-optimization benchmarks further crystallizes the progress made, shedding light on the tangible impact of our efforts. • This methodical approach empowers us with a clear understanding of the positive changes brought about by optimization.	2
Task Review and Planning	Engaged in a thorough and exhaustive review of the tasks accomplished during Week 4, assessing the trajectory of progress. Considering this assessment, I strategically formulated a task plan designed to navigate the challenges and capitalize on the opportunities presented in Week 5. <u>As a result</u>	1

Task	Description	Hours
	- Through the holistic review of Week 4's achievements, we've calibrated our understanding of the project's evolution. - The task plan crafted for Week 5 served as a navigational compass, steering us towards a successful culmination by strategically prioritizing objectives and fostering an environment of efficient work allocation and continuous forward momentum.	
Communication and Updates	Attended team meeting that aimed to communicate relevant updates about the ongoing project status and recent developments. During this collaborative session, I transparently shared insights about the accomplished milestones, candidly discussed challenges faced, and thoughtfully provided a preview of the tasks that lie ahead on the project's horizon. <u>As a result</u> Through the team meeting, we fostered a cohesive environment of shared knowledge and informed decision-making. By unveiling accomplished milestones, openly addressing challenges, we solidified our collective understanding and commitment to project success.	2
Model Performance Monitoring	Monitored essential performance metrics, such as response times, prediction accuracy, and user satisfaction. This diligent approach allowed me to derive a comprehensive understanding of the system's effectiveness and its alignment with expectations. By sensitively identifying any deviations from anticipated behavior and performance through ongoing monitoring, I could promptly react to any anomalies. Leveraging the invaluable insights gained from this continuous monitoring practice, I enacted prompt adjustments, strategic updates, and nuanced optimizations. <u>As a result</u> - The seamless amalgamation of real-world usage data and product team's feedback offers a window into the intricate nuances of our system's operation. - Through unwavering monitoring, we've established a proactive mechanism for detecting anomalies and optimizing performance. - This data-driven approach empowers us to make timely adjustments that align with product team's expectations, ensuring our model consistently thrives within real-world scenarios.	3

Task	Description	Hours
Model Updates	Analysis of the accumulated data and feedback identified areas where updates and refinements could have a meaningful impact. Strategically, crafted actionable recommendations that outlined changes poised to enhance the system's functional capabilities, elevate user experiences, and reinforce performance benchmarks. <u>As a result</u> By dissecting collected data and heeding teams' feedback, we've embarked on a journey of insight-driven evolution. Our formulated recommendations hold the promise of invigorating our system's attributes, resulting in a heightened user experience and optimized performance that aligns seamlessly with evolving expectations.	2

Table 8 - Week 4 (Product Recommendations & Consumer Basket Analysis)

Week 5: May 29th - May 31st, 2023 - Demo and Documentation:

- Conducting a demo of the project and update the documentation in Notion.

Task	Description	Hours
Demo Preparation	Thoroughly tested and validated each implemented enhancement and optimization to ensure their seamless functionality. Curated essential elements including data, visuals, and documentation, crafting a showcase that vividly exemplified the system's newfound capabilities. Conducted rehearsals of the demonstration, refining each detail to guarantee a fluid and articulate execution that effortlessly showcased the system's features. Additionally, prepared elucidative explanations, delving into the intricacies of each facet of the system. <u>As a result</u> - The rigorous testing and validation process instils unwavering confidence in the system's refined functionalities. - Our curated showcase, combined with well-prepared demonstrations and elucidative explanations, ensured that our achievements and advancements are impeccably communicated, leaving no room for ambiguity, and effectively celebrating the project's milestones.	2
Finalization and Wrap-Up	Reviewed the entire project, evaluating every aspect from its inception, including initial goals, to the ultimate outcomes achieved. Diligently, I gathered a wealth of pertinent documentation, comprehensive reports,	4

Task	Description	Hours
	and indispensable materials, skillfully fortifying the forthcoming presentation. With careful consideration, I crafted a structured presentation outline, ingeniously weaving together key points while ensuring a seamless and coherent flow. <u>As a result</u> • The comprehensive review process unfurled a tapestry that encapsulated the entire project's evolution, from conception to fruition. • By amassing a trove of supporting documentation and expertly crafting a structured presentation outline, we're poised to convey the project's narrative with precision, elegance, and unwavering clarity.	
Project Demo	Presented the culmination of the project's journey to stakeholders, adeptly capturing a concise yet compelling summary that traversed the path from the project's nascent goals to the tangible results achieved. Also, spotlighted the pivotal milestones, noteworthy enhancements, and the overarching impact that the project has had on the very goals and problem-solving imperatives that initiated its inception. <u>As a result</u> • Through a presentation, we've illuminated the project's voyage, deftly showcasing its transformation from aspirations to actuality. • By accentuating the key accomplishments and the profound improvements realized, we've etched a vivid portrayal of the project's resounding success and its definitive influence on the intended goals and challenges at hand.	1
Demo Reflection and Updates	Conducted a comprehensive analysis of the feedback gathered from the demo presentation, evaluating both the positive accolades and constructive suggestions for enhancement. I astutely pinpointed specific areas of feedback that centered on aspects such as clarity in communication, seamless functionality, and the overarching user experience. Leveraging the invaluable insights provided by the feedback as a compass, I navigated the course of refinement thoughtfully. <u>As a result</u> • Through a discerning review of the feedback landscape, we unveiled a treasure trove of actionable insights.	2

Task	Description	Hours
Documentation Update	Compiled a comprehensive collection of project-related documents, encompassing code documentation, design choices, and implementation specifics. It was my priority to ensure that the documentation is well-organized, thorough, and primed for archiving and future points of reference. <u>As a result</u> By expertly collating this comprehensive ensemble of project-related documents, we've woven a tapestry of knowledge that resonates with precision, order, and standard structure.	3
Notion Update	Initiated the establishment of a dedicated project page on Notion, serving as a central portal to update project progress and documentation. On this platform, I added the project description, objectives, and scope, reflecting the vibrant pulse of the latest developments. To ensure clarity and structure, I refined task lists and milestones, effectively differentiating various task areas. In the pursuit of enhancing the project's potency, I incorporated relevant links, attachments, and references to external resources, creating a comprehensive and easily navigable knowledge hub. Recognizing the impact of visual storytelling, I adorned the canvas with the latest data visualizations, crafted diagrams, and vivid images. Furthermore, I chronicled the cherished moments of the final presentation, carefully cataloguing the tapestry of feedback provided by the product team. <u>As a result,</u> Through this orchestrated symphony of edits and updates, we completed the documentation of the project.	8
Task Review and Planning	Reflected on the internship experience, considering accomplishments, challenges, and lessons learned. <u>As a result</u> - Through this contemplative voyage, I noted threads of accomplishments, challenges, and the wisdom that have been sown into my growth. - This introspection stands as a testament to the resilience, adaptability, and unquenchable thirst for knowledge that underscored the entirety of the internship journey.	2

Task	Description	Hours
Reflection and Learning	Reflecting on the internship, I've gained valuable skills beyond my initial expectations. Personal growth is evident in improved strengths and problem-solving abilities. Contributions to projects showcased collaborative solutions to challenges. This experience aligns with long-term career goals, merging technical skills with communication. The internship has been transformative, nurturing growth and aligning with future aspirations. <u>As a result</u> • This introspective process has provided me with a clearer perspective on my professional and personal development. • Through assessing my progress, acknowledging strengths, and recognizing areas for improvement, I've gained valuable insights to guide my future endeavors and aspirations.	2

Table 9 - Week 5 (Product Recommendations & Consumer Basket Analysis)

Project 2 – Robyn (MMM) Implementation with Integration of Python and R

Week 1: June 1st - June 2nd, 2023 - Project Initialization and Environment Setup:

- Setting up the project environment and preparing for implementation.

Task	Description	Hours
Kick-off Meeting and Goal Setting	Participated in a pivotal team meeting, jointly setting precise project goals, objectives, and deliverables. Through collaborative efforts, we fostered a shared understanding and alignment on the project's overarching direction and purpose. <u>As a result</u> • Through this collective effort, the project's foundation was solidified with well-defined goals, objectives, and deliverables. • The collaborative engagement fostered a shared clarity and alignment among team members, setting a strong course for the project's successful direction and purposeful execution.	1
Literature Review	Conducted thorough online research to gather insights on integrating Robyn (R) into both Python and R programming environments. Explored diverse sources, including documentation, tutorials, and community forums, to unravel the intricacies of the integration process. This encompassed understanding compatibility, dependencies, and guidelines for seamless Robyn integration. Equally, research was undertaken to ensure Robyn's integration with R, delving into relevant resources and best practices. <u>As a result</u> • The research efforts have yielded a comprehensive understanding of how to integrate Robyn into both Python and R environments. • The exploration of documentation, compatibility, and potential challenges equipped us with valuable insights for effective integration and utilization.	4
Software Installation and Setup	Initiated the project's Python development environment by installing the latest essential libraries. In parallel, the R environment was adeptly configured, involving the installation of the R programming language and relevant packages pivotal for analysis and seamless integration. <u>As a result</u> • The project's development environment is now equipped with the latest and essential Python libraries, along with the R programming	1

Task	Description	Hours
	language and relevant packages, enabling seamless analysis and integration capabilities.	
Data Gathering	Assembled sample data that mirrors the project's scope and specifications and verified that the gathered data encompasses diverse scenarios to thoroughly assess Python and R integration. <u>As a result</u> • A comprehensive collection of sample data has been curated to accurately reflect the project's scope and requirements. • This dataset covers a wide array of scenarios, ensuring robust testing and evaluation of the Python and R integration process.	1
Environment Configuration	Established the development environment by adeptly configuring crucial IDEs (Integrated Development Environments) for both Python and R programming. This setup was further fortified through the strategic integration of supplementary extensions, plugins, and packages. <u>As a result</u> • The development environment has been successfully configured with the necessary IDEs for Python and R, complemented by additional extensions, plugins, and packages that facilitate a streamlined and efficient integration process.	1
Initial Python Code Integration	Familiarized myself with Robyn's documentation and functionalities essential for Python integration. Integrated the necessary Robyn library and established crucial components within the Python environment. Created a Jupyter notebook to experiment hands-on with Robyn's features and produce preliminary reports. Injected the notebook with sample data, mirroring real-world scenarios to streamline report generation. Iteratively tested and refined the integration process, promptly resolving any encountered issues or errors for a smooth integration experience. <u>As a result</u> • A thorough understanding of Robyn's Python integration was achieved through the exploration of its documentation and functionalities. • By importing the required library, initializing components, and experimenting within a Jupyter notebook. • Sample data integration and iterative testing have contributed to refining the integration for optimal functionality.	4

Task	Description	Hours
Initial R Code Integration	Reviewed Robyn's documentation to understand how to integrate it with Python. Installed necessary R packages and dependencies to enable Robyn's features. Created an R-Python notebook to kickstart the integration and produce initial reports using Robyn. Incorporated sample data into the R environment to simulate real-world scenarios for easier report generation. Utilized Robyn's R functions to customize report templates and layouts according to project requirements. Continuously refined the integration through iterative testing, ensuring flawless performance and addressing any arising issues or errors. <u>As a result</u> • An understanding of Robyn's integration into Python has been acquired through documentation exploration. • The successful setup of R packages and dependencies has paved the way for integration. • An R-Python notebook has been utilized to initiate the integration process and generate preliminary reports. • By incorporating sample data, utilizing Robyn's functions, and addressing encountered challenges, a robust integration has been established for tailored report generation.	4

Table 10 - Week 1 (Robyn Implementation with Integration of Python and R)

Week 2: June 5th - June 9th, 2023 - Robyn Implementation and Testing:

- Implementing Robyn functionality and conducting initial testing.

Task	Description	Hours
Python-R Integration Refinement	Participated in collaborative discussions with team members to collectively assess the progress and initial outcomes of the Python-R integration. Identified potential areas for refinement and enhancement based on the status of the integration process. Verified the quality and alignment of generated reports with predetermined standards and project objectives. Effectively resolved any observed discrepancies or inconsistencies between outputs from the Python and R integration. Conducted rigorous benchmarking and comprehensive performance evaluations to gauge the effectiveness of the integrated solutions. Ensured seamless compatibility across a range of data types, formats, and use cases during the iterative refinement of the integration.	8

Task	Description	Hours
	Thoroughly tested the integration under diverse scenarios and edge cases to validate its robustness and reliability. <u>As a result</u> • A productive collaboration with team members yielded valuable insights into the progress and initial outcomes of the Python-R integration. • Through detailed analysis, potential areas for improvement were identified, contributing to the enhancement of the integration's effectiveness. • Discrepancies between Python and R outputs were addressed to guarantee consistency. Rigorous benchmarking and performance comparisons provided a clear understanding of the integrated solutions' efficiency. • The integration refinement process emphasized compatibility across various data types and use cases, bolstering its overall reliability. • Extensive testing validated the integration's robustness across diverse scenarios and edge cases. A comprehensive summary report was prepared, highlighting the refined Python-R integration, its advantages, and key lessons learned during the journey.	
Initial Testing	Implemented Python code snippets for the Robyn integration in a controlled environment. Ensured error-free execution of the code and validated the generation of the expected outputs. Evaluated the initial outcomes and reports to ensure they align with the anticipated results. Documented any identified issues or anomalies observed during the initial testing phase, with the intention of addressing them for further investigation and improvement. <u>As a result</u> • The code ran without errors, producing the desired outputs. Initial results and reports analyzed confirmed their consistency with expectations. • Issues and anomalies encountered during this phase were diligently documented, preparing them for subsequent investigation and refinement.	2
Debugging	After reviewing the outcomes of the initial testing, I pinpointed errors, discrepancies, and unexpected behaviors. This analysis encompassed a thorough examination of the code logic, configurations, and data inputs to ascertain the root causes of the identified issues. Upon identifying these underlying factors, I promptly implemented the necessary fixes	4

Task	Description	Hours
	and adjustments to the Python code, aiming to rectify the initial problems. Following these modifications, I conducted subsequent testing to confirm that the applied changes had effectively resolved the issues and restored the desired functionality. <u>As a result</u> • Thorough analysis of the initial testing outcomes led to the identification of errors, discrepancies, and unexpected behaviors. • The code logic, configurations, and data inputs were examined to determine the root causes of the issues. Appropriate fixes and adjustments were then applied to the Python code to rectify the initial problems. • Testing carried out, validated the efficacy of the implemented changes, resulting in smoother functionality.	
Initial R Code Integration	Expanded the scope of the Python-Robyn integration, delving into more intricate data scenarios and diverse use cases. Leveraging Robyn's advanced functionalities, I adeptly generated comprehensive and insightful reports seamlessly within the integrated Python environment. To enhance the integration's flexibility, I engineered Python functions and scripts that harmoniously interacted with Robyn's capabilities. Real-world data was skillfully incorporated into the Python environment to thoroughly assess the integration's practicality and efficiency. My effort extended to conducting rigorous testing, assuring that the expanded integration maintained its stability and precision. <u>As a result</u> • The Python-Robyn integration was extended to accommodate complex data scenarios and varied use cases. • Leveraging Robyn's advanced features, the integration enabled the generation of comprehensive and insightful reports within the Python environment. This was achieved through the development of Python functions and scripts that seamlessly interacted with Robyn's capabilities. • By incorporating real-world data, the integration's practicality and efficiency were evaluated, while rigorous testing ensured its stability and precision. • Collaborating with team members facilitated progress communication, feedback gathering, and ensured alignment with the project's overarching goals.	6

Task	Description	Hours
Integration Refinement	Reviewed the outcomes of the integrated code testing, methodically identifying errors, pinpointing opportunities for enhancement, understanding the root causes of issues, and acknowledging any encountered challenges. By strategically implementing essential code adjustments, I adeptly resolved bugs, discrepancies, and performance bottlenecks. To optimize efficiency, bolster reliability, and elevate the overall functionality of the integration, I enhanced its logic. I reinforced error-handling mechanisms, ensuring the integrated code adeptly manages unforeseen scenarios. To prevent the introduction of new issues, I conducted regression testing to validate the refinements. I verified the code's robustness across a diverse range of datasets, scenarios, and user inputs, ensuring its unwavering resilience. <u>As a result</u> • The review of the integrated code's testing outcomes encompassed the identification and resolution of errors, alongside opportunities for improvement and the addressing of root issues. • Through bug fixes, adjustments, and performance optimizations, the integration logic was refined to bolster efficiency, reliability, and overall functionality. • A focus on error handling mechanisms ensured graceful handling of unexpected scenarios. Rigorous regression testing was conducted to guarantee that the refinements didn't introduce new complications. • This robustness was further verified across diverse datasets, scenarios, and user inputs. • The collaborative efforts of colleagues and team members played a pivotal role in validating the effectiveness of the code enhancements. A summary report was prepared, summarizing the refined integrated code's advantages, improvements, and suitability for deployment.	8
Unit Testing	Created unit test cases for individual functions and modules within the integrated Python code. By structuring test scenarios to encompass a wide array of input combinations and edge cases, I ensured comprehensive coverage of potential scenarios. These unit tests were executed to validate the accuracy and reliability of the components within the Python code. In a similar vein, I applied unit testing protocols to the integrated R code, rigorously evaluating its functionalities and the outputs it generated. <u>As a result</u>	4

Task	Description	Hours
	- The creation of unit test cases targeted specific functions and modules within the integrated Python codebase. These tests were carefully designed to encompass a wide array of input combinations and edge cases, ensuring thorough coverage. - By executing these unit tests, the accuracy and reliability of the individual components were verified, bolstering the overall robustness of the Python code. The same methodical unit testing process was extended to the integrated R code, thus ensuring its functionalities and outputs were rigorously assessed as well.	
Performance Analysis	Conducted a meticulous examination of the results and reports generated by the integrated Python-Robyn solution. This involved a comprehensive evaluation of crucial performance metrics such as processing time, memory utilization, and resource consumption. By scrutinizing these metrics, I was able to identify performance bottlenecks, areas of inefficiency, and aspects in need of optimization. Furthermore, I assessed the scalability of the integrated solution, gauging its ability to efficiently handle larger datasets and increased user demands. <u>As a result</u> - Thoroughly examined the outputs stemming from the integrated Python-Robyn collaboration. This assessment encompassed crucial performance metrics like processing time, memory usage, and resource allocation. - Through this evaluation, pinpointed performance bottlenecks, identified inefficiencies, and detected areas necessitating optimization. The solution's scalability in handling larger datasets and increased user demands was also explored. - Extracting valuable insights from this preliminary performance analysis, these findings laid the groundwork for subsequent optimization efforts and further refinements.	4
Integration Debugging	Identified and addressed challenges related to the integration of project systems, specifically focusing on issues that hindered smooth communication between different components. By working closely with relevant team members, I engaged in collaborative efforts to diagnose and resolve integration obstacles. <u>As a result</u>	2

Task	Description	Hours
	- Recognized and addressed hurdles linked to the integration of project systems, effectively identifying factors that impeded the seamless communication between distinct components. - Leveraging teamwork, actively collaborated with relevant team members to investigate, and overcome these integration issues, resulting in a streamlined and unhindered flow of both data and functionalities.	
Task Review and Planning	Reviewed the tasks accomplished in Weeks 1 and 2, evaluating the progress achieved and the outcomes attained. This assessment was conducted with the goal of identifying any possible areas that could benefit from improvement and adjustment in the project's direction. Additionally, I actively participated in collaborative discussions to strategize and outline the tasks for Week 3. Integrating the feedback received, considering the evolving project priorities, and aligning with the remaining objectives, I contributed to the task allocation process among team members. <u>As a result</u> - Thoroughly reviewed the tasks accomplished in the initial two weeks, diligently evaluating the progress achieved and the outcomes attained. This comprehensive review allowed for the identification of potential areas that could benefit from improvement and adjustment. - Through collaborative efforts, planned and allocated tasks for the upcoming Week 3. By incorporating received feedback, considering shifting project priorities, and aligning with the remaining goals, a well-structured task distribution strategy was established to ensure successful execution.	1
Communication and Updates	Participated in the team meeting to deliver a comprehensive update on the ongoing project's advancement. <u>As a result</u> Articulated notable accomplishments, engaged in dialogues about encountered challenges, and underscored any impediments that emerged during the process.	1

Table 11 - Week 2 (Robyn Implementation with Integration of Python and R)

Week 3: June 12th - June 16th, 2023 - Testing and Optimization:

- Further testing and optimization of the integrated code.

Task	Description	Hours
Integration Testing	Performed comprehensive integration tests that encompassed a wide range of project components. These tests were designed to accurately simulate real-world scenarios, thereby identifying any potential issues that could disrupt the seamless interactions between different parts of the project. Documented the results of these integration tests, recording any bugs, discrepancies, or inconsistencies that were uncovered during the testing process. <u>As a result</u> • These tests were instrumental in identifying potential glitches and facilitating seamless interactions. • This documentation serves as an invaluable resource, guiding future improvements and enhancements.	2
Robyn Functionality Testing	Performed thorough functionality tests on the Robyn system, examining its features and capabilities to verify their alignment with the project's specific requirements. I collaborated closely with the testing team to devise comprehensive test scenarios that encompassed a diverse range of functionalities within the Robyn system. The results of these functionality tests were documented, detailing any identified discrepancies, errors, or deviations from the expected behavior. <u>As a result</u> • These functionality tests have ensured a comprehensive evaluation of the Robyn system's capabilities. • By collaborating with the testing team and documenting observed discrepancies and errors, we have taken significant steps toward refining the system's quality and ensuring its preparedness for deployment. • The insights gained from these tests will play a pivotal role in addressing issues and optimizing the system's functionality.	2
Python-R Interaction Testing	Tested the interactions between the Python and R components, ensuring seamless data exchange and compatibility. To achieve this, I designed comprehensive test cases that covered a wide range of scenarios involving data transfer, function calls, and code execution between Python and R scripts. The outcomes of these tests were documented,	2

Task	Description	Hours
	emphasizing successful interactions while also addressing any issues and limitations that were encountered during the testing process. <u>As a result</u> • Thoroughly testing the interactions between Python and R components has demonstrated the system's ability to facilitate seamless communication between these languages. • By documenting the outcomes, we have provided a clear overview of the success achieved, while also identifying areas for improvement.	
Bug Fixes and Improvements	Identified and documented various issues, including bugs, glitches, and performance concerns, that were reported. Collaborating closely with relevant team members to analyze the root causes of these issues and brainstorm effective solutions. Once solutions were identified, I implemented fixes and improvements, following a systematic approach to ensure that the solutions were both robust and in alignment with the project's goals. Rigorous testing was then conducted to verify that the implemented solutions successfully resolved the reported issues, while also ensuring that no new problems were introduced in the process. <u>As a result</u> • By diligently addressing and resolving reported issues, we have further enhanced the system's reliability and performance. • The collaborative effort in ensured that the project maintains its quality standards. • The systematic approach to testing the implemented fixes guaranteed that these solutions seamlessly integrate into the project without introducing new complications.	8
Code Review	Engaged in a thorough code review of the integrated components, carefully examining the codebase. Through collaborative efforts with fellow developers, we identified potential issues such as code duplication, suboptimal algorithms, and the absence of comments and documentation. Offering constructive feedback, I provided insightful suggestions for improvements aimed at enhancing the code quality, readability, and maintainability. <u>As a result</u> • This meticulous code review process has significantly contributed to the overall integrity and efficiency of the project's codebase.	2

Task	Description	Hours
	- Through collaborative efforts, we identified and rectified potential shortcomings, ensuring that the code adheres to industry best practices and maintains high standards of quality. - The provided feedback and suggestions have not only improved the immediate code components but also fostered a culture of continuous improvement within the development team. - The outcome is a codebase that was not only functional but also optimized for long-term maintainability and scalability.	
Performance Optimization	Conducted a thorough analysis of the integrated code, aiming to identify areas where optimization could lead to enhanced performance and efficiency. This involved pinpointing opportunities to reduce computational complexity and minimize resource usage. Through collaboration with the UI team, I implemented optimization techniques, which encompassed algorithmic improvements, memory management strategies, and parallel processing approaches. Following the implementation of optimizations, I engaged in rigorous testing to validate that the changes didn't introduce new issues and that the overall performance of the integrated code was indeed improved as intended. <u>As a result</u> - The thorough analysis and subsequent optimizations have led to substantial improvements in the performance and efficiency of the integrated code. - By identifying and addressing bottlenecks and resource-intensive operations, we have significantly enhanced the system's responsiveness and resource utilization. - The collaboration with the UI team has fostered cross-functional expertise, resulting in the implementation of effective optimization strategies. - The outcome is a system that not only meets functional requirements but also operates at its full potential, providing a seamless and efficient user experience.	4
Integration Debugging	Reviewed the integrated system to search for any lingering integration-related issues and bugs that might have eluded detection during the earlier testing phases. Through close collaboration with pertinent team members, I engaged in the diagnostic and debugging process, utilizing logs, error messages, and debugging tools to accurately trace the root causes of the identified issues. Following a thorough understanding of	2

Task	Description	Hours
	the issues, I implemented effective solution to ensure that the system operated seamlessly. <u>As a result</u> • The thorough review of the integrated system has allowed us to catch and address any remaining integration-related issues that might have otherwise affected the system's performance. • Collaborative efforts helped to identify their root causes accurately. • This process has further strengthened the system's robustness and ensured that product team have a smooth experience without any hindrances caused by integration problems.	
Testing Evaluation	Undertook a comprehensive assessment of the integration process, comparing the project's original objectives with the accomplished outcomes. Through close collaboration with the team, I effectively collected valuable feedback and insights regarding the integrated system's functionality, performance, and user-friendliness. By scrutinizing essential key performance indicators, user experience metrics, and any post-integration issues that emerged, I systematically evaluated the integration's triumph and pinpointed specific domains for potential enhancement and refinement. <u>As a result</u> • The thorough evaluation of the integration process has provided valuable insights into how effectively the initial goals of the project were met through the achieved outcomes. • We've been able to gauge the success of the integration and pinpoint specific areas that require further refinement and enhancement. • This evaluation process has not only validated the achievements of the integration but also paved the way for continuous improvement and optimization of the integrated system to better meet user needs and expectations.	2
Code Refinement	Reviewed the outcomes of our rigorous testing initiatives to pinpoint specific areas necessitating code refinement in response to the issues and inconsistencies uncovered during testing. Collaborating closely with the backend team, I executed code refinements, which encompassed essential bug fixes, enhancements in error handling, and adjustments to ensure seamless alignment with our desired outcomes. Following these refinements and retesting to validate that the identified issues were resolved, and that the system's functionality remained unharmed.	6

Task	Description	Hours
	<u>As a result</u> • The review of testing results has played a crucial role in pinpointing areas of the code that required refinement to resolve issues and discrepancies. • Collaborating with the backend team has contributed to enhancing the overall quality of the codebase. • Rigorous retesting after implementing these refinements has ensured that the identified issues have been effectively addressed and that the system's functionality remains robust. • This process of continuous refinement and improvement is essential for delivering a reliable and high-quality integrated system.	
Project Review and Planning	Engaged in a contemplative process aimed at evaluating the project's trajectory, thoroughly examining our achievements, challenges, and the lessons gleaned across its lifecycle. By collaborating closely with fellow team members, we collectively identified areas warranting improvement based on this introspection, fostering productive discussions to formulate strategies that address our identified shortcomings and enhance future phases of the project. Employing the insights garnered from this reflective analysis, I adeptly refined the project plan, intelligently adjusting our goals, timelines, and strategic approaches as needed. <u>As a result</u> • Further illuminated areas for improvement based on this assessment, allowing us to collectively address shortcomings and enhance project outcomes. • Utilizing the newfound insights, we refined the project plan, making necessary adjustments. • This iterative refinement ensures that the project's subsequent phases has guided by a more informed and strategic approach, ultimately contributing to the project's overall success and effectiveness.	4
Performance Analysis	Conducted in-depth analysis of the performance enhancements that were strategically implemented, embarking on a thorough comparison of system metrics before and after optimization. This diligent assessment was designed to quantitatively measure the advancements achieved in terms of speed, efficiency, and resource utilization. Collaborating harmoniously with performance analysts and pertinent team members, I expertly interpreted the gathered data, discerning critical insights such	3

Task	Description	Hours
	as bottlenecks, emerging trends, and potential areas ripe for further enhancement. I diligently assessed how these performance enhancements reverberated across the product experience and system stability, ensuring a congruence with the overarching project objectives and the discerning expectations of end-users. <u>As a result</u> • The thorough analysis of performance enhancements has provided a clear understanding of the system's improvements in terms of speed, efficiency, and resource utilization. • Collaborative discussions with performance analysts and team members have helped identifying crucial bottlenecks and additional enhancements. • The evaluation of the impact on functionality and system stability confirms that the enhancements align with project objectives. • Documenting these findings ensured transparency and provided a valuable resource for future monitoring and optimization efforts, ensuring the ongoing success of the project.	
Testing and Evaluation Review	Conducted a thorough review of the testing outcomes and the valuable insights gleaned throughout the testing phase. This encompassed a detailed analysis of the reported issues, a keen identification of patterns, and a holistic understanding of the overall behavior of the system under scrutiny. In collaboration with the dedicated testing team, I fostered engaging discussions to delve into the implications of these testing outcomes. <u>As a result</u> • The comprehensive review of testing provided a deeper understanding of the system's behavior. • Collaborating with the testing team facilitated a shared perspective on the system's quality and areas requiring enhancement. • This collaborative effort ensures that the system is well-prepared for the next phases of development and refinement.	2
Communication and Updates	Attended team meeting to ensure team was informed about the project's progress, discussing adjustments to the plan, and addressing concerns raised during the meeting. <u>As a result</u> • The team was aligned with projects' status and future adjusted plan.	1

Table 12 - Week 3 (Robyn Implementation with Integration of Python and R)

Week 4: June 19th - June 23rd, 2023 - Finalization in Development Environment:

- Finalizing the integration and preparing for deployment.

Task	Description	Hours
Integration Refinement	Analyzed the results of testing efforts to identify specific areas within the integrated code that demanded refinement to rectify the reported issues and elevate the overall quality of the system. To address these concerns, I engaged in close collaboration with the dedicated backend team, leveraging their expertise to implement the required code refinements. These refinements encompassed activities such as meticulous debugging, enhancing error handling procedures, and optimizing the code for enhanced performance. Post-refinement, I executed a rigorous retesting process to ensure that the previously reported issues had indeed been successfully resolved and that the system's functionality remained robust and dependable. <u>As a result</u> • This process of analysis, collaboration, and refinement has led to the enhancement of the integrated code's quality and functionality. • The collaboration with the backend team has ensured that identified issues were effectively addressed, and the rigorous retesting validates the success of these refinements in resolving previously reported problems.	4
Final Testing and Evaluation	Conducted a final round of comprehensive testing to assess the integrated system's functionality, performance, and reliability before its deployment. <u>As a result</u> • This thorough testing and collaboration with the testing and quality assurance teams have ensured that the integrated system is well-prepared for deployment. • This final testing phase served as a critical step to ensure that the system is ready to deliver a seamless and reliable experience to its users.	2

Task	Description	Hours
Bug Fixes and Improvements	Reviewed the conclusive testing outcomes to detect any lingering issues and anomalies that could potentially undermine the performance and functionality of the integrated system. Subsequently, I embarked on a focused mission to implement pragmatic solutions that effectively addressed these issues. For each identified problem, I ensured that the solution was implemented and that the integrated system's integrity remained intact. Rigorous testing was executed after each fix to validate the effectiveness of the solution and to verify that the problem had been thoroughly resolved. To culminate this process, I orchestrated a final validation testing phase, examining the integrated system to ascertain that all previously identified issues had been successfully resolved. <u>As a result</u> • This review of the final testing outcomes and collaboration with the team have been pivotal in ensuring the integrated system's robustness. • By final testing, the system's performance and functionality have been thoroughly vetted. The implementation of effective solutions and rigorous testing have contributed to the system's readiness for deployment, providing confidence in its reliability and the seamless experience it is poised to deliver.	8
Final Python-R Interaction Test	Conducted a comprehensive interaction test to ensure that all components of the integrated system interacted seamlessly and as intended. Designed test scenarios that covered a wide range of user interactions and system functionalities, identifying potential issues related to data flow, communication, and integration. <u>As a result</u> • By conducting thorough interaction tests and designing diverse test scenarios, the integrated system's components have been rigorously evaluated for seamless interactions. • This testing approach ensured that user interactions and system functionalities align with expectations and that potential issues has been addressed.	2
Performance Analysis	After making improvements to the code and optimizing its performance, I assessed how these changes affected the system's speed, efficiency, and use of resources. I worked closely with performance experts to understand the data and determine if the refinements had achieved the intended goals of making the system faster and more efficient.	2

Task	Description	Hours
	<u>As a result</u> The performance analysis conducted provided valuable insights into the impact of the changes on the system's speed, efficiency, and resource utilization.	
Deployment Preparation	Collected and arranged the improved code and all necessary resources to make sure the deployment process would be seamless. Working together with the team, we put together the code, dependencies, and essential configuration files in a way that makes deployment efficient and error-free. I carried out thorough testing on the deployment package to ensure that it worked properly and was compatible with the intended environment. If any issues cropped up during deployment testing, I addressed them to ensure a smooth deployment experience. <u>As a result</u> The compilation and organization of the refined codebase, along with the collaboration on packaging code and dependencies, ensured a streamlined and error-free deployment process.	2
Deployment to Test Environment	Deployed the prepared code in a controlled test environment, simulating the conditions of the actual production environment. Working collaboratively with the team, we carefully carried out the deployment process while closely monitoring its progress. This allowed us to swiftly identify and address any issues or discrepancies that emerged during this critical stage. <u>As a result</u> - The deployment of the prepared code and resources in a test environment allowed for a controlled assessment of the system's functionality and performance. - Collaborating with the team during this process facilitated the identification and resolution of any potential issues or inconsistencies that could affect the system's operation. - This proactive approach ensured that any deployment-related challenges are addressed before the system is rolled out to the production environment, contributing to a smoother and more reliable transition.	4

Task	Description	Hours
Testing and Validation	Worked closely with the testing and quality assurance teams to validate the deployed system's performance, ensuring its alignment with the specified requirements and expectations. After the testing phase, I carefully analyzed the test results and compared them to the anticipated outcomes. In cases where discrepancies were identified, I proactively addressed these issues by making necessary adjustments to the system. <u>As a result</u> • The deployed system's behavior and outcomes were rigorously evaluated to confirm its alignment with project requirements. • Got insights into any deviations from the expected outcomes, allowing for prompt adjustments and refinements.	2
Bug Fixes and Enhancements	Documented any deployment-related issues that surfaced during the testing phase in the test environment. This documentation served as a valuable reference for troubleshooting and improvement. Working closely with the deployment and core teams, I collaborated to identify the root causes of these deployment issues and developed effective solutions to address them. <u>As a result</u> • The identification and documentation of deployment-related issues in the test environment facilitated a collaborative approach towards addressing them. • By working together with the deployment and development teams, root causes were identified, leading to the implementation of effective solutions. • Rigorous retesting ensured that the previously encountered deployment issues were resolved, and the system's performance in the test environment was validated.	6
Accuracy Review	Engaged in productive collaborations with domain experts to gain insights into how the model's accuracy aligned with the project's objectives and requirements. Through these discussions, I aimed to comprehend the significance of the accuracy metric within the specific context of the project. I conducted an in-depth analysis of various factors that could impact the model's accuracy, including data quality, feature selection, overall functionality, performance considerations, and compatibility with the intended use cases. <u>As a result</u>	2

Task	Description	Hours
	- The collaborative effort with domain experts provided valuable insights into how the model's accuracy aligns with the project's goals. - This understanding enabled informed decision-making and optimization strategies to enhance the model's performance in accordance with the project's requirements.	
Task Review and Planning	Engaged in a reflective process to assess the project's overall progress, achievements, and challenges, gathering insights to inform future planning and decision-making. <u>As a result</u> - This reflective assessment has provided valuable insights into the project's trajectory, accomplishments, and obstacles. - These insights will guide future planning, helping to refine strategies, anticipate challenges, and make more informed decisions in upcoming projects.	1
Finalization Reflection	Conducted a comprehensive reflection on the overall project, evaluating its goals, outcomes, and lessons learned throughout its lifecycle. Collaborated with team members to share individual perspectives and insights, fostering a collective understanding of the project's successes, areas for improvement, and potential future directions. <u>As a result</u> This reflective process has provided a well-rounded view of the project's journey, encompassing its achievements, challenges, and the wisdom gained along the way.	2
Deployment to Production	After the completion of comprehensive testing and validation, the prepared code was deployed to the production environment. This crucial stage involved close collaboration with the team, orchestrating a seamless execution of the deployment process. My role included vigilant monitoring of the deployment process to swiftly address any potential issues that might arise. To ensure the system's stability and reliability, post-deployment testing was carried out. This testing phase aimed to confirm that the system performed as anticipated within the production environment and to ensure that the deployment process hadn't introduced any new issues. <u>As a result</u>	4

Task	Description	Hours
	- The successful deployment of the system to the production environment marks a significant milestone in the project's journey. - Through seamless collaboration and vigilant monitoring, any potential disruptions or challenges were swiftly addressed. - The post-deployment testing has provided reassurance that the system's functionality and stability align with the intended objectives.	
Real-Time Testing	Conduct tests to evaluate the real-time functionality of the deployed system, ensuring that it could handle and process data in real-time scenarios. After the tests, we look at the results to make sure the system meets the project's needs and works as intended in real-time situations. If we find any problems, we fix them to improve the system's performance. <u>As a result</u> - The rigorous testing of the system's real-time functionality has provided valuable insights into its capability to process data in dynamic scenarios. - By analyzing the test results, we were able to validate the system's responsiveness in real-time scenarios. Identified discrepancies and performance issues were prompted.	4
Communication and Updates	Engaged in the team meeting to provide updates on the project's status, sharing accomplishments, challenges, and milestones reached since the last update. <u>As a result</u> The team was well-informed about the project's progress and current developments, fostering effective communication and alignment among team members.	1

Table 13 - Week 4 (Robyn Implementation with Integration of Python and R)

Week 5: June 26th - June 30th, 2023 - Docker Performance and Python-R Integration Issue:

- Addressing performance issues in the Docker container and resolving code breaks in the Python-R integration.

Task	Description	Hours
Docker Performance Analysis	Collaborated with system administrators to diagnose the root causes of the performance issues, using tools like monitoring and profiling to gather relevant data. Propose and implement optimizations, adjustments to container configurations, and resource allocation changes to mitigate the identified performance issues and enhance overall system efficiency. <u>As a result</u> • Through collaborative efforts with system administrators, we successfully identified the root causes of the performance issues affecting the system.	4
Performance Optimization	Analyzed the existing Docker container setup to pinpoint opportunities for enhancing efficiency. This involved optimizing resource allocation, trimming down image size, and streamlining dependencies. Collaborating closely with Docker specialists, we executed various optimization techniques, such as refining container startup times, eliminating unnecessary layers, and enhancing container communication. To validate the improvements, I conducted comprehensive performance tests both before and after the optimization process. <u>As a result</u> • Our analysis of the existing Docker container setup allowed us to identify key areas for efficiency improvement. • We successfully implemented a range of optimization techniques, including optimizing container startup times, minimizing unnecessary layers, and improving container communication. • These optimizations led to significant improvements in resource usage, response times, and overall system efficiency. • By conducting performance tests before and after optimization, we were able to quantitatively measure the positive impact of these changes, further validating the effectiveness of our optimization efforts.	8

Task	Description	Hours
Troubleshooting Docker Issues	Identified issues related to Docker, such as container crashes, network glitches, and conflicts in resource allocation, by closely monitoring error logs and system behavior. Working in collaboration with Docker experts and system administrators, I delved into debugging tools and thorough analysis to uncover the underlying causes of these problems. Subsequently, I skillfully implemented solutions to rectify the Docker-related issues, which included optimizing resource allocation and updating Docker images. Rigorous testing followed the implementation of these solutions to ensure the effective resolution of the identified Docker problems and to confirm the seamless functionality of the system. <u>As a result</u> • Our proactive approach to identifying Docker-related problems allowed us to address issues such as container crashes, network glitches, and resource allocation conflicts. • Through collaboration with Docker experts and system administrators, we diagnosed the root causes of these problems using debugging tools and careful analysis. • By adjusting container configurations, optimizing resource allocation, and updating Docker images, we successfully resolved the identified issues. • Thorough testing after implementing these solutions confirmed the successful resolution of the Docker-related problems.	6
Python-R Integration Debugging	Investigated instances of code breaks by analyzing error messages, stack traces, and log files, aiming to pinpoint the underlying reasons behind the issues. Upon diagnosing the code breaks, I engaged in collaborative discussions to propose and deliberate potential solutions. Subsequently, I adeptly implemented the necessary fixes to restore the system's functionality. Rigorous testing ensued after the fixes were applied to guarantee that the code breaks were effectively resolved, and that the system operated without any regression issues, ensuring its proper functioning. <u>As a result</u> • Our rigorous investigation into code breaks, using error messages, stack traces, and log files, enabled us to identify the underlying causes of these issues. • We discussed potential solutions and effectively implemented fixes that restored the system's functionality. Subsequent thorough testing validated the success of our efforts, ensuring that the	4

Task	Description	Hours
	addressed code breaks were resolved, and that the system operated as intended without any regressions. This approach demonstrates our commitment to maintaining a reliable and fully functional system.	
Code Refinement	Thorough review of the existing Python-R integration code to identify opportunities for enhancing and optimizing the interaction between the two languages. Through collaborative efforts with the team, we refined the integration code for both Python and R. Our focus was on facilitating smoother communication, efficient data exchange, and the seamless execution of combined functionality. <u>As a result</u> - Our careful review of the existing Python-R integration code allowed us to pinpoint areas that needed improvement. - Through collaboration with the team, we diligently refined the code, leading to enhanced interaction between Python and R. This effort resulted in strengthening the integrity of our integrated system.	4
Docker Testing and Validation	Test the Docker container after implementing changes and optimizations, verifying that the modifications did not introduce new issues and that the container still functioned as expected. <u>As a result</u> - Our thorough testing of the Docker container ensured the stability and reliability of the modifications. - We confirmed that these adjustments did not introduce any new issues and that the container continued to function in alignment with our expectations.	1
Integration Testing	Created a comprehensive testing plan to evaluate the Python-R integration. We executed end-to-end tests using various datasets and scenarios to ensure the integration's functionality, accuracy, and performance across different use cases. <u>As a result</u> - The development of a comprehensive testing plan ensured a thorough evaluation of the Python-R integration. - By executing end-to-end tests, we were able to assess the robustness and reliability of the integrated code. - This rigorous testing approach guaranteed the functionality and effectiveness of the integration in diverse real-world situations.	2

Task	Description	Hours
Documentation Update	Added comprehensive details about the Python-R integration process to the project documentation. The documentation now includes step-by-step instructions, code snippets, and detailed explanations to guide future users through the setup and usage of the integration. <u>As a result</u> • The project documentation now provides comprehensive guidance on the Python-R integration process. • We ensured that future users can seamlessly navigate the integration setup and effectively utilize its capabilities. • This documentation serves as a valuable resource, facilitating smooth onboarding and utilization of the integration for various projects and purposes.	2
Code Review and Refactor	Identified areas within the code structure that could benefit from improvements in terms of modularity and maintainability. After careful analysis, I collaborated with the team to restructure code modules, functions, and classes. <u>As a result</u> • By identifying and addressing areas for enhancement, we've restructured code modules, functions, and classes to foster clearer communication and facilitate effective collaboration.	1
Task Review and Planning	Reviewed the tasks completed during the project in light of the initial goals and objectives. I assessed the progress achieved, including the successful completion of key milestones and the challenges that were encountered and overcome along the way. This reflection allowed me to gain insights into the lessons learned throughout the project's lifecycle and to identify areas for potential improvement in future endeavors. Additionally, I reflected on my personal growth and skill development, recognizing the strengths I've gained and areas where further improvement is possible for continued professional growth. <u>As a result</u> • The review process has provided valuable insights into the alignment between our initial goals and the project's outcomes. • By evaluating completed milestones, challenges overcome, and lessons learned, we've gained a holistic view of our journey.	1

Task	Description	Hours
Final Testing and Validation	Developed a comprehensive testing plan that encompassed a range of testing scenarios, data inputs, and expected outcomes. This plan included various types of testing to ensure the robustness and reliability of the project. Unit testing was conducted to verify the correctness of individual components, functions, and modules. Integration testing was performed to assess the interaction between different components and ensure seamless communication. Additionally, end-to-end testing was executed to simulate real-world user interactions and evaluate the entire system's functionality. <u>As a result</u> - The testing plan provided a structured approach to ensure the thorough evaluation of the system's functionality. - Final round of testing was done and verified.	2

Table 14 - Week 5 (Robyn Implementation with Integration of Python and R)

Week 6: July 3rd - July 7th, 2023 – Finalization, Reflection and Demo:

- Wrapping up the Docker container optimization and Python-R integration and reflecting on the project.

Task	Description	Hours
Docker Finalization	Analyzed the existing Docker container setup with a focus on identifying optimization opportunities. To simplify the container configuration, I worked on reducing the image size by removing unnecessary components and dependencies. By adhering to industry best practices, I revised the Dockerfile and effectively managed container resources to enhance overall efficiency. <u>As a result</u> - Through the analysis and optimization of the Docker container setup, significant enhancements in efficiency and resource utilization have been achieved. - The streamlined configuration has contributed to quicker container startup times and more effective resource management.	4
Integration Finalization	Conducted a thorough validation of the Python-R integration code to ensure its adherence to the project's specified requirements. This validation encompassed confirming the integrated solution's capability to generate accurate reports and achieve the expected outcomes. In	4

Task	Description	Hours
	the final testing phase, I rigorously tested the integration to assess its robustness and reliability under diverse scenarios. Any issues or refinements that emerged during this critical testing stage were promptly addressed to ensure the integration's readiness for deployment. <u>As a result</u> The thorough validation of the Python-R integration code has provided a high degree of confidence in its compliance with the project's established requirements.	
Bug Fixes and Enhancements	Reviewed the outstanding issues, bugs, and challenges that emerged during the testing phase. This involved a thorough analysis of the root causes behind these issues and an assessment of their potential impact on the integrated solution. Prioritizing based on severity; I focused on resolving critical issues first. I implemented the required code adjustments and fixes to address these challenges, aiming to enhance the system's stability and functionality. After implementing these fixes, I conducted rigorous testing to ensure that the identified issues were effectively resolved, and that the overall system integrity was maintained. <u>As a result</u> - By addressing the identified issues and bugs through careful analysis and targeted fixes, the integrated solution's reliability and functionality have been significantly enhanced. - The resolution of critical issues has led to a more stable and robust system, ensuring that end-users experience improved performance and a smoother interaction with the solution.	8
Code Review and Refactor	Conducted a comprehensive review of the codebase, paying close attention to code organization, readability, adherence to coding standards, and modularity. During this examination, I identified potential code smells, instances of redundancy, and opportunities for code optimization. To enhance the overall code quality, I provided constructive feedback and practical suggestions for improvement. <u>As a result</u> The careful analysis of code organization, readability, and adherence to coding standards has led to actionable insights for improving the codebase's quality and maintainability.	2

Task	Description	Hours
	The provided feedback and suggestions for improvement contribute to an enhanced codebase that is more readable, maintainable, and aligned with best practices.	
Final Docker Testing	Conducted a series of tests to comprehensively evaluate the functionality, scalability, and resource utilization of the container. This involved verifying that the containerized application exhibited consistent behavior across diverse environments and configurations. Additionally, I rigorously tested the container's resilience to different scenarios, such as high workloads and unexpected failures, ensuring that it maintained stability and continued functioning reliably. <u>As a result</u> - The comprehensive testing conducted on the containerized application has provided a clear understanding of its behavior, scalability, and robustness. - Rigorous testing of the container's resilience to high loads and unexpected failures has prepared it to handle real-world challenges and maintain stability even under adverse conditions.	6
Code Review and Refactor	Performed a comprehensive final review of the entire codebase, examining each individual component as well as their interactions within the system. This review aimed to identify any lingering issues, inconsistencies, or discrepancies that might have gone unnoticed during earlier testing phases. I addressed these issues by implementing necessary fixes and adjustments to ensure the codebase's overall quality, coherence, and functionality. <u>As a result</u> By scrutinizing individual components and their interactions, we have successfully identified and implemented improvements that contribute to better performance, enhanced readability, and long-term maintainability.	2
Task Review and Reflection	Reflect on the entire project journey, from inception to completion, considering achievements, challenges, and personal growth throughout the process. <u>As a result</u> The project's lifecycle has equipped me with new skills, improved my problem-solving abilities, and highlighted areas for continued development.	1

Task	Description	Hours
	This retrospective process enhanced my understanding of project management, collaboration, and technical execution, serving as a foundation for future endeavors.	
Project Handover	Compiled a comprehensive guide and manual intended to assist future maintainers and users in comprehending and effectively managing the project. This guide is organized to ensure easy access to project files, resources, and documentation in a structured manner. <u>As a result</u> - The compiled guide and organized resources offer a valuable resource for stakeholders. - This documentation ensured that the project's intricacies, functionalities, and maintenance procedures are well-documented and easily accessible.	2
Knowledge Sharing	Shared key insights from the project with the product team, highlighting lessons learned, challenges overcome, and successes achieved. I emphasized effective collaboration, robust testing practices, and the value of adopting best coding practices. Our proactive approach to optimization and close collaboration with experts led to significant efficiency gains. <u>As a result</u> Sharing these insights with product team fosters a collaborative learning environment and facilitates knowledge exchange.	2
Notion Update	Navigated to the project page on Notion and accessed the existing documentation. I diligently edited and updated the project details, outcomes, and recent changes to reflect the latest developments accurately. I added new sections and information pertaining to the final project status, integrations, and detailed testing results. <u>As a result</u> - The project documentation is now thoroughly revised and enriched with the latest information. - By updating details, outcomes, and integrating new sections, the Notion workspace is now a comprehensive resource reflecting the final project status, integrations, and testing outcomes.	6

Task	Description	Hours
Task Review and Reflection	Engaged in a thorough reflection on the entirety of the project, delving into its overarching goals, the challenges that were encountered along the way, the noteworthy accomplishments achieved, and the valuable lessons that were learned throughout the journey. This reflection enabled a holistic assessment of the project's impact and value, particularly in relation to its alignment with the initial objectives that were set at the outset. <u>As a result</u> • By evaluating the project's outcomes, challenges, and achievements, valuable insights I have been gained. • The project's impact and alignment with initial objectives have been carefully considered.	2
Demo	Presented a comprehensive demonstration showcasing the successful integration and utilization of Robyn within the project. Emphasized its features and the tangible benefits it brings to the table. <u>As a result</u> • The demo effectively conveyed the seamless integration of Robyn and highlighted its capabilities. By showcasing its features and benefits, stakeholders gained a clear understanding of its value in the project's context. • This presentation served as a testament to the successful implementation and utilization of Robyn.	1

Table 15 - Week 6 (Robyn Implementation with Integration of Python and R)

Project 3 - Geolift Implementation with Integration of Python and R

Week 1: July 10th - July 14th, 2023 - Geolift Implementation and Setup:

- Starting the Geolift implementation and setting up the required environment.

Task	Description	Hours
Geolift Planning	Crafted a comprehensive implementation strategy for the Geolift project, outlining key milestones and tasks needed for success. This involved planning, data collection, model development, testing, and deployment. <u>As a result</u> - This well-defined implementation strategy was designed. - By outlining key milestones and tasks, the project's progress can be tracked and managed effectively. - The established timeline provided a clear roadmap for achieving project goals in a timely manner.	1
Environment Setup	Installed and configured the essential software and tools for Geolift development, ensuring seamless compatibility and proper settings. To prepare the development environment, I created the necessary directories and structures, efficiently organizing project files for smooth workflow. <u>As a result</u> With the necessary software, tools, and organized project files, the foundation for efficient and organized development environment has been established.	2
Geolift Code Implementation	Initiated the coding process for the Geolift project, translating the established plan into foundational logic. Developing the core algorithms and functions, I focused on crafting the backbone of Geolift's key features. <u>As a result</u> - The fundamental building blocks of the Geolift project have been laid down. - Core algorithms and functions are now in place, including the initial code for handling geospatial data manipulation and analysis. - This forms the basis for further development and refinement.	8

Task	Description	Hours
Initial Integration	Explored various methods to effectively integrate Python and R within the Geolift project. By establishing communication channels between the two languages, I ensured a smooth exchange of data. I implemented initial code snippets to showcase the interaction between Python and R, laying the foundation for seamless collaboration between the languages within the project. <u>As a result</u> Initial code snippets have been developed to showcase the interaction and collaboration between Python and R, paving the way for more sophisticated integrations.	8
Geolift Coding	Extend the Geolift implementation by integrating a variety of advanced features to elevate its capabilities. Through iterative development cycles, I refined the existing codebase based on feedback obtained from testing and reviews. I addressed diverse edge cases and potential challenges that emerged during development, ensuring the geospatial algorithms' strength and dependability. Rigorous testing was conducted to validate the extended code's performance, accuracy, and reliability, with adjustments made where necessary. <u>As a result</u> - The Geolift project had to encompass a diverse set of advanced features, enhancing its capabilities for geospatial data manipulation and analysis. - The iterative development approach has contributed to a refined codebase and improved functionality. Addressing edge cases and challenges has bolstered the robustness of the geospatial algorithms. - Thorough testing has verified the extended code's performance and reliability, with adjustments ensuring optimal outcomes.	8
Integration Refinement	Improved the integration between Python and R components by implementing optimizations that notably enhanced overall performance. I streamlined data transfer mechanisms between the two languages, effectively minimizing overhead and latency to facilitate smoother communication. The integration underwent rigorous testing across diverse scenarios, ensuring reliable communication and consistent data interchange between Python and R. During testing, compatibility issues emerged and were promptly addressed to ensure seamless interaction between the languages.	8

Task	Description	Hours
	<u>As a result</u> - A dedicated effort to refine Python-R integration facilitated a more cohesive and efficient workflow, ultimately contributing to the overall success of the Geolift project. - The optimized integration has improved performance and data exchange between Python and R, enabling smoother collaboration and enhanced capabilities within the project.	
Task Review and Planning	Took moment to review the completed tasks and milestones that had been achieved. I evaluated whether the project was on track and assessed whether any adjustments to the plan were necessary. Additionally, I documented the lessons learned and insights gained during the implementation process. <u>As a result</u> - This review provided valuable insights into the progress of the project, ensuring alignment with the initial plan and goals. - Adjustments were made where necessary to enhance efficiency.	2
Code Review and Refactor	Conducted a comprehensive code review to identify potential bugs, errors, and inefficiencies within the codebase. After pinpointing areas for improvement, I proceeded to refactor the code to align with coding standards and enhance overall readability. <u>As a result</u> The code review and refactoring process led to improved code quality, increased maintainability, and reduced likelihood of bugs.	2
Communication and Updates	Maintained open communication with the project team, consistently providing updates on the progress of the Geolift implementation. During these discussions, I openly addressed challenges that had arisen and actively sought input from team members to collectively brainstorm potential solutions moving forward. <u>As a result</u> Effective communication with the project team ensured transparency, collaboration, and alignment, facilitating timely resolution of challenges, and promoting the successful implementation of Geolift.	1

Table 16 -Week 1 (Geolift Implementation with Integration of Python and R)

Week 2: July 17th - July 21st, 2023 - Geolift Functionality and Testing:

- Developing Geolift functionality and conducting thorough testing.

Task	Description	Hours
Geolift Coding	Made substantial strides in implementing new functionality within the Geolift project. I dedicated time to refining and expanding the codebase, ensuring that it could effectively accommodate the project's evolving requirements. A key focus was ensuring the smooth integration of the newly added features with the existing logic, maintaining a cohesive and functional codebase throughout the development process. <u>As a result</u> • The project's functionality was enhanced and extended to meet the changing needs, and integrated new developed feature.	6
Integration Testing	Conducted thorough testing to validate the communication and compatibility between the Python and R components. I placed special emphasis on optimizing data transfer mechanisms to ensure seamless interaction between the two languages. As part of this process, I addressed any issues that surfaced during testing, working to enhance the overall reliability and efficiency of the integration. <u>As a result</u> • The Python-R integration was thoroughly tested and refined, resulting in a robust and dependable communication between the two languages, bolstering the Geolift's functionality.	2
Geolift Enhancements	Expanded Geolift's capabilities by introducing new features that aligned with the project's objectives. These additional functionalities were seamlessly integrated into the existing code architecture to ensure a cohesive system. To ensure the correct implementation of the added features, I conducted preliminary testing, verifying their functionality and compatibility within the system. <u>As a result</u> • The project's functionalities were enhanced, providing a more comprehensive and valuable tool.	4
Testing Preparation	Developed a comprehensive testing plan that outlined various testing scenarios to ensure a thorough validation of the codebase. This plan included identifying key testing points and formulating a strategy to	1

Task	Description	Hours
	assess both functionality and stability. By covering different aspects such as unit testing, integration testing, and end-to-end testing, I aimed to ensure that the codebase was rigorously tested and validated to meet the highest standards of quality and performance. <u>As a result</u> • A well-defined testing plan was established to comprehensively evaluate the codebase's functionality and stability. • This contributed to a systematic approach to testing and ensured the project's quality and reliability.	
Geolift Coding	Continuing with the incremental development of Geolift functionality, I maintained a steady pace of progress to ensure consistent advancement. Throughout this process, I ensured that the ongoing code development remained aligned with the established goals and timelines. By following this approach, I aimed to achieve a well-structured and well-paced development process that allowed for thorough testing and refinement while staying on track to meet project objectives. <u>As a result</u> • The consistent and incremental development approach led to steady progress, ensuring that the project remained on track and within the defined timeline.	4
Comprehensive Testing	Executed a comprehensive testing regimen to thoroughly verify the functionality and stability of the recently added features in Geolift. During this testing phase, I diligently identified and addressed any bugs, glitches, and inconsistencies that surfaced. By rigorously testing the new features in various scenarios and use cases, I aimed to ensure that they operated as intended and integrated seamlessly with the existing codebase. <u>As a result</u> • The rigorous testing efforts contributed to the identification and resolution of potential issues, ensuring that the codebase was robust and well-prepared for subsequent stages of development. • This approach helped enhance the overall quality and reliability of the Geolift project.	2

Task	Description	Hours
Create New Docker Image	Designed a Docker image tailored to encapsulate all necessary components and configurations for parallel processing in the Geolift project. This involved selecting suitable base images and defining dependencies to establish a consistent and reliable environment. Within the Docker image, I seamlessly integrated parallelization libraries and tools, enabling the project to efficiently execute tasks in parallel. Additionally, I fine-tuned container settings to optimize resource allocation and utilization, ensuring that parallel processing was as efficient as possible. To enhance reproducibility, I thoroughly documented the Docker image, enabling seamless deployment and consistent parallel processing capabilities within the Geolift project. <u>As a result</u> • The Docker image was successfully created and configured to support parallel processing in the Geolift project. • This image offers a reliable and efficient environment for executing tasks concurrently, contributing to improved performance and resource utilization. • The detailed documentation ensures that the setup process can be easily replicated and maintained.	8
Test New Instance	Conducted a series of diverse test scenarios on the newly developed Docker instance, specifically designed for parallel processing in the Geolift project. These tests were designed to emulate real-world workloads, providing a comprehensive assessment of the efficiency and scalability of the parallelization approach. Throughout testing, I diligently collected and analyzed performance metrics, including processing time and resource utilization, to gauge the effectiveness of the Docker setup. By closely examining the test results, I identified potential bottlenecks, performance issues, and areas for improvement within the parallel processing environment. Leveraging these insights, I fine-tuned the Docker configuration, implementing optimizations to streamline the parallel processing workflow and ultimately enhance overall performance. <u>As a result</u> • The comprehensive testing of the Docker instance revealed valuable insights into the efficiency and scalability of the parallel processing approach. • Performance metrics provided a clear understanding of the system's behavior under various workloads.	8

Task	Description	Hours
	- Identified bottlenecks and issues were addressed through careful configuration adjustments, resulting in an optimized parallel processing environment. - This optimization contributes to faster execution times and improved utilization of resources, aligning with the project's goals of efficiency and responsiveness.	
Testing Analysis	Reviewed the outcomes of the comprehensive testing conducted on Geolift's functionality, as well as the analysis of the Docker instance designed for parallel processing. In doing so, I identified several opportunities for further enhancing the performance and efficiency of Geolift. These insights helped me pinpoint areas where optimizations could lead to significant improvements in processing time, resource utilization, and overall system efficiency. With a clear understanding of these potential enhancements, I developed a strategy to implement targeted optimizations that align with the project's goals and requirements. <u>As a result</u> - The review of testing outcomes and Docker analysis provided a clear understanding of the system's strengths and areas for improvement. - By identifying performance bottlenecks and analyzing data-driven insights, we gained valuable guidance for optimization strategies. - This informed approach ensures that optimization efforts are targeted and effective, ultimately contributing to a more efficient and responsive Geolift system.	2
Bug Fixes and Enhancements	Allocated dedicated time to enhance the overall quality and stability of the Geolift codebase. This involved addressing issues that were identified during testing and analysis, including the optimization of performance bottlenecks to ensure smoother execution. To further elevate the codebase's quality, I engaged in code refactoring and made necessary adjustments to ensure alignment with industry best practices and coding standards. <u>As a result</u> - The focused effort on code improvement led to a more stable and efficient codebase. - By addressing identified issues, optimizing performance, and adhering to best practices, the codebase's quality was enhanced.	4

Task	Description	Hours
	These refinements contribute to a more maintainable, readable, and reliable Geolift project, ensuring smoother development and future updates.	
Task Review and Planning	Reflected on the completed tasks and overall progress of the Geolift project, ensuring alignment with the plan and making necessary adjustments. I strategized for the final stages, leveraging lessons learned and gained insights for a successful conclusion. <u>As a result</u> - The reflective process provided valuable insights into the project's trajectory. - Adjustments were made to align with the evolving requirements and challenges. - By considering lessons learned and insights gained, the final stages of the project were approached with a refined strategy, ensuring a more effective and successful project culmination.	1
Communication and Updates	Participated in a team meeting to discuss findings and share insights from recent development and testing phases. Delivered updates on progress and addressed questions and concerns raised by fellow team members. <u>As a result</u> The collaborative team meeting facilitated effective communication and alignment among team members.	1

Table 17 - Week 2 (Geolift Implementation with Integration of Python and R)

Week 3: July 24th - July 28th, 2023 - Finalization and Deployment Preparation:

- Finalizing Geolift implementation and preparing for deployment.

Task	Description	Hours
Geolift Refinement	Dedicated time to enhancing the Geolift codebase, prioritizing improvements for enhanced quality and stability. Conducted thorough code reviews and implemented optimizations guided by best practices. Fine-tuned algorithms and functionalities to ensure streamlined execution and maintainability. Addressed identified bugs and inconsistencies through meticulous refinement efforts.	8

Task	Description	Hours
	<u>As a result</u> The concerted efforts to enhance the codebase have led to improved code quality, stability, and efficiency. By conducting code reviews, optimizing algorithms, and addressing issues, the project has become more resilient and easier to maintain.	
Integration Testing	Performed rigorous integration tests to confirm the smooth interaction between Python and R components. Validated the accuracy of communication and data exchange mechanisms. Ensured that the Python-R integration aligned with the project's specifications and requirements. Swiftly identified and resolved any compatibility issues that surfaced during testing, ensuring seamless interaction between the two languages. <u>As a result</u> - The thorough integration testing process confirmed the successful interaction between Python and R components. - By verifying data exchange mechanisms and adherence to project requirements, the integration's reliability was established. - Addressing compatibility issues ensures a smooth collaborative workflow between the two languages, contributing to the overall effectiveness of the Geolift project.	4
Bug Fixes and Enhancements	Proactively identified and resolved issues within the Geolift project. Collaborated with team members to prioritize critical problems and expedite their resolution. Employed thorough debugging and troubleshooting techniques to pinpoint the root causes of reported issues. Implemented precise fixes and solutions to rectify bugs, inconsistencies, and errors in the codebase. Ensured rigorous testing and validation of the resolved issues to prevent future recurrences and maintain the project's stability. <u>As a result</u> - The systematic approach to issue identification and resolution significantly improved the stability and functionality of the Geolift project. - Collaborative efforts and effective problem-solving contributed to overcoming obstacles and maintaining the project's momentum. - Through rigorous testing and validation, the project's codebase was fortified against recurring issues, ensuring a more robust and reliable outcome.	8

Task	Description	Hours
Deployment Preparation	Organized project resources and the codebase in anticipation of deploying Geolift to the Vertex AI platform. Ensured that essential dependencies and configurations were accurately configured for seamless deployment. Conducted thorough checks to validate the consistency and readiness of the deployment environment, guaranteeing a smooth transition to the Vertex AI platform. <u>As a result</u> • The project resources and codebase were streamlined and organized, facilitating a smooth deployment process. • With properly configured dependencies and a consistent environment, the readiness of the deployment setup was confirmed. • This preparation sets the stage for a successful deployment of Geolift to the Vertex AI platform, ensuring that the project's capabilities can be utilized effectively.	3
Final Testing	Conducted an exhaustive series of tests to thoroughly validate the end-to-end functionality of Geolift. Ensured the seamless integration of new features while rigorously verifying the preservation of existing functionality. Scrutinized the project's behavior against predefined use cases and requirements to confirm alignment with project goals. <u>As a result</u> • The thorough testing process provided a clear understanding of Geolift's end-to-end functionality. • New features integration and confirming the preservation of existing functionality, the project's quality and reliability were upheld.	4
Deployment to Test Environment	Successfully deployed the refined Geolift project to a designated test environment, meticulously ensuring a smooth and error-free deployment process. Thoroughly verified the proper functioning of all components within the test environment, confirming their alignment with project expectations and requirements. <u>As a result</u> • This deployment marked a significant milestone in the project's development, confirming that the refined Geolift system could be successfully set up and operated in a controlled environment.	2

Task	Description	Hours
Testing and Validation	Conducted a targeted evaluation of the deployed Geolift functionality to validate its anticipated behavior. Carefully verified that the project's performance aligned seamlessly with the intended outcomes and satisfactorily fulfilled the established project requirements. Effectively addressed any identified discrepancies and rectified issues that emerged during the rigorous testing phase. <u>As a result</u> - The thorough evaluation of the deployed Geolift functionality confirmed its alignment with the intended objectives and project requirements. - This process ensures that the deployed solution maintains its functionality and meets user expectations, contributing to the successful execution of the Geolift project.	2
API Testing	Thoroughly validated the functionality of API endpoints within the Geolift project. Executed meticulous tests to guarantee that the APIs reliably and accurately responded in accordance with their intended functionalities. <u>As a result</u> - Through this validation process, the proper functioning and reliability of the API endpoints in the Geolift project were confirmed. - This step was crucial to ensure that the system's APIs perform as expected and meet the project's requirements, ultimately contributing to the successful implementation of Geolift's functionalities.	2
Notion Updates	Logged the project's advancements and developments in the Notion documentation. Ensured that the relevant sections were revised to provide an accurate representation of the project's status. <u>As a result</u> By maintaining accurate and up-to-date documentation, the project's evolution and current status were effectively captured, facilitating easy access to crucial information for team members and stakeholders.	2
Knowledge Sharing	Shared significant insights, findings, and lessons learned with the team regarding the Geolift project. Discussed experiences related to the	2

Task	Description	Hours
	challenges encountered, the effective solutions implemented, and valuable best practices discovered during the project's lifecycle. <u>As a result</u> • The presentation provided an opportunity to reflect on the project's journey, highlighting successes, addressing challenges, and sharing valuable insights gained. • By sharing experiences and lessons learned, the team was better equipped to apply these learnings in future projects and endeavors.	
Task Review and Reflection	Engaged in a comprehensive reflection to evaluate the entirety of the Geolift project's progress. Summarized notable achievements attained and challenges that were encountered over the course of the project's lifecycle. Carefully assessed the project's alignment with the initial goals set at the outset, while also evaluating any deviations that may have occurred during its execution. <u>As a result</u> • This reflective process offered valuable insights into the project's trajectory, highlighting successful milestones, addressing encountered challenges, and evaluating the project's overall alignment with its initial objectives.	2
Demo	Conducted an extensive demonstration of the Geolift project's features and functionalities, showcasing its capabilities to stakeholders. Emphasized the project's benefits and value proposition, effectively communicating its potential impact and advantages to key stakeholders. <u>As a result</u> • This demonstration provided stakeholders with a firsthand experience of Geolift's capabilities, showcasing its features and illustrating its potential benefits and value.	1

Table 18 - Week 3 (Geolift Implementation with Integration of Python and R)

Project 4 - Customer Lifetime Value

Week 1: July 31st – August 4th, 2023 - Project Initialization and Environment Setup

Task	Description	Hours
Requirement Gathering	Conducted meetings with team members and relevant departments to discuss and understand their expectations and needs for the project. Held these meetings to ensure alignment between the project's goals and the stakeholders' vision. Clarified key deliverables, such as reports, models, or systems, to ensure everyone had a clear understanding of what the project would produce. Identified data sources and required features for the Customer Lifetime Value (CLV) model to determine the necessary data sources, including customer databases, transaction records, and behavioural data. Identified the essential features and variables needed for the model, such as customer demographics, purchase history, and customer interaction data, to calculate CLV accurately. <u>As a result</u> Enhanced clarity and alignment on project goals and stakeholder expectations were achieved through targeted meetings. The identification of key deliverables and data sources provides a solid foundation for the successful development of the Customer Lifetime Value model, ensuring accuracy in calculations and meaningful insights.	8
Python Environment Setup	Installed Python and necessary packages (e.g., NumPy, Pandas, Matplotlib, Scikit-Learn) for data analysis and modeling. Ensured that the environment had the appropriate version and compatibility for the project. Installed Jupyter Notebook and set it up to create an interactive and easily shareable environment for data analysis. Configured Jupyter Notebook extensions and themes for improved productivity. <u>As a result</u> - The installation of essential Python packages and the setup of a compatible environment provide a solid foundation for data analysis and modeling. - The incorporation of Jupyter Notebook enhances the interactive and collaborative aspects of the project, promoting efficient data analysis and sharing.	4

Task	Description	Hours
Explore Packages	Researched the "lifetimes" and "PyMC" Python packages to determine their applicability for building a CLV model. Focused on understanding how these packages could contribute to the project's objectives. Examined the official documentation for "lifetimes" and "PyMC" to gain insights into their functionalities, methods, and capabilities. Looked for information that could help in integrating these packages effectively. Sought out code examples and practical demonstrations of CLV modeling using "lifetimes" and "PyMC." Analyzed these examples to understand how the packages can be applied in real-world scenarios. <u>As a result</u> - The research on "lifetimes" and "PyMC" has provided a comprehensive understanding of their applicability and potential contributions to the CLV model. Insights gained from the official documentation enhance the ability to effectively integrate these packages into the project. - The exploration of code examples and practical demonstrations offers valuable knowledge on applying "lifetimes" and "PyMC" in real-world CLV modeling scenarios. This research lays the groundwork for informed decision-making in selecting and implementing the most suitable packages for the project's objectives.	8
Finalization of Package	Investigated user reviews, forum discussions, and community feedback regarding experiences with "lifetimes" and "PyMC." Considered the overall sentiment, satisfaction, and any identified challenges. <u>As a result</u> - The investigation into user reviews, forum discussions, and community feedback has provided valuable insights into the experiences of users with "lifetimes" and "PyMC." - This understanding of overall sentiment, satisfaction levels, and identified challenges contributes to a more informed decision-making process regarding the selection and implementation of these packages for the CLV model.	4

Task	Description	Hours
Data Exploration in BigQuery	Utilized BigQuery, a cloud-based data warehouse, to access and examine the dataset. Executed SQL queries to gain insights into the dataset's structure, including the tables, columns, and relationships, as well as the types of data it contains. Scrutinized the dataset for issues such as missing values, duplicate records, outliers, or inconsistent data formats. Detected anomalies or discrepancies that might affect the reliability and accuracy of the dataset. Recognized specific areas within the dataset that required data preprocessing, including cleaning, transformation, or feature engineering. Determined the necessary data preparation steps to improve data quality and readiness for analysis or modeling. <u>As a result</u> • The utilization of BigQuery and the thorough examination of the dataset have provided a comprehensive understanding of its structure and potential issues. • The insights gained from this process enable informed decisions regarding data preprocessing steps, ensuring that the dataset is clean, reliable, and well-prepared for subsequent analysis or modeling tasks.	8
Feature Extraction	Examined the dataset to pinpoint key features and attributes that could offer valuable insights into customer behavior. Focused on aspects such as purchase history, browsing patterns, interaction frequencies, and any other relevant data points. Applied data transformation methods to convert raw data into a format suitable for analysis. Aggregated and structured data to facilitate the extraction of meaningful customer behavior information. Investigated advanced analysis techniques, including cohort analysis to group customers, sessionization to track user interactions, and user segmentation to categorize customers based on behavior. Employed these techniques to gain deeper insights into customer behavior patterns and trends. Worked closely with data engineers to streamline the extraction process. Addressed any technical challenges that arose during data extraction, ensuring data was accessible and efficiently processed. Created documentation outlining the methods used for data extraction. Detailed the data sources accessed and any preprocessing steps applied. Ensured transparency and reproducibility for future analysis and to share insights with other team members. <u>As a result</u> • The examination and extraction of key features from the dataset, coupled with the application of advanced analysis	6

Task	Description	Hours
	techniques, have yielded valuable insights into customer behavior. The structured and transformed data is now readily available for analysis, and the documentation ensures transparency and reproducibility for future reference and collaboration within the team.	
Task Review and Planning	Took time to reflect on the completed tasks and the overall progress of the project. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. Strategized for the next stages of work, considering any lessons learned and insights gained. <u>As a result</u> The reflective process provided valuable insights into the project's progress and allowed for adjustments to the plan as needed. This strategic evaluation ensures that the project is aligned with its goals, incorporating lessons learned for more effective future stages.	1
Communication and Updates	Engaged in a team meeting to discuss findings and share insights from exploration. Provided updates on progress and addressed any questions or concerns raised by team members. <u>As a result</u> The team meeting facilitated open communication, allowing for the exchange of findings and insights.	1

Table 19 - Week 1 (Customer Lifetime Value)

Week 2: August 7th – August 11th, 2023 - Data Manipulation and Query Optimization

Task	Description	Hours
Data Transformation	Accessed the custom attributes table, ensuring data is available for analysis. Cleaned the data by addressing any missing values, duplicates, or inconsistencies to ensure data quality and accuracy. Determined the most recent transaction date for each customer. This represents the "Recency" value, indicating how recently a customer made a purchase. Counted the number of transactions or purchases made by each customer. This represents the "Frequency" value, indicating how often a customer makes purchases. Calculated the total monetary value of transactions for each customer, which represents the "Monetary" value. This shows the cumulative spending of each customer. Determined the time duration between the first purchase made by each customer	8

Task	Description	Hours
	and the present date. This represents the "T" value, indicating the duration of the customer's relationship with the business. Combined the calculated R, F, M, and T values into a single dataset, with each row representing a unique customer. <u>As a result</u> A comprehensive dataset was created, incorporating key customer metrics (Recency, Frequency, Monetary, and Time) that provide valuable insights into customer behavior and engagement. This dataset serves as a foundation for further analysis and modeling in the Customer Lifetime Value (CLV) estimation.	
Create Query	Analyzed the structure of the custom attributes table to understand its schema, including column names and data types. Identified which columns contain information related to customer transactions, such as purchase dates and transaction amounts. Wrote an initial SQL query that retrieves the necessary data for calculating RFM and T values. This query includes columns for customer identifiers, purchase dates, transaction amounts, and any other relevant information. Extended the SQL query to calculate the RFM and T values based on the extracted data, using SQL functions and operations to determine Recency (R), Frequency (F), Monetary Value (M), and Time since the first purchase (T). Executed the query to ensure that it accurately calculates RFM and T values. Verified the correctness of the results by comparing them to expected outcomes. <u>As a result</u> The SQL query successfully extracted and calculated the RFM and T values, providing a comprehensive dataset with key metrics that offer insights into customer behavior and engagement. This dataset serves as a crucial foundation for further analysis and the estimation of Customer Lifetime Value (CLV).	6

Task	Description	Hours
Query Optimization	Reviewed the query execution plan and identified potential performance bottlenecks. Explored opportunities to optimize the query for reduced execution time and resource consumption. If the custom attributes table is large, considered creating appropriate indexes and partitions on the columns involved in the query to significantly improve query performance. Applied filters or WHERE clauses to limit the data retrieved from the table to only what is necessary for RFM and T calculations, aiming to enhance query performance by reducing the dataset size. Minimized the use of subqueries and complex joins in the query, opting for simpler, straightforward SQL statements to mitigate potential impacts on performance. Utilized caching mechanisms or materialized views when appropriate to store precomputed results, reducing the need to rerun the query frequently. Executed the optimized query and compared its performance with the initial query, ensuring efficient and timely retrieval of results. <u>As a result</u> • The optimization efforts led to improved query performance, addressing potential bottlenecks. • Filtering data for RFM and T calculations resulted in a more streamlined query process, further optimizing resource utilization. • The optimized query demonstrated efficient and timely retrieval of results, validating the success of the performance improvement measures.	6
Initialise PyMC model	Set up the data and defined the variables used in the model, incorporating observed data, prior distributions, and any latent variables. Specified the prior distributions for the model parameters, defining the beliefs about the model's parameters before observing the data. Wrote the likelihood function that described the relationship between the model parameters and the observed data, quantifying how well the model fit the data. Developed model custom utility functions and transformations, writing these functions to preprocess the data or handle specific aspects of the modeling process. <u>As a result:</u> • The data was effectively prepared for modeling, with variables, observed data, and latent variables clearly defined. • Clear prior distributions were specified, reflecting the beliefs about the model's parameters before data observation.	8

Task	Description	Hours
Develop PyMC model builder	Used PyMC to construct the model, combining prior distributions and likelihood functions to create a probabilistic model. Chose the appropriate inference method for the model, specifying the method that suited the modeling needs, considering options like MCMC (Markov Chain Monte Carlo) or variational inference. Executed the chosen inference method to estimate the posterior distribution of the model parameters, sampling from the posterior using PyMC's sampling functions. After running inference, analyzed the results, including posterior distributions, trace plots, and summary statistics. Diagnosed the model's convergence and checked for any issues in the inference process. <u>As a result</u> - The model's construction and inference processes were successfully executed using PyMC. - Analysis of the results, including posterior distributions and diagnostic checks, provided insights into the model's performance, and guided potential refinements.	8
Comparison of Different models	Researched these models in academic literature and relevant resources to understand the theory and assumptions behind each model. Prepared data, including purchase history, customer behavior, and transaction data, which is crucial for parameter estimation and model fitting. Researched the Beta-Geometric model, commonly used to model purchase patterns, noting its assumptions, including the probability of being "alive" and making repeated purchases. Explored the Pareto/NBD model, another popular CLTV model, understanding its assumptions, such as the transaction process and customer dropout. Learned about the Gamma-Gamma model, which focuses on modeling customer spending behavior, familiarizing myself with its assumptions and how it estimates monetary value. Investigated methods for estimating the parameters of these models using the data, typically involving maximum likelihood estimation or Bayesian methods. <u>As a result</u> - Acquired a comprehensive understanding of the theory and assumptions behind each model, enhancing the foundation for subsequent modeling steps. - Prepared relevant data essential for parameter estimation and model fitting, ensuring a robust basis for analysis. - Gained insight into specific models such as the Beta-Geometric, Pareto/NBD, and Gamma-Gamma, understanding their assumptions and applications in modeling customer behavior.	6

Task	Description	Hours
	Explored methods for parameter estimation, laying the groundwork for the practical application of these models to the provided data.	
Task Review and Planning	Took time to reflect on the completed tasks and the overall progress of the project. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. Strategized for the next stages of work, considering any lessons learned and insights gained. <u>As a result</u> - Identified areas of success and challenges, fostering a deeper understanding of the project's dynamics. - Adjusted the project plan to ensure alignment with goals, maintaining a proactive and adaptable approach. - Strategically planned for upcoming stages, leveraging lessons learned and insights for enhanced project execution.	1
Communication and Updates	Engaged in a team meeting to discuss findings and share insights from exploration. Provided updates on progress and addressed any questions or concerns raised by team members. <u>As a result</u> - Fostered a collaborative environment through team meetings, encouraging the exchange of findings and insights. - Ensured transparent communication by providing updates on project progress and addressing any questions or concerns raised by team members.	1

Table 20- Week 2 (Customer Lifetime Value)

Week 3: August 14th – August 18th, 2023 – Beta-Geo Modeling

Task	Description	Hours
Develop Code for Beta-Geo Model	Specified the prior distributions for the model parameters, utilizing Beta distributions in a Beta-Geo model to represent prior beliefs about parameters such as the probability of making repeat purchases. Wrote the likelihood function that described the relationship between the model parameters and the observed data. In the Beta-Geo model, the likelihood function was based on the Negative Binomial distribution. Used PyMC to construct the Beta-Geo model, combining the specified prior distributions and the likelihood function to create a probabilistic model for customer purchase behavior. Chose the appropriate inference method for the model, utilizing PyMC's built-in sampling methods, specifically NUTS (No-U-Turn Sampler), to estimate the posterior distribution. <u>As a result</u> - Established a comprehensive probabilistic model for customer purchase behavior using PyMC, integrating prior distributions and a likelihood function. - Applied the NUTS sampling method to estimate the posterior distribution, ensuring a robust and accurate inference process for the Beta-Geo model.	8
Develop dependent function of Beta-Geo for CLV	Created a function that calculated the probability that a customer had "churned" or stopped making purchases. This function was often complementary to the alive probability function. Developed a function that estimated the probability of a customer dropping out within a specific time frame. This function was often based on the churn probability and considered various factors like attrition rates. Created a function to estimate the expected number of futures purchases a customer was likely to make. This function used a combination of alive probability, churn probability, and other model-specific parameters. Developed a function that estimated the number of new customers who were expected to join in each period. This function may have considered acquisition rates, marketing efforts, or other relevant factors. If the functions depended on model parameters, ensured there was a mechanism to estimate these parameters using Bayesian techniques, maximum likelihood estimation, or other appropriate methods. <u>As a result</u> - Established comprehensive functions for calculating customer churn probability, dropout probability, expected future purchases, and the expected number of new customers. - Enhanced the predictive capabilities of the model by incorporating various factors and model-specific parameters into the functions.	8

Task	Description	Hours
	Implemented mechanisms to estimate model parameters using suitable techniques, ensuring the functions remained accurate and adaptable to changes in the underlying data.	
Bug Fixes and Enhancements	Reviewed any outstanding issues, bugs, or challenges identified during testing. Analyzed the root causes of these issues and their potential impact on the integrated solution. Prioritized the resolution of critical issues and implemented necessary code adjustments or fixes. Conducted thorough testing after implementing fixes to ensure the issues were effectively resolved. <u>As a result</u> - Addressed outstanding issues, bugs, or challenges identified during testing, ensuring a more stable and reliable integrated solution. - Analyzed root causes to understand the underlying issues and their potential impact on the overall system. - Conducted thorough testing post-implementation to verify the effectiveness of the fixes and ensure a higher level of quality in the integrated solution.	6
Locally Test Enhancement	Defined a set of test cases that covered the interactions between different components or modules of the CLV module. Executed the tests to validate that the components worked together seamlessly and produced the expected outcomes after the bug fixes. <u>As a result</u> - Ensured comprehensive test coverage for interactions between different components or modules within the CLV module. - Validated the seamless functioning of components after bug fixes through the execution of the defined test cases, confirming the reliability and effectiveness of the implemented solutions.	2
Code Review and Refactor	Examined code organization, readability, adherence to coding standards, and modularity. Identified any potential code smells, redundancy, or opportunities for code optimization. Provided constructive feedback and suggestions for improvement to enhance the overall code quality. <u>As a result</u> Ensured the code adhered to organizational standards, promoting readability and modularity.	2

Task	Description	Hours
	- Identified and addressed potential code smells, redundancy, and opportunities for optimization, optimizing the codebase for better performance. - Offered constructive feedback and suggestions for improvement, contributing to overall code quality and maintainability.	
Integration Testing	Conducted comprehensive testing to validate the communication and compatibility between Beta-Geo and its functional components. Ensured that chain mechanisms were optimized and that priors interacted smoothly. Addressed any issues that emerged during testing, enhancing the overall reliability of the integration. <u>As a result</u> - Validated the communication and compatibility between Beta-Geo and its functional components through thorough testing, ensuring a robust and cohesive system. - Optimized chain mechanisms and ensured smooth interactions of priors, contributing to the efficiency of the integration. - Addressed and resolved issues identified during testing, enhancing the overall reliability and performance of the integrated system.	2
Code Enhancements	Expanded the capabilities of CLV by introducing new features to meet the project's objectives. Integrated these additional functionalities in a way that complemented the existing code architecture. Conducted preliminary testing to verify the correct implementation of the added features. <u>As a result</u> - The capabilities of CLV were enhanced with the introduction of new features, aligning with the project's objectives. - Preliminary testing confirmed the correct implementation of the added features, contributing to the overall reliability and functionality of CLV.	4
Testing Preparation	Developed a plan for testing scenarios and an approach to ensure thorough validation of the codebase. Identified key testing points and formulated a strategy for assessing functionality and stability. <u>As a result</u> Established a comprehensive plan for testing scenarios, ensuring a systematic and thorough validation of the codebase.	1

Task	Description	Hours
	Identified key testing points and formulated a strategy to assess functionality and stability, contributing to the overall effectiveness of the testing process.	
Beta-geo Coding	Continued with the incremental development of functionality, maintaining a steady pace of progress. Ensured that the ongoing code development aligned with the established goals and timelines. <u>As a result</u> - The incremental development approach was sustained, allowing for a consistent and steady progression of functionality. - Ongoing code development remained aligned with established goals and timelines, ensuring a systematic and goal-oriented advancement of the project.	3
Comprehensive Testing	Executed a comprehensive testing regimen to verify the functionality and stability of the recently added features. Identified and addressed any bugs or inconsistencies that surfaced during the testing phase. Ensured that the codebase was well-prepared for subsequent stages of development. <u>As a result</u> - Verified the functionality and stability of recently added features through a comprehensive testing regimen. - Addressed identified bugs or inconsistencies during the testing phase, enhancing the overall reliability of the codebase. - Prepared the codebase for subsequent stages of development, ensuring a solid foundation for ongoing work.	2
Code Review and Refactor	Performed a comprehensive code review to identify any potential bugs, errors, or inefficiencies. Refactored code to adhere to coding standards and improve overall readability. Addressed any issues raised during the review process. <u>As a result</u> - Identified and addressed potential bugs, errors, or inefficiencies through a thorough code review. - Improved overall code quality by refactoring to adhere to coding standards and enhance readability. - Addressed issues raised during the review process, ensuring the codebase met quality standards and was more maintainable.	1

Task	Description	Hours
Communication and Updates	Communicated with the project team to provide updates on the progress of eta-geo implementation. Discussed any challenges faced and potential solutions moving forward. Aligned everyone's understanding of the status and next steps. <u>As a result</u> - Maintained effective communication within the project team, providing updates on the progress of eta-geo implementation. - Addressed challenges by discussing potential solutions, fostering a collaborative problem-solving approach. - Ensured a shared understanding of the status and next steps, promoting alignment within the project team.	1

Table 21 - Week 3 (Customer Lifetime Value)

Week 4: August 21st – August 25th, 2023 – Pareto-NBD Modeling

Task	Description	Hours
Develop Code for Pareto-NBD Model	Defined the prior distributions for the model parameters, employing Pareto distributions in the context of a Pareto-NBD model to represent prior beliefs about parameters like the shape and scale of the distribution. Formulated the likelihood function that characterized the relationship between the model parameters and the observed data. In the Pareto-NBD model, the likelihood function was grounded in the Negative Binomial distribution. Utilized PyMC to construct the Pareto-NBD model, amalgamating the specified prior distributions and the likelihood function, creating a probabilistic model for customer purchase behavior. Selected the appropriate inference method for the model, utilizing PyMC's built-in sampling methods, such as NUTS (No-U-Turn Sampler), for estimating the posterior distribution. <u>As a result</u> - Established a comprehensive probabilistic model for customer purchase behavior using PyMC, integrating prior distributions and a likelihood function. - Applied the NUTS sampling method to estimate the posterior distribution, ensuring a robust and accurate inference process for the Pareto-NBD model.	8
Develop dependent functions of Pareto-NBD for CLV	Formulated functions for assessing the likelihood of a customer having "churned" or ceased making purchases, often serving as a complement to the alive probability function. Created a predictive function to gauge the probability of a customer discontinuing their engagement within a specified timeframe, considering factors	8

Task	Description	Hours
	such as churn probability and attrition rates. Designed a function for forecasting the anticipated number of futures purchases a customer is expected to make, utilizing a blend of alive probability, churn probability, and other model-specific parameters. Constructed a function for predicting the quantity of new customers anticipated to join during each period, potentially factoring in acquisition rates, marketing efforts, and other pertinent variables. When these functions rely on model parameters, established a mechanism for parameter estimation employing Bayesian techniques, maximum likelihood estimation, or other suitable methodologies. <u>As a result</u> • Developed comprehensive functions for evaluating customer churn likelihood, predicting customer discontinuation probabilities, forecasting future purchases, and estimating the quantity of new customers. • Enhanced the predictive capabilities of the model by incorporating various factors and model-specific parameters into the functions. • Implemented mechanisms for parameter estimation using suitable techniques, ensuring the functions remained accurate and adaptable to changes in the underlying data.	
Bug Fixes and Enhancements	Assessed any unresolved matters, bugs, or obstacles that surfaced during testing. Investigated the underlying reasons for these issues and evaluated their potential repercussions on the integrated solution. Gave priority to resolving critical problems and carried out essential code modifications or remedies. Performed comprehensive testing after the implementation of fixes to confirm the successful resolution of the issues. <u>As a result</u> • Identified and assessed unresolved matters, bugs, or obstacles, ensuring a thorough understanding of the challenges encountered during testing. • Prioritized and implemented fixes for critical problems, ensuring the stability and reliability of the integrated solution.	6

Task	Description	Hours
Locally Test Enhancements	Established a suite of test cases that encompassed the interactions among various components or modules within the CLV module. Conducted the tests to verify that the components harmoniously collaborated and generated the anticipated results following the bug fixes. <u>As a result</u> - Ensured comprehensive test coverage for interactions among different components or modules within the CLV module. - Validated the harmonious collaboration of components and the generation of anticipated results after implementing bug fixes, confirming the effectiveness of the solutions.	2
Code Review and Refactor	Evaluated the code's structure, clarity, conformity to coding standards, and level of modularity. Detected any possible code issues, redundancy, or prospects for code enhancement. Furnished constructive feedback and recommendations for refining the overall code quality. <u>As a result</u> - Ensured that the code adhered to organizational standards, promoting clarity and modularity. - Provided constructive feedback and recommendations, contributing to the improvement of overall code quality and maintainability.	2
Integration Testing	Carried out thorough testing to confirm the effective communication and compatibility between the Pareto-NBD model and its functional components. Verified the optimization of chain mechanisms and the seamless interaction of priors. Resolved any issues identified during testing, thereby improving the overall reliability of the integration. <u>As a result</u> - Confirmed the effective communication and compatibility between the Pareto-NBD model and its functional components through thorough testing. - Resolved identified issues during testing, enhancing the overall reliability and performance of the integrated system.	2

Task	Description	Hours
Code Enhancements	Enhanced the capabilities of CLV by introducing novel features to align with the project's goals. Integrated these supplementary functionalities in a manner that harmonized with the current code architecture. Performed initial testing to confirm the accurate implementation of the newly added features. <u>As a result</u> - Integrated supplementary functionalities seamlessly into the existing code architecture, maintaining a cohesive system. - Confirmed the accurate implementation of the newly added features through initial testing, contributing to the overall reliability and functionality of CLV.	4
Testing Preparation	Developed a plan for testing scenarios and an approach to ensure thorough validation of the codebase. Identified key testing points and formulated a strategy for assessing functionality and stability. <u>As a result</u> Identified key testing points and formulated a strategy to assess functionality and stability, contributing to the overall effectiveness of the testing process.	1
Beta-NBD Coding	Continued with the incremental development of functionality, maintaining a steady pace of progress. Ensured that the ongoing code development aligned with the established goals and timelines. <u>As a result</u> Sustained an incremental development approach, facilitating a consistent and steady progression of functionality.	3
Comprehensive Testing	Carried out an exhaustive testing process to validate the functionality and robustness of the newly introduced features. Recognized and resolved any bugs or discrepancies that emerged during the testing phase. Secured the codebase for smooth progression into subsequent development stages. <u>As a result</u> - Validated the functionality and robustness of the newly introduced features through an exhaustive testing process. - Secured the codebase, ensuring a stable foundation for smooth progression into subsequent development stages.	2

Task	Description	Hours
Code Review and Refactor	Conducted an extensive code review to detect possible bugs, errors, or inefficiencies. Restructured the code to comply with coding standards and enhance its overall readability. Rectified any concerns or problems brought to light during the review procedure. <u>As a result</u> - Improved overall code quality by restructuring it to comply with coding standards and enhance readability. - Addressed concerns or problems brought to light during the review procedure, ensuring the code met quality standards and was more maintainable.	1
Communication and Updates	Maintained open communication with the project team to relay updates regarding the advancement of the Pareto-NBD model implementation. Engaged in discussions about encountered challenges and proposed solutions for the way ahead. Ensured that everyone had a shared understanding of the present status and the forthcoming steps in the project. <u>As a result</u> - Addressed challenges through collaborative discussions, proposing solutions and fostering a problem-solving approach. - Ensured a shared understanding of the present status and the forthcoming steps, promoting alignment within the project team.	1

Table 22 - Week 4 (Customer Lifetime Value)

Week 5: August 28th – September 1st, 2023 – Gamma-Gamma Modeling

Task	Description	Hours
Develop Code for Gamma-Gamma Model	Specified the prior distributions for the model parameters, utilizing Gamma distributions in the context of a Gamma-Gamma model to represent prior beliefs about parameters like the shape and scale of the distribution. Wrote the likelihood function that described the relationship between the model parameters and the observed data. In the Gamma-Gamma model, the likelihood function was based on the Gamma distribution. Used PyMC to construct the Gamma-Gamma model, combining the specified prior distributions and the likelihood function to create a probabilistic model for customer purchase behavior. Chose the appropriate inference method for the model, opting for PyMC's NUTS (No-U-Turn Sampler) for estimating the posterior distribution.	8

Task	Description	Hours
	As a result: - Established a comprehensive probabilistic model for customer purchase behavior using PyMC, integrating prior distributions and a likelihood function. - Applied the NUTS sampling method to estimate the posterior distribution, ensuring a robust and accurate inference process for the Gamma-Gamma model.	
Develop dependent functions of Gamma-Gamma for CLV	Created functions to calculate various aspects of customer behavior, such as the expected monetary value (EMV) and the probability of making repeat purchases. Developed functions to estimate the monetary value of customers and their lifetime value. Ensured these functions were based on the parameters of the Gamma-Gamma model. Used Bayesian techniques or maximum likelihood estimation to estimate model parameters if needed. <u>As a result</u> - Established comprehensive functions for calculating various aspects of customer behavior, contributing to a thorough analysis of customer interactions. - Utilized Bayesian techniques or maximum likelihood estimation to estimate model parameters when necessary, ensuring accurate and reliable parameter values for the functions.	8
Bug Fixes and Enhancements	Reviewed outstanding issues, bugs, or challenges identified during testing. Analyzed the root causes of these issues and their impact on the integrated solution. Prioritized and implemented necessary code adjustments or fixes. Conducted comprehensive testing to ensure the issues were effectively resolved. <u>As a result</u> - Gained insights into the root causes of issues, facilitating a better understanding of the system's intricacies. - Confirmed the effectiveness of the fixes through comprehensive testing, validating the improvements made to the system.	6
Locally Test Enhancements	Defined a set of test cases to cover interactions between different components or modules of the CLV model. Executed tests to validate that components work together seamlessly and produce expected outcomes after bug fixes. As a result:	2

Task	Description	Hours
	Validated the seamless collaboration of components and verified the expected outcomes after implementing bug fixes, ensuring the robustness and reliability of the CLV model.	
Code Review and Refactor	Examined code organization, readability, adherence to coding standards, and modularity. Identified potential code smells, redundancy, or opportunities for code optimization. Provided constructive feedback and suggestions for improvement to enhance overall code quality. <u>As a result</u> - Identified and addressed potential code issues, redundancy, and opportunities for optimization, enhancing the overall efficiency of the codebase. - Provided constructive feedback and suggestions for improvement, contributing to the enhancement of overall code quality and maintainability.	2
Integration Testing	Conducted comprehensive testing to validate the communication and compatibility between the Gamma-Gamma model and its functional components. Ensured that parameters and priors interacted smoothly. Addressed any issues emerging during testing to enhance overall integration reliability. <u>As a result</u> - Validated the communication and compatibility between the Gamma-Gamma model and its functional components through comprehensive testing. - Addressed and resolved issues identified during testing, contributing to the overall reliability and performance of the integration.	2
Code Enhancements	Expanded the capabilities of CLV by introducing new features aligned with the project's objectives. Integrated these additional functionalities in a way that complements the existing code architecture. Conducted preliminary testing to verify the correct implementation of the added features. <u>As a result</u> Verified the correct implementation of the added features through preliminary testing, contributing to the overall reliability and functionality of CLV.	4

Task	Description	Hours
Testing Preparation	Developed a plan for testing scenarios and an approach to ensure thorough validation of the codebase. Identified key testing points and formulated a strategy for assessing functionality and stability. <u>As a result</u> Established a comprehensive plan for testing scenarios, ensuring a systematic and thorough validation of the codebase.	1
Gamma-Gamma Coding	Continued with incremental development of functionality, maintaining progress aligned with established goals and timelines. Ensured ongoing code development aligns with project objectives. <u>As a result</u> Maintained alignment of ongoing code development with established goals and timelines, facilitating systematic and goal-oriented advancement in the project.	3
Comprehensive Testing	Executed a comprehensive testing regimen to verify functionality and stability of the recently added features. Identified and addressed any bugs or inconsistencies surfaced during testing. Ensured the codebase was well-prepared for subsequent development stages. <u>As a result</u> Verified the functionality and stability of the recently added features through a comprehensive testing regimen.	2
Code Review and Refactor	Performed a comprehensive code review to identify potential bugs, errors, or inefficiencies. Refactored code to adhere to coding standards and improve overall readability. Addressed issues raised during the review process. <u>As a result</u> Improved overall code quality by refactoring to adhere to coding standards and enhance readability.	1
Communication and Updates	Communicated with the project team to provide updates on the progress of the Gamma-Gamma model implementation. Discussed challenges faced and potential solutions moving forward. Aligned everyone's understanding of the status and next steps. <u>As a result</u> Addressed challenges through collaborative discussions, proposing solutions and fostering a problem-solving approach.	1

Table 23 - Week 5(Customer Lifetime Value)

Week 6: September 4th – September 8th, 2023 – Integration of Models and Deployment

Task	Description	Hours
Integration of Models	Integrated the Beta-Geo, Pareto-NBD, and Gamma-Gamma models into a unified framework, ensuring seamless collaboration for a comprehensive customer lifetime value (CLV) prediction. Developed a data pipeline to efficiently feed customer data into the integrated model, facilitating a smooth and continuous flow of information. <u>As a result</u> - Achieved a seamless integration of the Beta-Geo, Pareto-NBD, and Gamma-Gamma models, providing a unified and comprehensive framework for accurate CLV prediction. - Established a robust data pipeline that enhances the efficiency of feeding customer data into the integrated model, contributing to the accuracy and reliability of CLV predictions.	8
API Development	Designed and developed a RESTful API that exposes endpoints for predicting CLV using the integrated models. Implemented error handling and request validation to ensure the API is robust and user-friendly. <u>As a result</u> Established a RESTful API with endpoints dedicated to predicting CLV, providing a user-friendly interface for model integration.	6
Add API Validations	Checked for required fields to ensure that all mandatory fields are present in the input data. Validated data types to confirm that they match the expected types (e.g., strings, numbers, dates). Conducted range and boundary checks to validate that numerical values fall within acceptable ranges. Performed format checks to ensure that data adheres to specific formats (e.g., email addresses, phone numbers). Implemented cross-field validations to check for consistency and relationships between different fields. Utilized whitelisting or blacklisting to validate data against predefined lists of acceptable or prohibited values. <u>As a result</u> - Ensured data integrity and completeness by checking for required fields in the input data. - Established consistency and relationships between different fields, contributing to a more reliable dataset. - Validated data against predefined lists, enhancing data accuracy and adherence to specified criteria.	2

Task	Description	Hours
API Testing and Validation	Conducted thorough testing of the API to validate its functionality and reliability. Created test cases to cover various use cases and edge scenarios. Ensured that the API uploads accurate CLV predictions based on input data. <u>As a result</u> - Developed test cases that cover various use cases and edge scenarios, providing a robust evaluation of the API's capabilities. - Ensured that the API consistently uploads accurate CLV predictions based on input data, enhancing the reliability of the predictions provided by the system.	6
Deployment to Vertex AI	Prepared the integrated model and API for deployment in Google Vertex AI. Configured the necessary resources and environment for deployment. Deployed the API to Vertex AI, ensuring scalability and availability. <u>As a result</u> - Successfully prepared the integrated model and API for deployment in Google Vertex AI, ensuring compatibility with the platform. - Configured the required resources and environment to facilitate a seamless deployment process. - Deployed the API to Vertex AI, ensuring scalability to handle varying workloads and guaranteeing high availability for users.	2
Testing in Vertex AI Environment	Conducted testing of the deployed API within the Vertex AI environment, ensuring its capability to handle real-time requests and data. Verified that the models performed as expected in a production setting. <u>As a result</u> - Validated the functionality of the deployed API in the Vertex AI environment, confirming its ability to handle real-time requests and data effectively. - Ensured that the integrated models performed as expected in a production setting, meeting the performance and accuracy criteria established during development.	8
Demo Preparation	Prepared a demo environment with sample data and use cases. Created a presentation to showcase the capabilities of the integrated CLV prediction system and the deployed API. <u>As a result</u>	2

Task	Description	Hours
	- Established a demo environment with representative sample data and use cases, providing a realistic demonstration of the integrated CLV prediction system. - Developed a presentation that effectively highlighted the capabilities of the integrated system and the deployed API, ensuring a clear and compelling showcase for stakeholders and users.	
CLV Prediction Demo	Conducted a live demonstration of the CLV prediction system to stakeholders and team members. Showcased the API's capabilities in making accurate CLV predictions using real and simulated data. <u>As a result</u> - Engaged stakeholders and team members through a live demonstration, providing them with a firsthand experience of the CLV prediction system's functionalities. - Effectively showcased the API's capabilities in generating accurate CLV predictions, demonstrating its reliability and performance with both real and simulated data.	1
Post-Demo Adjustments and Final Checks	Conducted a live demonstration of the CLV prediction system to stakeholders and team members. Showcased the API's capabilities in making accurate CLV predictions using real and simulated data. As a result: - Engaged stakeholders and team members through a live demonstration, providing them with a firsthand experience of the CLV prediction system's functionalities. - Effectively showcased the API's capabilities in generating accurate CLV predictions, demonstrating its reliability and performance with both real and simulated data.	2
Documentation and Knowledge Sharing	Created comprehensive documentation for the integrated system, API endpoints, and model usage. Shared knowledge with the team to ensure they can maintain and further develop the system. As a result: - Developed thorough documentation that provides insights into the integrated system, API endpoints, and guidelines for model usage, enabling effective understanding and reference. - Shared knowledge with the team, empowering them to maintain and enhance the system independently, fostering a collaborative and knowledgeable team environment.	3

Table 24- Week 6(Customer Lifetime Value)

Project 5 – Next Purchase Day

Week 1: September 11th - September 15th, 2023 – Project Initialization for Next Purchase Day

Task	Description	Hours
Requirement Gathering	Met with stakeholders to gather detailed requirements for the next purchase day prediction system. Identified key features, user expectations, and any specific business rules that need to be considered. <u>As a result</u> Identified key features and business rules that are crucial for the success of the project, setting the foundation for a well-defined and effective development process.	2
Environment Setup	Set up the development environment, including necessary software, libraries, and databases. Configured version control and established collaboration tools for the development team. <u>As a result</u> Established a functional development environment with all required software, libraries, and databases, ensuring a smooth workflow for the development team.	4
BigQuery Connection	Connected successfully to BigQuery and explored data fields required for exploration and EDA (Exploratory Data Analysis). <u>As a result</u> Explored relevant data fields required for exploration and EDA, laying the groundwork for a thorough analysis and understanding of the dataset.	2
Data Collection and Cleaning	Acquired relevant historical data on customer purchase patterns. Cleaned and pre-processed the data to remove any inconsistencies or missing values. <u>As a result</u> Pre-processed the data, eliminating inconsistencies and addressing missing values to enhance the quality and reliability of the dataset.	3
Exploratory Data Analysis (EDA)	Performed exploratory data analysis (EDA) to gain insights into the distribution of purchase intervals. Utilized statistical measures and visualizations to analyze the central tendencies and variations in purchase interval data. Identified patterns in the purchase intervals, such as recurring trends or seasonality. Explored potential outliers or anomalies in the data that may influence the	5

Task	Description	Hours
	distribution of purchase intervals. Documented key findings and observations to inform further analysis and model development. <u>As a result</u> - Identified patterns, recurring trends, and seasonality in the purchase intervals, enhancing the understanding of customer behavior. - Identified outliers or anomalies, ensuring a thorough examination of factors that may influence the distribution of purchase intervals.	
Research on Models and Methods Exploration	Explored various machine learning models for predicting the "next purchase day." Researched methods and approaches for time-series prediction. Investigated relevant literature and resources to understand best practices in the field. Explored potential algorithms and techniques specifically tailored for the task at hand. <u>As a result</u> - Reviewed relevant literature and resources to understand best practices in the field, ensuring a comprehensive understanding of the state-of-the-art techniques. - Explored algorithms and techniques specifically tailored for predicting the "next purchase day," laying the groundwork for informed model selection and development.	8
Choosing the Best Methods and Start Coding	Evaluated insights from Exploratory Data Analysis (EDA) to inform the selection of methods for predicting the "next purchase day." Initiated the coding process for the initial version of the prediction model using the chosen methods. Implemented basic functionality for data input, feature extraction, and model training in the coding process. <u>As a result</u> - Initiated the coding process, laying the foundation for the development of the prediction model and ensuring alignment with the chosen methods. - Implemented basic functionality for data input, feature extraction, and model training, establishing the initial framework for the prediction model's functionality.	8
Continue Coding for Next Purchase Day	Began coding the foundational logic for the functions and utils based on the established plan. Developed the core algorithms and functions that drove the main features of the next purchase. Start initial code for handling customer data manipulation and analysis. <u>As a result</u>	6

Task	Description	Hours
	- Initiated the coding process by implementing the foundational logic for functions and utilities, aligning with the established plan. - Developed core algorithms and functions essential for driving the main features of predicting the next purchase, ensuring the model's fundamental components were in place.	
Task Review and Planning	Took time to reflect on the completed tasks and the overall progress of the project. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. Strategized for the next stages of work, considering any lessons learned and insights gained. <u>As a result</u> Evaluation of the project's progress ensured alignment with project goals and timelines.	1
Communication and Updates	Engaged in a team meeting to discuss findings and share insights from exploration. Provided updates on progress and addressed any questions or concerns raised by team members. <u>As a result</u> The team meeting facilitated open communication and collaboration, allowing for the exchange of insights and findings from the exploration phase.	1

Table 25 - Week 1 (Next Purchase Day)

Week 2: September 18th – September 22nd, 2023 – API Creation and Deployment

Task	Description	Hours
Code Review and Refactor	Examined code organization, readability, adherence to coding standards, and modularity. Identified any potential code smells, redundancy, or opportunities for code optimization. Provided constructive feedback and suggestions for improvement to enhance the overall code quality. <u>As a result</u> Identification of potential code smells, redundancy, and opportunities for optimization highlighted areas where the code could be refined for better efficiency.	2
Integration of Function and Utils	Integrated the min, max, average, and monetary bucket models into a unified framework, ensuring seamless collaboration for a comprehensive next purchase day (NPD) prediction. Developed a data pipeline to feed customer data into the integrated model.	2

Task	Description	Hours
	<u>As a result</u> Achieved integrated and unified framework for accurate and comprehensive next purchase day (NPD) prediction.	
Testing	Conducted initial testing on the developed model to promptly identify any issues or errors. Checked for compatibility and performance concerns during the testing phase. Adjusted the code as necessary based on the results and findings from the initial testing. <u>As a result</u> Modified the code based on the results and findings from the initial testing, improving the overall reliability and performance of the developed model.	4
API Creation	Developed an API to make the "next purchase day" model accessible. Implemented endpoints to facilitate the reception of input data and the delivery of predictions. Ensured comprehensive documentation for the proper usage of the API. <u>As a result</u> Enhanced user interaction by implementing efficient endpoints for input data reception and prediction delivery.	4
API Testing	Conducted testing on the API to ensure its proper functionality. Checked for potential security vulnerabilities and addressed any identified issues. Tested various input scenarios to validate the model's responses and enhance robustness. <u>As a result</u> - Verified the proper functionality of the API through rigorous testing, ensuring reliable performance. - Improved model robustness by testing various input scenarios and validating corresponding responses.	4
Integration with Existing Prediction Container	Integrated the "next purchase day" model with the existing prediction container or environment. Ensured seamless communication and compatibility with other components in the system. Addressed any challenges related to integration that emerged during the process. <u>As a result</u> - Established seamless communication and compatibility with other components, fostering a cohesive system. - Ensured a smooth and efficient collaboration between the model and other components.	6

Task	Description	Hours
Integration Testing	Conducted comprehensive testing on the integrated system to assess data flow, communication, and overall system performance. Checked for any issues or anomalies identified during the integration testing phase. Addressed and resolved issues to ensure the smooth functioning of the integrated system. <u>As a result</u> - Identified and addressed anomalies detected during the integration testing phase, ensuring a robust system. - Improved the overall functionality of the integrated system by resolving issues and enhancing its smooth operation.	4
Optimization and Testing	Optimized both the code and model to enhance overall performance. Conducted additional testing to verify that the optimized version-maintained accuracy. Addressed any remaining performance or efficiency concerns identified during testing. Fine-tuned the implementation to achieve better computational efficiency. <u>As a result</u> - Achieved better computational efficiency through fine-tuning the implementation. - Ensured that the optimized code and model continued to meet accuracy standards, maintaining a balance between performance and precision.	6
Bug Fixes	Addressed and resolved any bugs or issues uncovered during testing. Conducted thorough debugging processes to enhance the reliability of the model. Implemented necessary fixes for the issues identified during the debugging phase. <u>As a result</u> Enhanced the reliability of the model and implemented necessary fixes for the issues identified during the debugging phase, ensuring a robust and error-free model.	5
Deployment on Vertex AI	Deployed the "next purchase day" model on Google Cloud Vertex AI. Configured essential settings for a seamless deployment experience. Ensured proper integration with Vertex AI services to optimize functionality and performance. <u>As a result</u> Successfully deployed the "next purchase day" model on Google Cloud Vertex AI, making it accessible for utilization.	3

Table 26 - Week 2 (Next Purchase Day)

Week 3: September 25th, 2023 – Demo and Documentation

Task	Description	Hours
Demo	Conducted a demonstration of the deployed "next purchase day" model. Showcased key features and functionalities to stakeholders during the demonstration. Gathered feedback from stakeholders and addressed any questions or concerns raised during the demonstration. <u>As a result</u> • Provided a comprehensive demonstration of the deployed "next purchase day" model, highlighting its functionality and capabilities.	2
Documentation	Developed comprehensive documentation for the "next purchase day" model. Documented the model architecture, API endpoints, usage instructions, and troubleshooting steps. Ensured the documentation was clear, concise, and accessible for ease of understanding and reference. <u>As a result</u> • Created comprehensive documentation for the "next purchase day" model, providing detailed insights into its architecture and functionality. • Documented essential information, including API endpoints, usage instructions, and troubleshooting steps, facilitating seamless user interaction.	6

Table 27 - Week 3 (Next Purchase Day)

Project 6 – Affinity to Buy

Week 1: September 26th – September 29th, 2023 – Project Initialization for Affinity to Buy

Task	Description	Hours
Requirement Gathering	Conducted meetings with relevant stakeholders to collect requirements for the "Affinity to Buy" model. Discussed expectations, project goals, and key deliverables with stakeholders during the meetings. Clarified and documented the necessary features and functionalities essential for the successful implementation of the model. <u>As a result</u> • Gathered comprehensive requirements for the "Affinity to Buy" model through collaborative meetings with relevant stakeholders.	2

Task	Description	Hours
	Clarified and documented essential features and functionalities necessary for the successful implementation of the "Affinity to Buy" model.	
Virtual Env Setup	Installed and configured the virtual environment for the project. Set up essential Python packages, such as XGBoost and Hyperopt, to facilitate model development. Ensured compatibility and version control within the virtual environment. Verified the proper functioning of the configured environment for seamless model development. <u>As a result</u> Established a well-configured virtual environment for the project, providing a controlled and isolated development environment.	4
Doubt Clarification	Sought clarification from stakeholders to ensure a thorough understanding of gathered requirements. Addressed any uncertainties or doubts regarding the project requirements through further discussions with stakeholders. <u>As a result</u> - Enhanced understanding of gathered requirements by actively seeking clarification from stakeholders. - Addressed doubts regarding project requirements through additional discussions, ensuring a clear and shared understanding among all stakeholders.	2
Research on Different Models	Conducted a comparative analysis of various models, including neural networks, logistic regression, and XGBoost, to determine their effectiveness in predicting affinity to buy. Utilized dummy data for testing and benchmarking different models to evaluate their performance under similar conditions. Examined and compared the strengths and weaknesses of each model in the context of the affinity to buy prediction task. Considered metrics such as accuracy, precision, recall, and F1 score to assess and contrast the models' performance on the dummy data. <u>As a result</u> Utilized dummy data to test and benchmark various models, ensuring a fair and consistent evaluation of their performance.	5

Task	Description	Hours
Extraction of Customer Data from BigQuery	Leveraged BigQuery to extract pertinent customer data for the "Affinity to Buy" task. Executed SQL queries to retrieve both features and labels required for training the model. Checked and cleaned the extracted data to identify and rectify any inconsistencies or missing values, ensuring data quality for subsequent modeling. <u>As a result</u> Successfully utilized BigQuery to extract relevant customer data, providing a robust foundation for the "Affinity to Buy" task.	3
Research on XGBoost and Hyperopt	Explored XGBoost as a potential classifier for the "Affinity to Buy" task. Conducted research on the Hyperopt library, focusing on its capabilities in hyperparameter optimization. Investigated best practices for tuning XGBoost using Hyperopt to enhance model performance. Gathered information on key hyperparameters relevant to optimizing XGBoost for the specific task. Compiled insights on the interplay between XGBoost, Hyperopt, and hyperparameter tuning best practices for informed model development. <u>As a result</u> - Identified XGBoost as a suitable classifier for the "Affinity to Buy" task, considering its potential strengths. - Enhanced model performance by implementing best practices for tuning XGBoost using Hyperopt. - Utilized key hyperparameters to optimize XGBoost, improving its effectiveness for the specific task.	8
EDA on the Extracted Data	Conducted exploratory data analysis (EDA) on the extracted customer data to gain insights into its characteristics. Examined data distributions to understand the spread and central tendencies of key features. Identified patterns within the data through various statistical analyses during the EDA. Visualized key features to provide a comprehensive understanding of the dataset and pre-processed the data for input into the XGBoost model. <u>As a result</u> - Gained valuable insights into the characteristics of the extracted customer data through comprehensive exploratory data analysis (EDA). - Provided a comprehensive understanding of the dataset by visualizing key features, facilitating informed preprocessing for input into the XGBoost model.	5

Task	Description	Hours
Task Review and Planning	Took time to reflect on the completed tasks and the overall progress of the project. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. Strategized for the next stages of work, considering any lessons learned and insights gained. <u>As a result</u> Optimized future work by strategizing for the next stages, incorporating lessons learned and insights gained during the reflective process.	1
Communication and Updates	Engaged in a team meeting to discuss findings and share insights from exploration. Provided updates on progress and addressed any questions or concerns raised by team members. <u>As a result</u> - Fostered collaborative communication within the team by actively engaging in discussions and sharing insights from exploration. - Maintained transparency and ensured a shared understanding of project developments by providing updates on progress. - Addressed team members' questions and concerns, promoting a supportive and collaborative team environment.	1

Table 28 - Week 1 (Affinity to Buy)

Week 2: October 2nd – October 6th, 2023 – Testing and Deployment

Task	Description	Hours
XGBoost Model Development	Implemented the XGBoost classifier for the "Affinity to Buy" task. Tuned initial hyperparameters using default settings as a starting point for optimization. Conducted training of the XGBoost model on the pre-processed data to learn patterns and relationships. Monitored the training process to ensure convergence and assessed preliminary model performance. Evaluated the model's performance metrics to gauge its effectiveness in predicting affinity to buy. Iteratively refined hyperparameters and retrained the model as needed to enhance predictive accuracy. <u>As a result</u> - Successfully implemented the XGBoost classifier, laying the foundation for the "Affinity to Buy" task. - Trained the XGBoost model on pre-processed data, enabling it to learn patterns and relationships within the dataset.	8

Task	Description	Hours
	Achieved enhanced predictive accuracy through iterative refinement of hyperparameters and retraining of the model.	
Hyperparameter Optimization with Hyperopt	Employed Hyperopt for hyperparameter optimization of the XGBoost model. Defined the search space, specifying the range of values for hyperparameters subject to optimization. Conducted multiple iterations using Hyperopt to systematically explore the search space and identify the optimal hyperparameter configuration. Leveraged the results of the optimization process to fine-tune the XGBoost model for improved performance on the "Affinity to Buy" task. <u>As a result</u> - Utilized Hyperopt for effective hyperparameter optimization, enhancing the overall performance of the XGBoost model. - Conducted multiple iterations using Hyperopt, systematically exploring the search space to identify the optimal hyperparameter configuration. - Fine-tuned the XGBoost model based on the results of the optimization process.	8
Testing	Conducted testing on the developed XGBoost model to assess its performance. Evaluated the model's performance using dedicated test data to ensure generalizability. Adjusted model settings based on testing results to enhance predictive accuracy and robustness. Iteratively refined the model through testing and adjustment cycles to achieve optimal performance on the "Affinity to Buy" task. <u>As a result</u> - Ensured the generalizability of the model by evaluating its performance using dedicated test data. - Achieved optimal performance on the "Affinity to Buy" task through iterative refinement cycles involving testing and adjustments.	3
API Creation	Developed an API to facilitate accessibility to the "Affinity to Buy" model. Implemented endpoints within the API to handle the reception of input data and the delivery of predictions. Ensured comprehensive documentation was provided for the proper usage of the API, including input formats, and expected outputs. <u>As a result</u> Enabled convenient accessibility to the "Affinity to Buy" model through the development of a dedicated API.	5

Task	Description	Hours
API Testing	Conducted testing on the API to verify its proper functionality and responsiveness. Checked for potential security vulnerabilities, addressing any identified issues to enhance the API's security. Tested various input scenarios to validate the XGBoost model's responses and ensure the robustness of the API. <u>As a result</u> • Verified the proper functionality and responsiveness of the API through comprehensive testing. • Ensured the robustness of the API by testing various input scenarios and validating the XGBoost model's responses.	2
Integration with Existing Prediction Container	Integrated the "Affinity to Buy" XGBoost model with the existing prediction container or environment. Ensured seamless communication and compatibility with other components within the system. Addressed any challenges related to integration that emerged during the process, ensuring a cohesive and functional integration. <u>As a result</u> • Successfully integrated the "Affinity to Buy" XGBoost model with the existing prediction container or environment providing seamless communication and compatibility with other components within the system.	4
Integration Testing	Conducted comprehensive testing on the integrated system to assess data flow, communication, and overall system performance. Checked for any issues or irregularities identified during the integration testing phase. Addressed and resolved issues to ensure the smooth functioning and reliability of the integrated system. <u>As a result</u> • Ensured the smooth functioning and reliability of the integrated system by resolving issues and implementing necessary adjustments.	2
Optimization and Testing	Optimized the XGBoost model to enhance overall performance. Conducted additional testing to validate that the optimized version-maintained accuracy. Addressed any remaining performance or efficiency concerns identified during testing to ensure the optimal functioning of the model. <u>As a result</u> • Improved the overall performance of the XGBoost model through optimization efforts.	2

Task	Description	Hours
	Ensured optimal functioning of the model by addressing any remaining performance or efficiency concerns identified during testing.	
Bug Fixes	Addressed and resolved any bugs or issues identified during testing to enhance model reliability. Conducted thorough debugging processes to ensure the XGBoost model's robustness. Implemented necessary fixes for the issues identified during the debugging phase. Ensured the final version of the model is free from critical issues and operates reliably. <u>As a result</u> Implementing necessary fixes for issues identified during debugging, contributing to the overall reliability of the final model version.	4
Deployment on Vertex AI	Deployed the "Affinity to Buy" XGBoost model on Google Cloud Vertex AI. Configured essential settings for a seamless deployment experience. Ensured proper integration with Vertex AI services to optimize functionality and performance. <u>As a result</u> Successfully deployed the "Affinity to Buy" XGBoost model on Google Cloud Vertex AI.	2

Table 29 - Week 2 (Affinity to Buy)

Week 3: October 9th, 2023 – Demo and Documentation

Task	Description	Hours
Demo	Conducted a demonstration of the deployed XGBoost model. Showcased key features and functionalities to stakeholders during the demonstration. Gathered feedback from stakeholders and addressed any questions or concerns raised during the demonstration. Ensured a comprehensive understanding and satisfaction among stakeholders regarding the XGBoost model's capabilities and performance. <u>As a result</u> Successfully demonstrated the capabilities of the deployed XGBoost model to stakeholders.	2

Task	Description	Hours
Documentation	Developed comprehensive documentation for the "Affinity to Buy" XGBoost model. Documented the model architecture, API endpoints, hyperparameter settings, usage instructions, and troubleshooting steps. Ensured the documentation was clear, concise, and easily accessible to facilitate understanding and reference. <u>As a result</u> - Documented essential components such as model architecture, API endpoints, hyperparameter settings, usage instructions, and troubleshooting steps. - Ensured the documentation was clear, concise, and easily accessible for user understanding and reference.	6

Table 30 - Week 3 (Affinity to Buy)

Project 7 – Propensity to Buy

Week 1: October 10th - October 13th, 2023 - Project Initialisation and Development

Task	Description	Hours
Requirement Gathering	Conducted meetings with relevant stakeholders to gather requirements for the "Propensity to Buy" model. Engaged in discussions to understand expectations, project goals, and key deliverables with stakeholders during the meetings. Clarified and documented the necessary features and functionalities essential for the successful implementation of the "Propensity to Buy" model. <u>As a result</u> Obtained a clear understanding of expectations, project goals, and key deliverables from stakeholder discussions.	2
Virtual Env Setup	Installed and configured the virtual environment for the project, ensuring a streamlined and organized development setup. Set up essential Python packages necessary for model development within this environment, promoting a cohesive and efficient workflow. Implemented version control to maintain consistency and reproducibility, facilitating seamless collaboration and future modifications. <u>As a result</u> The project benefits from a well-organized and standardized development environment, enhancing overall efficiency.	4

Task	Description	Hours
	The use of version control ensures traceability and reproducibility, providing a robust foundation for collaboration and future modifications.	
Research on Propensity to Buy	Explored relevant literature and resources to gain valuable insights into building models for "Propensity to Buy." Investigated various machine learning algorithms suitable for the task, considering their individual strengths and weaknesses. Conducted research on best practices for feature engineering to enhance the predictive capabilities of the model. Explored recommended approaches for model training to optimize performance in predicting propensity to buy. <u>As a result</u> - Acquired insights into effective feature engineering strategies to improve the model's predictive capabilities. - Gained a clear understanding of best practices for model training to optimize overall performance in predicting propensity to buy.	8
Extraction of Customer Data from BigQuery	Utilized BigQuery to extract relevant customer data for the "Propensity to Buy" task. Executed SQL queries to retrieve both features and labels necessary for training the model. Checked and cleaned the extracted data to identify and rectify any inconsistencies or missing values, ensuring data quality for subsequent model training. <u>As a result</u> - Acquired a comprehensive dataset tailored for training the "Propensity to Buy" model. - Ensured data quality by addressing inconsistencies and missing values, laying a solid foundation for accurate model training.	4
EDA on the Extracted Data	Conducted exploratory data analysis (EDA) on the extracted customer data to understand its characteristics. Examined data distributions, identified patterns, and visualized key features during the EDA process. <u>As a result</u> Pre-processed the data, ensuring it meets the necessary format for optimal input into the "Propensity to Buy" model.	4

Task	Description	Hours
Study for Best Model	Explored literature and resources to identify the best models for predicting propensity to buy. Investigated machine learning algorithms known for their effectiveness in propensity modeling, such as logistic regression, decision trees, and ensemble methods. Examined recent research papers and industry practices to stay updated on emerging models and techniques for propensity prediction. Explored the use of advanced techniques like gradient boosting or deep learning for propensity modeling, considering their potential advantages. Reviewed case studies and success stories to understand real-world applications and performance of different models in predicting propensity to buy. Considered the scalability, interpretability, and ease of implementation of various models to determine their suitability for the specific context of propensity to buy prediction. <u>As a result</u> - Stayed informed about state-of-the-art algorithms and emerging techniques in propensity modeling. - Evaluated advanced techniques for their potential benefits in the context of propensity prediction. - Considered real-world applications and performance through case studies.	6
Task Review and Planning	Took time to reflect on the completed tasks and the overall progress of the project. Evaluated whether the project was on track and made any necessary adjustments to the remaining plan. Strategized for the next stages of work, considering any lessons learned and insights gained. <u>As a result</u> Strategically planned the next stages of work to align with project goals and timelines.	1
Communication and Updates	Engaged in a team meeting to discuss findings and share insights from exploration. Provided updates on progress and addressed any questions or concerns raised by team members. <u>As a result</u> Enhanced collaboration and shared understanding among team members.	1

Table 31 - Week 1 (Propensity to Buy)

Task	Description	Hours
Model Development	Implemented the machine learning model for "Propensity to Buy," emphasizing feature engineering to enhance predictive capabilities. Conducted thorough model training on the pre-processed data to learn patterns and relationships. Utilized hyperparameter tuning to optimize the model's performance, adjusting parameters for better predictive accuracy. Iteratively refined hyperparameters based on model evaluation metrics to achieve optimal results. Considered cross-validation techniques to ensure the robustness of the model across different subsets of the data. Evaluated the impact of various feature engineering strategies on model performance. Explored different algorithms and ensemble methods to identify the most effective approach for the "Propensity to Buy" task. Documented the implemented model's architecture, hyperparameter settings, and key considerations for future reference. <u>As a result</u> - Developed a robust and optimized "Propensity to Buy" model with enhanced predictive capabilities. - Established a comprehensive documentation resource for future model maintenance and improvements.	8
Continue Coding	Developed code for data preprocessing, handling missing values, and encoding categorical variables to prepare the data for model training. Implemented feature engineering techniques, such as creating new variables or transforming existing ones, to capture relevant patterns for propensity prediction. Utilized a machine learning library, such as scikit-learn, to build and train the propensity model. Incorporated cross-validation strategies to assess the model's generalization performance and mitigate overfitting. Implemented a well-designed pipeline, encapsulating data preprocessing, model training, and evaluation steps, for clarity and reproducibility in future iterations. <u>As a result</u> - Established a robust data preprocessing and feature engineering pipeline for preparing the data for propensity model training. - Developed a well-organized and reproducible machine learning model implementation using scikit-learn. - Mitigated overfitting through the incorporation of cross-validation strategies during model development.	6

Task	Description	Hours
Testing	Conducted thorough testing on the developed propensity model to assess its performance. Evaluated key metrics such as accuracy, precision, recall, and other relevant indicators to gauge predictive effectiveness. Adjusted model settings, including hyperparameters or feature engineering strategies, based on testing results to enhance overall performance. Iteratively refined the model through testing and adjustment cycles to achieve optimal predictive accuracy for the "Propensity to Buy" task. <u>As a result</u> - Optimized the propensity model based on comprehensive testing and performance evaluation. - Achieved enhanced predictive accuracy through iterative refinement of model settings and adjustments.	2
API Creation	Developed an API to facilitate accessibility to the "Propensity to Buy" model. Implemented endpoints within the API to handle the reception of input data and the delivery of predictions. Ensured comprehensive documentation was provided for the proper usage of the API, including input formats, and expected outputs. <u>As a result</u> Established endpoints for seamless communication, allowing users to submit input data and receive predictions.	2
Integration with Existing Prediction Container	Integrated the "Propensity to Buy" model with the existing prediction container or environment. Ensured seamless communication and compatibility with other components within the system. Addressed any challenges related to integration that emerged during the process, ensuring a cohesive and functional integration. <u>As a result</u> - Successfully incorporated the "Propensity to Buy" model into the existing prediction environment. - Addressed and resolved challenges, ensuring the integrated system functions seamlessly.	4
Integration Testing	Conducted comprehensive testing on the integrated system to assess data flow, communication, and overall system performance. Checked for any issues or irregularities identified during the integration testing phase. Addressed and resolved issues to ensure the smooth functioning and reliability of the integrated system. <u>As a result</u>	2

Task	Description	Hours
	Ensured the smooth functioning and reliability of the integrated system through issue resolution.	
Optimization and Testing	Optimized the "Propensity to Buy" model to enhance overall performance. Conducted additional testing to validate that the optimized version-maintained accuracy. Addressed any remaining performance or efficiency concerns identified during testing to ensure optimal functioning of the model. <u>As a result</u> - Enhanced the overall performance of the "Propensity to Buy" model through optimization. - Addressed and resolved any remaining concerns for optimal performance and efficiency.	4
Bug Fixes	Addressed and resolved any bugs or issues identified during testing to enhance model reliability. Conducted thorough debugging processes to ensure the "Propensity to Buy" model's robustness. Implemented necessary fixes for the issues identified during the debugging phase. <u>As a result</u> - Improved the reliability of the "Propensity to Buy" model by addressing and resolving identified bugs. - Ensured the robustness of the model through thorough debugging processes.	4
Deployment on Vertex AI	Deployed the "Propensity to Buy" model on Google Cloud Vertex AI. Configured essential settings for a seamless deployment experience. Ensured proper integration with Vertex AI services to optimize functionality and performance. <u>As a result</u> Successfully deployed the "Propensity to Buy" model on Google Cloud Vertex AI.	2
Demo	Conducted a demonstration of the deployed "Propensity to Buy" model. Showcased key features and functionalities to stakeholders during the demonstration. Gathered feedback from stakeholders and addressed any questions or concerns raised during the demonstration. <u>As a result</u> Provided stakeholders with a clear understanding of the "Propensity to Buy" model's capabilities.	2

Task	Description	Hours
	Addressed questions and concerns, ensuring stakeholder satisfaction.	
Documentation	Developed comprehensive documentation for the "Propensity to Buy" model. Documented the model architecture, API endpoints, feature engineering strategies, usage instructions, and troubleshooting steps. Ensured the documentation was clear, concise, and easily accessible for stakeholders, facilitating understanding and reference. <u>As a result</u> - Enabled easy access to information on model architecture, API usage, and troubleshooting. - Supported stakeholders in understanding and effectively utilizing the model.	4

Table 32 - Week 2 (Propensity to Buy)

ACHIEVEMENTS

During this enriching internship, I had the privilege of contributing to the development and implementation of advanced machine learning models for e-commerce analytics, utilizing cutting-edge AI technologies. The accomplishments garnered throughout this internship span a spectrum of technical, analytical, and collaborative achievements, underscoring the successful attainment of project goals and personal growth:

➤ **Customer Lifetime Value Prediction Model:**

- Successfully designed, trained, and deployed a robust machine learning model to predict customer lifetime value, enabling Lifesight Technologies to proactively target high-value customers for personalized engagement.

➤ **Customer Churn Prediction Model:**

- Developed an accurate customer churn prediction model that identified potential churners, allowing the company to implement retention strategies and enhance customer loyalty.

➤ **Product Recommendations & Consumer Basket Analysis:**

- Engineered a recommendation system that provided personalized product suggestions to customers, resulting in increased cross-selling and upselling opportunities.
- Conducted comprehensive consumer basket analysis, uncovering purchase patterns and enabling targeted marketing campaigns.

➤ **Sales Prediction and Time Series Analytics:**

- Constructed time series forecasting models that accurately predicted sales trends, aiding Lifesight Technologies in inventory management and demand forecasting.

➤ **Scientific Attribution Models:**

- Successfully implemented scientific attribution models to assign credit to various touchpoints along the customer journey, enabling data-driven marketing decisions.

➤ **Integration of Advanced Language Models (LLMs):**

- Seamlessly integrated GPT-3 and BERT to enhance the interactivity of the e-commerce platform, fostering natural language understanding and improving user engagement.

➤ **Google Cloud Platform Utilization:**

- Leveraged Google Cloud Storage for efficient data management and Google BigQuery for structured data storage and retrieval.
- Utilized Google Cloud's Vertex AI for model deployment and serving, ensuring scalable and real-time predictions.

➤ **Data Visualization and Interpretation:**

- Employed Matplotlib, Seaborn, and Plotty to create compelling visualizations, enabling stakeholders to gain actionable insights from complex data.

➤ **Exploration of BigQuery:**

- Explored and harnessed the capabilities of Google BigQuery, enhancing data exploration, querying, and analysis capabilities, and contributing to enriched data-driven decision-making.

➤ **Collaboration and Communication:**

- Actively collaborated with cross-functional teams, fostering effective communication and contributing to a synergistic work environment.
- Presented project progress, findings, and insights through clear and engaging reports and presentations.

➤ **Documentation and Knowledge Sharing:**

- Produced comprehensive documentation encompassing model architecture, implementation details, and usage guidelines, facilitating knowledge transfer within the organization.

The culmination of these achievements not only underscores the successful realization of the project's objectives but also signifies personal and professional growth. The experience gained throughout this internship has further equipped me with a profound understanding of advanced AI technologies, machine learning applications, and their tangible impact on real-world challenges.

CONCLUSION

Summary

The internship journey with Lifesight Technologies has been an exceptional and transformative experience, propelling me into the dynamic realm of advanced machine learning, artificial intelligence, and e-commerce analytics. Over the course of this internship, I had the privilege of contributing to the development of sophisticated machine learning models and harnessing cutting-edge AI technologies to unravel insights from e-commerce data. This summary encapsulates the key aspects, accomplishments, and learnings that have defined this enriching internship:

Throughout the internship, the overarching objective was to create advanced machine learning models capable of providing insightful e-commerce analytics. This encompassed a diverse range of objectives, from predicting customer behaviors and sales trends to enhancing user engagement through natural language understanding. The project's multifaceted scope served as a platform for comprehensive skill development and hands-on application of theoretical concepts.

The achievements of this internship have been manifold. Notably, the successful creation and deployment of machine learning models for predicting customer lifetime value, identifying potential churners, offering personalized product recommendations, and forecasting sales trends have demonstrated the tangible impact of data-driven decision-making. The seamless integration of advanced language models, such as GPT-3 and BERT, has enhanced user interactivity and engagement, enriching the overall e-commerce experience.

The utilization of Google Cloud Platform's versatile tools, including Google Cloud Storage, Google BigQuery, and Vertex AI, streamlined data management, storage, and model deployment, fostering scalability and efficiency. The exploration of Google BigQuery for data analysis has broadened the scope of analytical capabilities, enabling deeper insights into the e-commerce landscape.

Collaboration emerged as a cornerstone of this internship, as active engagement with cross-functional teams nurtured effective communication, diverse perspectives, and collective problem-solving. Regular interactions with mentors, colleagues, and stakeholders fostered an environment of knowledge sharing and growth.

Furthermore, the internship underscored the importance of meticulous documentation, enabling transparent project tracking, effective knowledge transfer, and future reference. The process of

documenting methodologies, codebases, and results has not only contributed to project success but has also instilled a culture of professionalism and accountability.

In conclusion, the internship with Lifesight Technologies has been an illuminating voyage into the intricacies of advanced machine learning, artificial intelligence, and data analytics in the context of e-commerce. The amalgamation of technical prowess, analytical acumen, collaborative spirit, and a commitment to innovation has resulted in the realization of project goals and personal enrichment. As I reflect on the journey, I am confident that the skills acquired, and insights gained during this internship will serve as a solid foundation for my future endeavours in the field of AI and machine learning.

Skills Learned

The internship with Lifesight Technologies provided an immersive learning experience that enriched my skill set and equipped me with a diverse range of technical, analytical, and interpersonal competencies. The dynamic nature of the project and the integration of advanced AI technologies facilitated the acquisition of valuable skills that are integral to the field of machine learning, artificial intelligence, and e-commerce analytics. The following is a comprehensive overview of the key skills I have acquired and honed during this internship:

- **Advanced Machine Learning Techniques:** Through the hands-on development of machine learning models, I gained proficiency in employing advanced techniques such as deep learning, ensemble methods, and time series analysis to extract meaningful insights from complex e-commerce data.
- **Data Pre-processing and Feature Engineering:** I acquired the ability to clean, pre-process, and transform raw data into structured formats suitable for analysis. Feature engineering techniques allowed me to extract relevant information and enhance model performance.
- **Model Development and Training:** By designing, implementing, and training machine learning models, I developed a strong foundation in crafting algorithms, tuning hyperparameters, and optimizing model architectures to achieve accurate predictions and recommendations.
- **Integration of Advanced Language Models (LLMs):** The integration of GPT-3 and BERT enriched my understanding of incorporating pre-trained language models into applications, enhancing natural language understanding and interaction.

- **Cloud Computing and Deployment:** I acquired practical experience in leveraging Google Cloud Platform's resources for data storage, querying, and deploying machine learning models using Vertex AI while ensuring scalability and efficient resource management.
- **Data Visualization and Interpretation:** Proficiency in visualization libraries such as Matplotlib, Seaborn, and Plotty enabled me to effectively communicate complex data insights through visually compelling representations.
- **Scientific Attribution Modeling:** I gained expertise in designing and implementing attribution models to allocate credit across touchpoints, offering valuable insights into customer engagement and marketing effectiveness.
- **Collaboration and Communication:** Active participation in cross-functional teams enhanced my collaboration and communication skills, enabling effective knowledge sharing, idea exchange, and collective problem-solving.
- **Documentation and Reporting:** Meticulous documentation of methodologies, code, and results fostered a disciplined approach to project management, communication, and knowledge dissemination.
- **Exploration of BigQuery:** I explored Google BigQuery to conduct advanced data analysis, honing skills in querying and leveraging powerful analytical tools for comprehensive insights.
- **Project Management and Agile Methodology:** Through iterative development cycles, I acquired an understanding of Agile project management principles, emphasizing adaptability, prioritization, and incremental progress.
- **Ethical Considerations in AI:** The internship instilled a heightened awareness of ethical considerations in AI and machine learning, emphasizing responsible data usage, privacy, and bias mitigation.

These acquired skills collectively form a robust toolkit that will undoubtedly contribute to my professional growth and future endeavours in the realm of machine learning, artificial intelligence, and data analytics. The internship has provided a transformative platform for skill development, empowering me to tackle complex challenges and innovate in a rapidly evolving technological landscape.

No.	Goals	Tasks	Completed
1	Advanced Machine Learning Techniques	Gain understanding of deep learning, ensemble methods, and real-world applications	Yes
2	E-commerce Analytics	Develop expertise in customer behavior analysis, sales prediction, recommendation systems	Yes
3	Natural Language Processing (NLP)	Delve into NLP techniques, integrate GPT-3 and BERT for enhanced interactivity	Yes
4	Artificial Intelligence Fundamentals	Grasp foundational AI concepts, applications, ethics, and modern technology roles	Yes
5	Cloud Computing and Deployment	Proficiency in deploying ML models on Google Cloud's Vertex AI, scalability, resource management	Yes
6	Collaboration and Communication	Enhance teamwork, communication, idea sharing, and technical concept presentation	Yes
7	Development of Advanced Analytics Models	Design and implement ML models for e-commerce insights: CLV prediction, churn prediction, etc.	Yes
8	Integration of Language Models	Successfully integrate GPT-3 and BERT for improved e-commerce platform interactivity	Yes
9	Utilization of Cloud Services	Deploy ML models on Google Cloud's Vertex AI for efficient serving, scalability	Yes
10	Data Visualization and Interpretation	Use Matplotlib, Seaborn, Plotty for clear presentation of analytical findings	Yes

Table 33 - Skill Learned

Hours Spent

Month	Hours
April	32
May	176
June	168
July	176
August	168
September	168
October	168
Total Hours	1056

Table 34 - Hours Spent

Challenges Faced and Overcome

Throughout the duration of my internship with Lifesight Technologies, I encountered various challenges that tested my skills, adaptability, and problem-solving abilities. These challenges, while demanding, provided invaluable learning opportunities, and helped shape my growth as a professional in the fields of machine learning, data analytics, and artificial intelligence. The following are some of the prominent challenges I encountered during the internship and the strategies I employed to overcome them:

- **The complexity of Advanced Machine Learning Techniques:** Addressing the intricacies of advanced machine learning algorithms, including deep learning and ensemble methods, posed a significant challenge. The complexity of model architectures, parameter tuning, and optimization required thorough research and experimentation.

Strategy: I dedicated ample time to studying relevant literature, attending online courses, and engaging in practical exercises to build a strong foundation in advanced machine

learning techniques. Collaborative discussions with mentors and peers also played a crucial role in clarifying concepts and refining my approach.

- **Integration of Language Models (GPT-3 and BERT):** Integrating advanced language models such as GPT-3 and BERT into the e-commerce platform presented a technical challenge due to the need for seamless interaction, natural language understanding, and model compatibility.

Strategy: I approached this challenge methodically by breaking down the integration process into smaller steps. I began with a thorough documentation study, followed by experimentation in a controlled environment. Regular feedback from mentors and iterative testing allowed me to fine-tune the integration and address technical obstacles.

- **Data Quality and Pre-processing:** Dealing with real-world e-commerce data introduced data quality and pre-processing challenges, including missing values, noisy data, and data inconsistencies. Ensuring the integrity of the dataset for accurate model training was crucial.

Strategy: I adopted a systematic approach to data pre-processing, implementing data cleaning techniques, handling missing values, and performing feature engineering. Collaboration with domain experts within the team facilitated a deeper understanding of the data and effective strategies to handle data challenges.

- **Cloud Deployment and Scalability:** Deploying machine learning models on Google Cloud's Vertex AI, while ensuring scalability and resource optimization, required meticulous planning and configuration. Optimizing model performance while controlling costs was a balancing act.

Strategy: I conducted thorough research on cloud deployment best practices, closely following Google Cloud's documentation and guidelines. I engaged with cloud architecture experts to design an efficient deployment strategy that addressed scalability concerns while managing resources effectively.

- **Interdisciplinary Collaboration:** Collaborating with cross-functional teams demanded effective communication and collaboration across different domains. Translating technical concepts into understandable insights for non-technical stakeholders was a challenge.

Strategy: I honed my communication skills through regular interactions, status updates, and presentations to convey complex technical concepts clearly and concisely. Actively seeking feedback from team members and adapting my communication style based on their backgrounds proved to be invaluable.

- **Time Management and Prioritization:** Balancing multiple project goals, learning objectives, and deadlines required adept time management and prioritization skills. Ensuring that each task received adequate attention posed a continuous challenge.

Strategy: I embraced agile project management principles, breaking down tasks into manageable sprints, setting realistic goals, and regularly assessing progress. Collaborative discussions with mentors and colleagues helped in adjusting priorities and aligning efforts effectively.

In conclusion, the challenges encountered during the internship were integral to my growth as a professional. Through determination, proactive learning, and effective collaboration, I was able to navigate these challenges and emerge with valuable experiences, enhanced skills, and a deeper understanding of the intricate interplay between machine learning, data analytics, and artificial intelligence. These challenges have reinforced my resilience and adaptability, qualities that are crucial for success in the rapidly evolving landscape of technology and innovation.

Future Work

The internship with Lifesight Technologies provided a solid foundation for the development of advanced machine learning models for e-commerce analytics. The achievements and insights gained during this internship lay the groundwork for future endeavours that can further enhance the capabilities and impact of the e-commerce platform. Several avenues for future work and exploration have emerged, focusing on refining, and extending the existing framework:

- **Enhanced Sales Prediction and Time Series Analytics:** Building upon the developed time series forecasting models, future work can delve deeper into refining the accuracy and robustness of sales predictions. Exploring more sophisticated algorithms and integrating external factors such as market trends, seasonal variations, and economic indicators could lead to more precise sales forecasts.
- **Advanced Scientific Attribution Models:** The scientific attribution models implemented during the internship can be extended to encompass additional touchpoints and channels within the customer journey. Investigating the incorporation of multi-touch attribution models and experimenting with alternative algorithms can provide a more comprehensive understanding of the impact of marketing efforts on customer engagement.

- **Optimized Adoption of Advanced Language Models (LLMs):** While GPT-3 and BERT integration has enriched user interactivity, future work could involve fine-tuning the language models specifically for e-commerce contexts. Tailoring the models to provide contextually relevant and domain-specific responses could further enhance user engagement and satisfaction.
- **BigQuery Databases for Large Language Models (LLMs):** The exploration of BigQuery in the deployment of Large Language Models (LLMs) presents an exciting avenue for future research. Investigating the optimization of vector storage, retrieval mechanisms, and real-time interactions can contribute to more efficient and responsive user experiences.
- **Real-time Interactive Personalization (Lifesight and Store):** The adoption of advanced language models can be extended to develop real-time interactive personalization features within the Lifesight platform and the e-commerce store. Creating intelligent virtual assistants capable of understanding and responding to user queries and preferences can elevate user engagement and satisfaction.
- **Enriched Data Visualization and Interpretation:** Future work can focus on enhancing the data visualization capabilities to provide more interactive and dynamic visual representations. Integration with interactive visualization libraries, augmented reality, and virtual reality technologies could empower stakeholders to interact with data in innovative ways.
- **Integration of External Data Sources:** Expanding the scope of data analysis by incorporating external data sources, such as social media trends, macroeconomic indicators, and competitor insights, could provide a more holistic view of the e-commerce landscape and lead to more informed decision-making.
- **Ethical Considerations and Bias Mitigation:** Addressing ethical considerations and bias mitigation within machine learning models is a critical area for future work. Implementing fairness-aware algorithms and continuous monitoring of model outputs to identify and rectify potential biases is essential to ensure equitable and unbiased results.

In conclusion, the achievements of this internship to date serve as a strong foundation for future work that aims to refine, expand, and innovate within the realm of e-commerce analytics. By capitalizing on the insights gained and building upon the developed models and frameworks, the future work outlined above has the potential to drive advancements, enrich user experiences, and contribute to Lifesight Technologies' continued success in the rapidly evolving landscape of advanced AI technologies.

References

The successful completion of the internship was made possible with the guidance, insights, and resources provided by a range of reputable sources. The following references have been instrumental in shaping the methodologies, approaches, and understanding employed in the development of machine learning models for e-commerce analytics:

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Please note that these references provide a foundational understanding and have informed the approach taken in this internship report. They serve as a starting point for further exploration and validation of the methodologies employed in the development of customer lifetime value prediction, customer churn prediction, and product recommendations & consumer basket analysis.